

Identifying regulators of associative learning using a protein-labelling approach in *C. elegans*

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Aelon Rahmani, Anna McMillen, Ericka Allen, Radwan Ansaar, Renee Green, Michaela E Johnson, Anne Poljak, Yee Lian Chew 

Flinders Health and Medical Research Institute, College of Medicine and Public Health, Flinders University, Bedford Park, Australia • Bioanalytical Mass Spectrometry Facility, Mark Wainwright Analytical Centre, University of New South Wales, Sydney, Australia

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This **valuable** work defines a "learning proteome" for a *C. elegans* gustatory associative learning paradigm. These results provide the field with a new set of genes to further explore their roles in learning and memory, provide new tools for other labs to employ in their investigations of behavior, and molecular pathways relevant for *C. elegans* learning and memory. The methodological evidence and the quality of the dataset are **convincing**. The results will be of interest to neuroscientists and developmental biologists seeking to understand the self-assembly and operation of neural circuits for learning and memory.

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Abstract

The ability to learn and form memories is critical for animals to make choices that promote their survival. The biological processes underlying learning and memory are mediated by a variety of genes in the nervous system, acting at specific times during memory encoding, consolidation, and retrieval. Many studies have utilised candidate gene approaches or random mutagenesis screens in model animals to explore the key molecular drivers for learning and memory. We propose a complementary approach to identify this network of learning regulators: the proximity-labelling tool TurboID, which promiscuously biotinylates neighbouring proteins, to snapshot the proteomic profile of neurons during learning. To do this, we expressed the TurboID enzyme in the entire nervous system of *C. elegans* and exposed animals to biotin only during the training step of an appetitive gustatory learning paradigm. Our approach revealed hundreds of proteins specific to 'trained' worms, including components of molecular pathways previously implicated in memory in multiple species such as insulin signalling, G protein-coupled receptor signalling, and MAP kinase signalling. Most (87-95%) of the proteins identified are neuronal, with relatively high representation for neuron classes involved in locomotion and learning. We validated several novel regulators of

learning, including cholinergic receptors (ACC-1, ACC-3, LGC-46) and putative arginine kinase F46H5.3. These previously uncharacterised learning regulators all showed a clear impact on appetitive gustatory learning, with F46H5.3 showing an additional effect on aversive gustatory memory. Overall, we show that proximity labelling can be used in the brain of a small animal as a feasible and effective method to advance our knowledge on the biology of learning.

Introduction

All animals with a brain have the capacity to change their behaviour in response to changes in the environment. This capacity – to learn and remember – is essential for survival. There are numerous structural and molecular changes in the brain that modulate learning and memory in specific brain regions, occurring in a time and context-dependent manner (examples in (Huckleberry et al., 2016 [↗](#), Lin et al., 2010 [↗](#), Peixoto et al., 2015 [↗](#), Watteyne et al., 2020 [↗](#)); reviewed in (Bailey et al., 2015 [↗](#))). Research using model organisms has been essential towards understanding the key regulatory mechanisms underlying learning, many of which involve neurotransmitter signalling, neuromodulator signalling, signal transduction pathways, and cytoskeletal dynamics (Rahmani and Chew, 2021 [↗](#), Peng et al., 2011 [↗](#), Lamprecht, 2014 [↗](#)). Importantly, many of these mechanisms appear to be conserved across diverse species (Bailey et al., 2015 [↗](#), Matsumoto et al., 2018 [↗](#), Rahmani and Chew, 2021 [↗](#)).

Multiple studies have demonstrated that changes in the neuronal proteome are required for learning and memory formation (Inberg et al., 2013 [↗](#), Barzilai et al., 1989 [↗](#), Rosenberg et al., 2014 [↗](#)). New protein synthesis appears to be critical in several contexts, as the addition of a protein synthesis inhibitor (e.g. cycloheximide) has been shown to abolish long-term memory (Chen et al., 2012 [↗](#), Pedreira et al., 1995 [↗](#), Hernandez and Abel, 2008 [↗](#)). Moreover, protein degradation together with new protein synthesis has been strongly implicated in synaptic plasticity and memory formation (Lee et al., 2008 [↗](#), Fazeli et al., 1993 [↗](#), Park and Kaang, 2019 [↗](#)). There is also evidence that local translation in neurons, specifically the synthesis of specific proteins in dendritic regions (thereby altering local proteome composition), plays a key role in learning (Bradshaw et al., 2003 [↗](#), Das et al., 2023 [↗](#), Smith et al., 2005 [↗](#), Sutton and Schuman, 2006 [↗](#)). Additionally, several key regulatory proteins have been shown to be required at specific timepoints, such as during the training/learning step, to trigger memory formation (Stefanoska et al., 2023 [↗](#), Watteyne et al., 2020 [↗](#)). Taken together, these findings suggest that the spatiotemporal regulation of protein composition within neurons is critical for learning and memory formation.

The molecular requirements for learning have primarily been identified by combining genetic approaches with behavioural paradigms to test learnt associations, typically through assaying candidate genetic mutants or performing a forward genetics screen. These strategies have been extremely insightful; however, they have some limitations. The first being that candidate genetic screens are time-consuming and labour intensive, and require subjective selection of which candidate genes to test (for example (Hukema, 2006 [↗](#), Stein and Murphy, 2014 [↗](#))). The second is that large-scale screens tend to only reveal genes that have the strongest phenotypes, so genes that have more subtle phenotypes (Hiroki and Iino, 2022 [↗](#), Lindsay et al., 2022 [↗](#)), or act in redundant pathways (Feng et al., 2010 [↗](#), Gyurkó et al., 2015 [↗](#), Shahmoradi, 2015 [↗](#)), may not be identified using these approaches despite their contributions to learning.

To overcome these limitations, and to gain a holistic view of the molecular pathways that contribute to learning, we used an objective proteomics approach to snapshot the protein-level changes that occur specifically during learning. To do this, we expressed the proximity-labelling tool TurboID in the entire *C. elegans* nervous system, and used this to identify the proteins present in neurons during the training step of an associative learning paradigm we call ‘salt associative learning’. TurboID is an enzyme based on the BirA* biotin ligase, engineered to provide greater

catalytic efficiency (Branon et al., 2018 [↗](#)) compared with the original BirA* enzyme used in BioID experiments (Roux et al., 2012 [↗](#)). TurboID catalyses a reaction where biotin is covalently added onto lysine residues – as this process requires biotin, its timing can be controlled by depleting tissues of biotin, then adding it exogenously only at specific time points. Additionally, spatial control can be provided by regulating the site of TurboID expression using cell-specific transgenes. TurboID has been used in multiple studies for identification of protein-protein interactions, usually by tagging a ‘bait’ protein N- or C-terminally with the TurboID enzyme, allowing for rapid biotinylation of bait interactors. Through this approach, TurboID has been used for protein-tagging experiments in *C. elegans* (Artan et al., 2022 [↗](#), Sanchez et al., 2021 [↗](#), Holzer et al., 2021 [↗](#), Hiroki et al., 2022 [↗](#)). For example, this approach identified cytoskeletal proteins in *C. elegans* proximal to the microtubule-binding protein PTRN-1 (Sanchez et al., 2021 [↗](#)), and detected interactors for ELKS-1, which localises other proteins to the presynaptic active zone in the nervous system (Artan et al., 2021 [↗](#)).

In our study, rather than focussing on specific protein-protein interactions, we expressed TurboID that was not tagged with any bait protein in the entire nervous system of *C. elegans*, to identify as many proteins as possible within the cytoplasm. Using this approach, we identified hundreds of proteins specific to ‘trained’ worms, which we refer to here as the *learning proteome*, including those in molecular pathways previously shown to contribute to learning and memory formation in worms and other organisms. In addition, we validated several novel regulators of gustatory learning, including cholinergic receptors (ACC-1, ACC-3, and LGC-46), Protein kinase A regulator KIN-2, and putative arginine kinase F46H5.3. These proteins all show a clear impact on appetitive gustatory learning. F46H5.3 showed an additional effect on aversive gustatory learning, suggesting a more general role for this kinase in memory encoding. In summary, we have demonstrated that our approach to use proximity labelling to snapshot the brain of a small animal during training is a feasible and effective method to further our understanding of the biology of learning.

Results

TurboID expression in the nervous system of *C. elegans* successfully labels proteins during learning

To model learning in *C. elegans*, we used a simple yet robust associative learning paradigm called salt associative learning. Briefly, this assay involves training worms to associate the absence of salt (NaCl) with the presence of food. *C. elegans* are typically grown in the presence of salt (usually ~ 50 mM) and display an attraction toward this concentration when assayed for chemotaxis behaviour on a salt gradient (Kunitomo et al., 2013 [↗](#), Luo et al., 2014 [↗](#)). Training/conditioning with ‘no salt + food’ partially attenuates this attraction (group referred to ‘trained’). This is because the presence of abundant food (unconditioned stimulus) is a strong innate attractive cue, and pairing this with ‘no salt’ (the conditioned stimulus) leads to the animals showing the same behaviour towards the conditioned stimulus as they do to the unconditioned stimulus, i.e., attraction towards no salt, reflected as a preference for lower salt concentrations (Hiroki et al., 2022 [↗](#), Nagashima et al., 2019 [↗](#)). Similar behavioural paradigms involving pairings between salt/no salt and food/no food have been previously described in the literature (Nagashima et al., 2019 [↗](#)). Here, learning experiments were performed by conditioning worms with either ‘no salt + food’ (referred to as ‘salt associative learning’) or ‘salt + no food’ (called ‘salt aversive learning’).

To identify the learning proteome, we adapted this learning paradigm to incorporate TurboID-catalysed biotinylation of proteins specifically during the learning/conditioning step. We did this by (1) performing the salt associative learning assay on transgenic animals expressing TurboID in the entire nervous system (*Prab-3::TurboID*) and (2) adding biotin only when the worms are being trained (i.e., exposed to both food and ‘no salt’ in the ‘trained’ group, or to food and ‘high salt’ concentrations in the ‘high-salt control’ group). As an additional control, we performed the same

assay on non-transgenic (non-Tg) animals that do not express TurboID (**Figures 1A** & **1B**). We then isolated proteins from >3000 whole worms per group for both ‘high-salt control’ and ‘trained’ groups, most of which were subjected to a sample preparation pipeline for mass spectrometry, and some of which were probed via western blotting to confirm the presence of biotinylated proteins. The same pipeline was used to generate five biological replicates.

Validation of TurboID-catalysed biotinylation was performed in two ways: First, we compared total protein from naïve/untrained animals that are non-Tg versus TurboID-encoding by western blot and probed for V5-tagged TurboID: as expected, we observed expression in transgenic worms only at the predicted size (39 kDa) (**Figure 1C**). Secondly, we tested if exposure to biotin increased the biotinylation signal in a TurboID-dependent manner. To do this, we quantified the biotinylation signal in (1) naïve non-Tg worms not exposed to biotin (2) non-Tg *C. elegans* exposed to biotin for 6 h, (3) naïve TurboID worms not exposed to biotin, and (4) TurboID animals exposed to biotin for 6 h. Although background biotinylation was present in worms not treated with biotin, we found that biotin exposure increased the signal 1.3-fold for non-Tg and 1.7-fold for TurboID *C. elegans* (**Figure S1**). Taken together, these findings indicate that there is increased biotinylation of proteins in the presence of both biotin and the TurboID enzyme.

Mass spectrometry experiments were performed with the following experimental groups per replicate: (1) non-transgenic/non-Tg high-salt control, (2) non-Tg trained, (3) TurboID high-salt control, and (4) TurboID trained. We did not include no-biotin treatment controls due to the practical challenges of handling >4 groups in the combined learning assay/mass spectrometry pipeline, for which >3000 worms are required per group. Therefore, all groups were exposed to biotin during the 6 h-exposure period to food and either high salt (for control) or no salt (for trained) (**Figure 1A**).

To confirm that each experimental group displayed the expected phenotype after training, a portion of worms from all groups were tested using a chemotaxis assay. The chemotaxis index (CI) was used as a readout of learning performance: a positive CI reflects high salt preference, a CI close to 0 represents a more neutral response, and a negative CI represents low salt preference (**Figure S2**). We confirmed after each learning assay that naïve/untrained worms had a strongly positive CI (~0.7-0.9), whereas trained animals showed a lower CI (~0.0). We also performed a learning control (indicated as high-salt ‘control’) in which the presence of food (the US) is paired with high salt concentrations – worms in this group are attracted to high salt and showed a strongly positive CI (~0.7-0.9) (**Figure 1B**), displaying a similar behaviour to naïve worms. This behavioural change seen in trained animals, versus the naïve and high-salt control groups, represented successful learning as seen in previous studies (Hiroki et al., 2022, Nagashima et al., 2019). This was observed in both non-Tg and transgenic animals, confirming that introducing the transgene did not perturb learning (**Figure 1B**).

We also confirmed by western blotting that biotinylated proteins could be observed in TurboID-expressing high-salt control and trained groups (**Figure 1C**). As in other *C. elegans* studies utilising TurboID, we saw background biotinylation in non-Tg controls; however, this is visually lower compared with groups from TurboID transgenic worms (**Figure 1C**) (Artan et al., 2021, Sanchez et al., 2021). Quantification of the signal within entire lanes showed a 1.1-fold increase in the ‘TurboID, control’ lane compared with the ‘non-Tg, control’ lane, and a 1.9-fold increase in the ‘TurboID, trained’ lane compared with the ‘non-Tg, trained’ lane. For all replicates, we determined that biotinylated proteins could be observed from total TurboID-positive worm lysate by western blotting before proceeding with downstream proteomic experiments (**Figure S3** & Table S2).

Our sample preparation methodology for mass spectrometry is based on similar protocols used in *C. elegans* and other systems (Artan et al., 2022, Sanchez et al., 2021, Prikas et al., 2020). We performed five biological replicates, in line with other *C. elegans* studies (Artan et al., 2022,

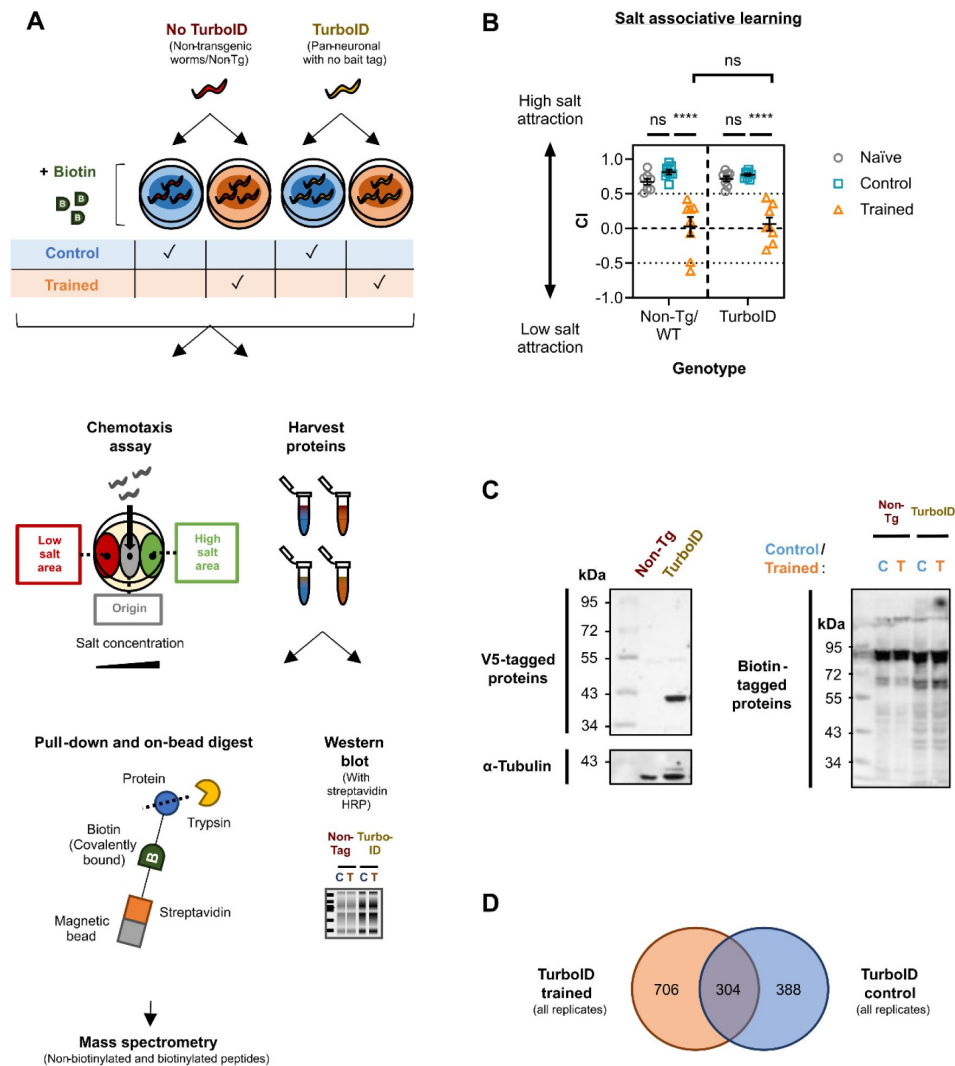


Figure 1

Summary of TurboID approach for protein labelling in all *C. elegans* neurons during learning.

(A) Workflow for mass spectrometry-based analysis. Biotin-depleted animals without (non-transgenic/Non-Tg, red) or with TurboID (transgenic, yellow) were exposed to 1 mM of exogenous biotin during conditioning by pairing food with 'no salt' (orange - trained) or 'high salt' (blue - control). >3000 worms were used per group /biological replicate (n = 5) – a small proportion of each group was tested in a chemotaxis assay to assess learning capacity, while the rest was subjected to sample preparation steps for mass spectrometry (see **Experimental Procedures**). Some harvested protein was probed via western blot for the presence of biotinylated proteins or V5-tagged TurboID (see **panel C** for representative image from replicate 1). **(B)** The graph shows chemotaxis assay data for Non-Tg/wild type (WT) and transgenic *C. elegans* following salt associative learning. Each data point represents a 'chemotaxis index' (CI) value for one biological replicate (n = 8). Each biological replicate includes three technical replicates (26-260 worms /technical replicate). **Statistical analysis:** Two-way ANOVA & Tukey's multiple comparisons test (**** ≤ 0.0001 ; ns = non-significant). Error bars = mean $\pm$ SEM. **(C)** Western blots to visualise V5-tagged TurboID and biotinylated proteins. The left side shows V5-tagged TurboID visualised using 18 μ g total protein from naïve worms per lane (39 kDa). Non-Tg protein lysates acted as a negative control. α tubulin was probed as a loading control. The right side shows biotinylated proteins visualised from 25 μ g total protein per lane from control (C) or trained (T) worms with streptavidin-horseradish peroxidase (HRP). **(D)** Venn diagram comparing all proteins assigned an identity by MASCOT from peptides detected by mass spectrometry from transgenic worms. Values represent the number of proteins listed as detected in 'TurboID, control' (blue) and 'TurboID, trained' (orange). These lists were generated by first subtracting proteins identified in corresponding Non-Tg lists, and then comparing both control and trained TurboID lists. The overlap represents proteins unique to 'TurboID, trained' worms in ≥ 1 replicate/s that were also detected in 'TurboID, control' worms in ≥ 1 other replicate/s.

Holzer et al., 2021 [↗](#)). To examine the learning proteome, we first subtracted proteins from ‘TurboID, trained’ groups also present in ‘Non-Tg, trained’ samples to generate a protein list specific to ‘TurboID, trained’ animals for each biological replicate. We next subtracted from ‘TurboID, control’ lists any proteins that appeared in ‘Non-Tg, control’ samples to generate a revised ‘TurboID, control’ protein list specific to each replicate. We then compared revised protein lists for ‘trained’ and ‘control’ worms from all biological replicates and examined both unique and shared proteins between these two groups. We found 304 proteins that were shared between ‘trained’ and ‘control’ TurboID groups, 706 proteins unique to the ‘TurboID, trained’ group, and 388 proteins unique to the ‘TurboID, control’ group (**Figure 1D** [↗](#)). We refer to the *learning proteome* as proteins unique to samples for ‘TurboID, trained’ worms. When generating the learning proteome, we categorised proteins as ‘assigned hits’ based on the criteria that at least one unique peptide was identified by the MASCOT search engine for the protein identity from at least one biological replicate. We also examined peptide sequences in our peak lists that were considered ‘unassigned’ by MASCOT, as these sequences were not detected as unique for any protein by the software, but specific protein identities could be found by performing a Basic Local Alignment Search Tool (BLAST) query (<https://blast.ncbi.nlm.nih.gov/Blast.cgi> [↗](#); see **Experimental Procedures** for details). The Venn diagram in **Figure 1D** [↗](#) shows assigned hits only. Learning proteome lists for both assigned and unassigned hits are in **Tables S3 & S4**.

Examination of the learning proteome reveals known regulators of learning and memory

Our initial analysis of the learning proteome sought to validate our TurboID-based approach by identifying components of biological pathways previously implicated in learning. We then performed a gene ontology (GO) term analysis of ‘cellular component’ to obtain a broad overview of the subcellular localisation of proteins identified in trained animals (**Figure S4A** [↗](#)). To do this, we generated protein-protein interaction (PPI) networks of assigned protein hits within the learning proteome for subcellular components of interest (**Figure S4B-G** [↗](#)), using data from STRING and curated with the Cytoscape ClueGO tool (**Tables S5 & S6**) (Bindea et al., 2009 [↗](#)). We found that a majority of proteins were categorised as ‘cytoplasmic’ (28.1%) as expected from our approach, which utilised the TurboID enzyme not tagged to any bait protein; this means that we would anticipate the enzyme to be present relatively evenly across the cell body and to catalyse biotinylation of proteins in this space. We saw that an unexpectedly high proportion of proteins were nuclear (18.1%), despite the presence of a nuclear export signal in our TurboID transgene, which should prevent TurboID from entering the nucleus and biotinylating nuclear proteins – this could be due to some ‘leaky’ entry of the enzyme or biotin reactive species into the nucleus, or that some proteins are categorised solely as ‘nuclear’ in the ClueGO database when they are also present in other cellular components. Importantly, we found that a proportion of proteins are categorised as present in neuronal compartments – the pre-synapse (0.5%), cilia/dendrites (2.7%), and in the axon/(synaptic) vesicles (4.0%) – as expected from transgenic expression of TurboID in the nervous system.

Learning and memory formation in organisms with brains of varying sizes have been shown to involve key regulatory pathways including signalling via neurotransmitters/neuromodulators, G protein-coupled receptors (GPCR), the mitogen-activated protein kinase (MAPK) pathway, and the insulin/insulin growth factor-like pathway (Matsumoto et al., 2018 [↗](#), Myhrer, 2003 [↗](#), Rahmani and Chew, 2021 [↗](#)). We next categorised proteins within the learning proteome (**Figure 2** [↗](#)) based on their known roles within these signalling pathways. This included (1) several GPCR components, including the $G_{i/o}$ protein subunit GOA-1 and G_q protein subunit GPA-2, (2) regulators of insulin signalling including the DAF-2 insulin receptor, phosphoinositide 3-kinase AGE-1, and serine/threonine protein kinases AKT-1 and SGK-1, which were previously reported to modulate salt-based learning in the worm (Tomioka et al., 2006 [↗](#), Sakai et al., 2017 [↗](#)), (3) MAPK signalling components including NSY-1/MAPKKK, MEK-2/MAPKK, and MPK-1/MAPK/ERK, (4) cAMP/PKA (protein kinase A) signalling regulators such as the regulatory PKA subunit KIN-2, and (5) multiple

components that modulate synaptic vesicle release, including N-ethyl-maleimide sensitive fusion protein NSF1/NSF-1 and syntaxin/SYX-2. In addition, we identified several proteins relevant to glutamate, acetylcholine, and GABAergic signalling. Several components involved in protein synthesis and degradation were also detected in our learning proteome, in line with studies that suggest changes in total protein composition following memory formation (Inberg et al., 2013 [↗](#), Barzilai et al., 1989 [↗](#)). These data are summarised in **Figure 2** [↗](#), **Figure S5** [↗](#), and Table S6. In summary, the learning proteome includes both known learning regulators and potentially novel candidates that warrant further study. We focused on proteins functioning within these pathways of interest in our subsequent investigations (highlighted nodes in **Figures 2** [↗](#), **S4** [↗](#), & **S5** [↗](#)).

In addition, we consistently observed enrichment of two metabolic pathways, fatty acid metabolism via the TCA cycle and carbohydrate metabolism (gluconeogenesis and glycolysis), in multiple biological replicates of mass spectrometry data, uniquely in TurboID-trained animals (**Figure 3** [↗](#)). These metabolic pathways play essential roles, including in cellular energy production and macromolecule biosynthesis (Krebs and Johnson, 1937 [↗](#), Lu and Goetsch, 1993 [↗](#)). Consequently, their disruption can severely impair animal health. For example, knock-down of mitochondrial components involved in the TCA cycle led to larval arrest and/or severely reduced lifespan in *C. elegans* (Artan et al., 2022 [↗](#), Liao et al., 2022 [↗](#)). This limits the capacity to assess these processes in learning using single gene mutants or knockdown tools. Therefore, our TurboID approach reveals biological pathways potentially involved in memory formation that are not detectable through conventional forward or reverse genetic screens.

Exploring neuron class representation within the learning proteome

Aside from identifying relevant biological networks, we also used the data from the learning proteome to identify potential neuron classes involved in memory formation. We based this analysis on two databases, the Wormbase Tissue Enrichment Analysis (TEA) Tool (Angeles-Albores et al., 2016 [↗](#)), based on Anatomy Ontology (AO) terms, and single-cell transcriptomics data from the *C. elegans* Neuronal Gene Expression Network (CeNGEN) (Taylor et al., 2021 [↗](#)). Firstly, we employed the CeNGEN database to confirm that 87-95% of assigned hits and 89-92% of all hits (assigned and unassigned hits) from the learning proteome show neuronal expression, i.e. were found in at least one neuron in the database (**Table 1** [↗](#)). Secondly, we assessed which tissues and neuron classes are most highly represented within the learning proteome. We used the Wormbase TEA tool to search for gene lists corresponding to proteins encoded by (1) assigned hits only and (2) both assigned and unassigned hits within the learning proteome. Anatomical terms were considered enriched when they had a q value < 1. We observed enriched terms for pharyngeal neurons (M1, M2, M5, NSM, & I4), sensory neurons (PVD), interneurons (ADA and RIG), ventral nerve cord (VNC) motor neurons (VB2, VB3, VB4, VB5, VB6, VB7, VB8, VB9, VB10, & VB11), and CAN cells from both gene lists. RIS interneurons and DD motor neurons were also enriched when including unassigned hits. Several of these neurons have previously been implicated in learning: RIG interneurons (Zhou et al., 2023 [↗](#)) and NSM neurons in butanone olfactory learning (Fadda et al., 2020 [↗](#)), VNC neurons through changes in glutamate receptor GLR-1 expression during touch habituation (Rose et al., 2003 [↗](#)) and diacetyl aversive learning (Vukojevic et al., 2012 [↗](#)), and RIS interneurons in salt aversive learning (Wang et al., 2025 [↗](#)). Therefore, neurons enriched within the learning proteome include those known to be required for learning; other neurons not previously identified in this context, such as pharyngeal neurons, may warrant further study.

We complemented this analysis by using the CeNGEN database to search for gene lists encoding proteins (assigned hits only, minus non-transgenic controls) identified in control worms (388 genes) versus trained animals (706 genes) from **Figure 1D** [↗](#) (Taylor et al., 2021 [↗](#)). Using this database, we determined the number of genes from each list that are expressed in a specific neuron type. Values for the trained gene list were normalized to account for the ~1.8-fold increase

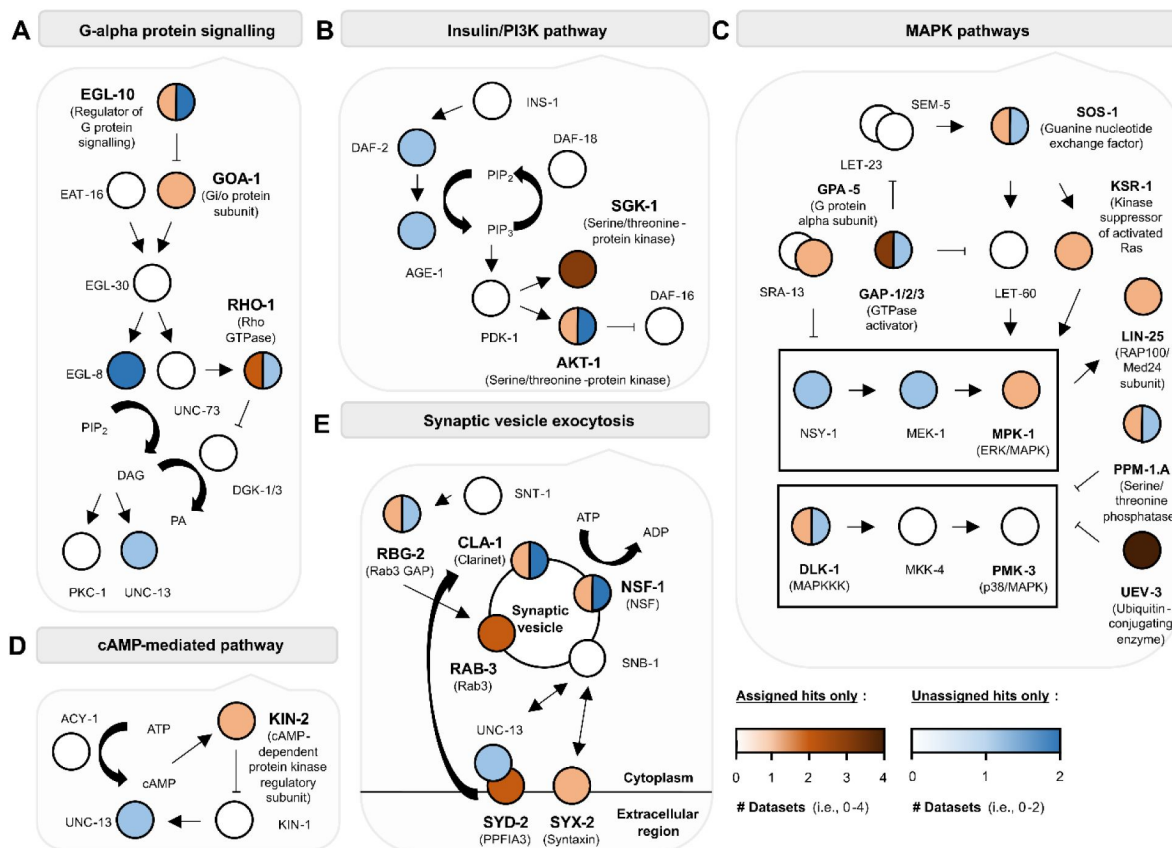


Figure 2

Molecular pathways previously implicated in associative learning are detected in our learning proteome.

Proteins detected from 'TurboID, trained' worm lysates by mass spectrometry are in bold with circles coloured as orange ('assigned hits' assigned protein identities by MASCOT) and/or blue ('unassigned hits' given protein identities by bulk BLAST searching, but not MASCOT). Darker colours mean the protein has been detected in more than one biological replicate (see legend).

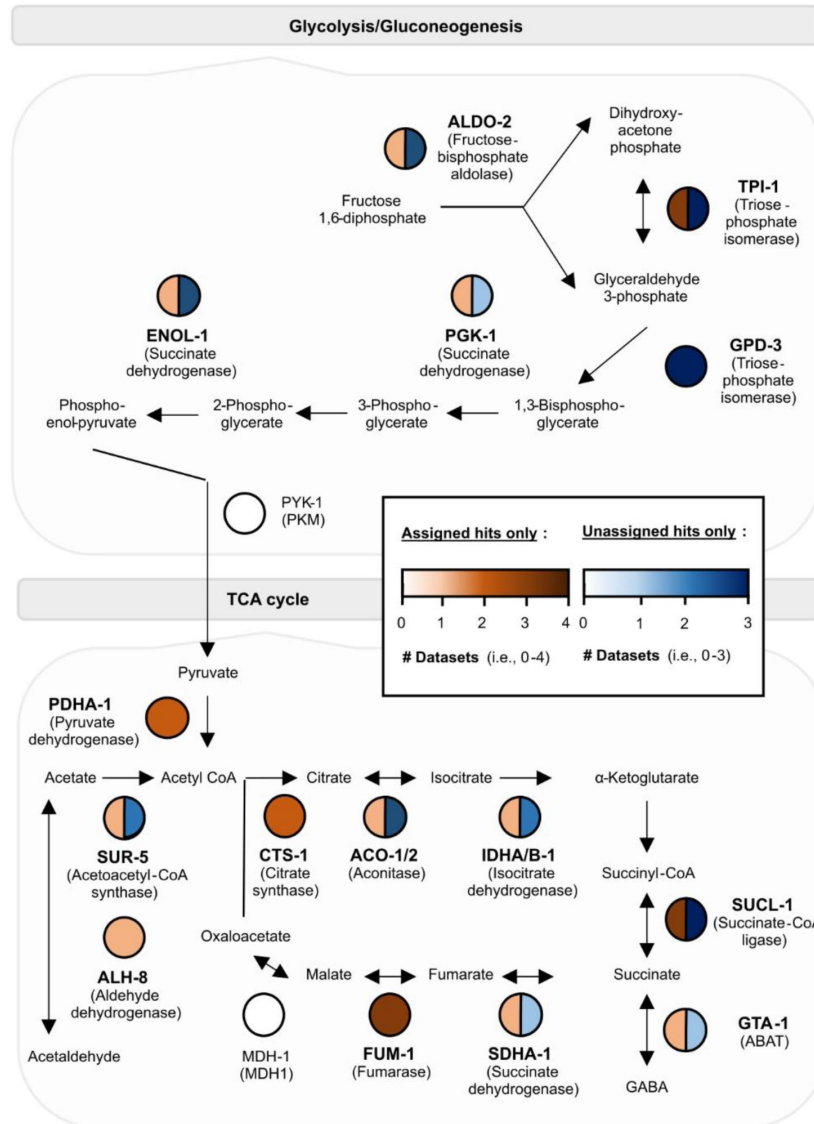


Figure 3

Schematics for metabolic processes represented in the learning proteome.

The molecular pathways above are **(A)** carbohydrate metabolism (glycolysis & gluconeogenesis) and **(B)** fatty acid metabolism (via the tricarboxylic acid or TCA cycle). Each protein is a node in white (not detected by TurboID during learning), orange (an 'assigned hit'), and/or blue (an 'unassigned hit') based on mass spectrometry data from 'TurboID; trained' worms. Darker colours mean the protein has been detected in more than one biological replicate (see legend).

Biological replicate	1	2	3a	3b	4	5
Mass spectrometer used	QE	QE	QE	Exploris	Exploris	Exploris
#Assigned hits	364	159	97	237	202	274
#All hits	675	516	279	455	708	578
% Neuronal for assigned hits	93	91	95	91	87	91
% Neuronal for all hits	91	89	92	90	89	89

Table 1

Neuron-specific expression within the learning proteome.

Mass spectrometry runs (n = 5) were performed with the ThermoFisher Scientific Q-Exactive Orbitrap ('QE') and/or ThermoFisher Scientific Orbitrap Exploris ('Exploris'), for technical reasons. There are six lists because replicate #3 was run on both mass spectrometers: corresponding protein lists are annotated as '3a' and '3b' respectively. The total '#Assigned hits' versus '#All hits' (assigned + unassigned hits) are shown in rows are listed above. The CeNGEN database (threshold = 2) was used to determine corresponding percentages for assigned hits versus all hits as '% Neuronal for assigned hits' versus '% Neuronal for all hits' (Taylor et al., 2021 [DOI](#)). The average percentages across all replicates were 91% for assigned hits only versus 89% for all hits.

in the number of proteins detected in trained samples compared to the high-salt control. For each neuron class that appeared in both datasets (128 in total), we calculated fold-change values between the number of genes from trained vs control gene lists. Neurons were ranked in descending order of fold-change. This ranked list is based on the relative enrichment of training-associated genes compared to control, with higher ranks suggesting neurons that may be more transcriptionally responsive or involved during training. These data are summarised in Table S7.

Cholinergic and glutamatergic neurons constituted 46% and 31% of these neurons, respectively. Glutamate signalling components previously have been implicated in *C. elegans* learning paradigms involving salt (e.g., NMDA-type glutamate receptor subunits *nmr-1* and *nmr-2*) (Kano et al., 2008 [↗](#)). Acetylcholine has not been explored extensively in *C. elegans* for its involvement in learning, but has been described in other animal models and in humans (reviewed in (Huang et al., 2022 [↗](#))). Other neuron classes identified have previously been implicated in salt-based associative learning (ranks in brackets): AVK interneurons (rank #7) (Beets et al., 2012 [↗](#)), RIS interneurons (rank #14) (Wang et al., 2025 [↗](#)), salt-sensing neuron ASEL (rank #18) (Beets et al., 2012 [↗](#)), CEP and ADE dopaminergic sensory neurons (ranks #22 & #39 respectively) (Vogliss and Tavernarakis, 2008 [↗](#)), and AIB interneurons (rank #21) (Sato et al., 2021 [↗](#)). Interestingly, unlike its counterpart ASEL, the salt-sensing neuron ASER was ranked only #104/128 (Table S7). ASER becomes activated in response to a decrease in salt concentration (Suzuki et al., 2008 [↗](#)) and its downstream targets likely function to redirect worms toward higher salt concentrations (Appleby, 2012 [↗](#)). This activation is suppressed after training that reduces attraction to high salt levels (Sato et al., 2021 [↗](#), Wang et al., 2025 [↗](#)). It is possible that this learning-dependent suppression of ASER activity may explain its lower fold-change in trained versus control groups.

Some neurons identified were not previously implicated in learning: IL1 polymodal head neuron class (rank #1), motor neuron DA9 (rank #2), and interneuron DVC (rank #5). IL1 releases glutamate (Pereira et al., 2015 [↗](#)) and mainly functions in regulating foraging behaviour (Hart et al., 1995 [↗](#)), potentially indicating a role in food-based responses. Separately, cholinergic neuron DA9 and glutamatergic neuron DVC are involved in backward locomotion (Pereira et al., 2015 [↗](#), Ardiel and Rankin, 2015 [↗](#), Chalfie et al., 1985 [↗](#)). Changes in locomotion are critical for learning-dependent modulation of chemotaxis: the incidence of sharp turns or ‘pirouette’ movements in *C. elegans* is influenced by prior experience in salt-based gustatory learning (Kunitomo et al., 2013 [↗](#)). IL1 may influence salt-based learning by signalling through interneurons AVE (rank #24) and PVR (rank #114) to the DA neurons (Bhatla, 2009 [↗](#)), potentially modulating backward locomotion as part of the chemotaxis response. We also identified pharyngeal neurons I3 (rank #4) and I6 (rank #5), which have not previously been implicated in learning. **Figure 4** [↗](#) provides a summary for the neural circuits implicated from these analyses, where neuron classes are highly connected to each other. Investigating the role of specific genes within these circuits opens new avenues for future research into gustatory learning.

Validating the requirement of learning proteome components in salt associative learning through single gene studies

Our initial analysis of learning proteome data indicates that there are multiple hits present in biological pathways important for neuron function, and that are potentially relevant to learning and memory formation. To test this directly, we performed salt associative learning experiments on selected learning proteome hits (**Figures 5** [↗](#), **6**, **S7**, & **S8**). We used the following general rules to interpret our data: if the average chemotaxis indices (CIs) for ‘trained’ worms were *higher* in a particular strain compared with wild-type, this strain was considered learning-defective, as this reflects a reduced magnitude of the expected behaviour change (an increased preference for low salt demonstrated by CIs closer to 0 or negative CIs). If the average CI for ‘trained’ worms was *lower* in a strain compared with wild-type, then this strain was considered to display ‘better’ learning, as the lower CI reflects an increased magnitude of the expected behaviour change. In

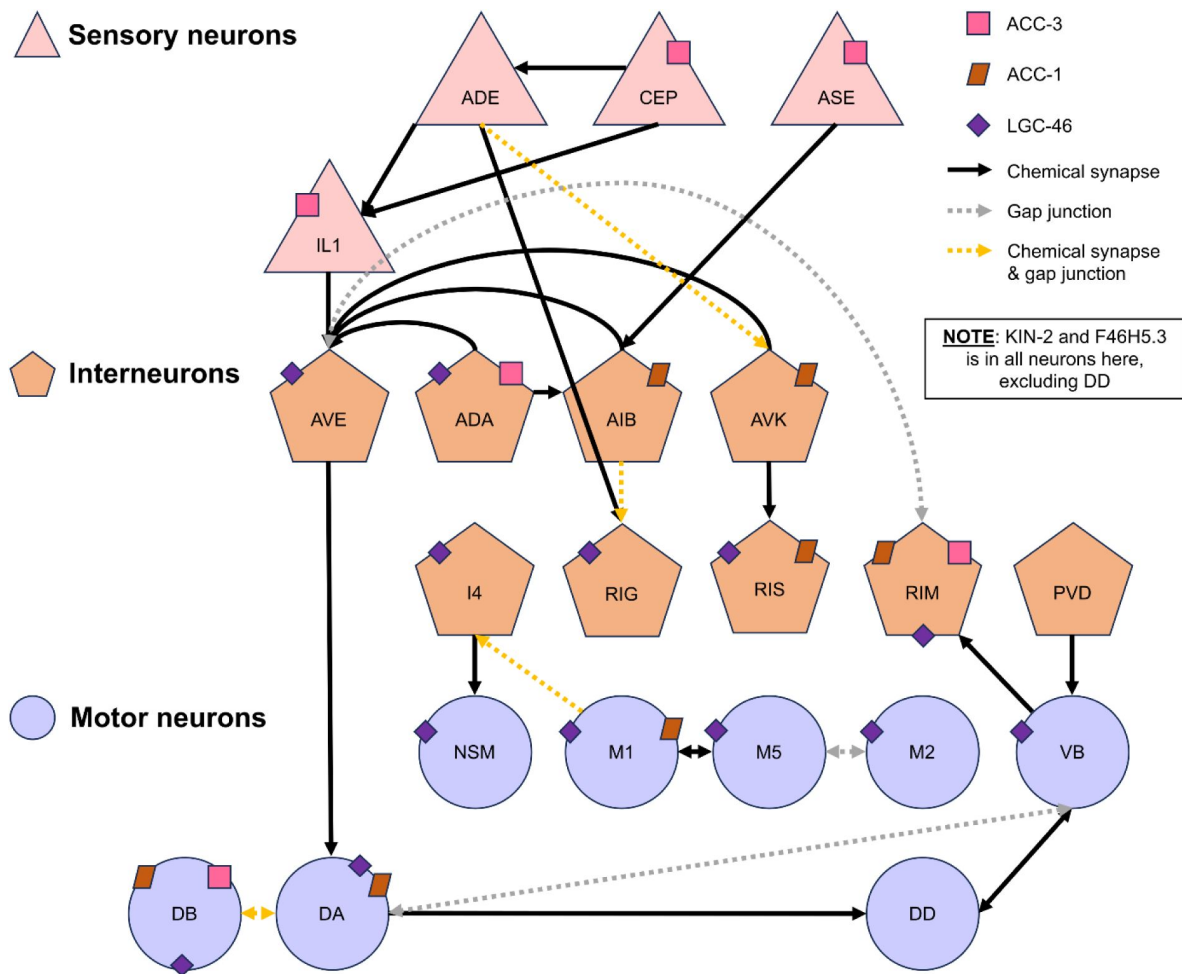


Figure 4

Schematic for neural circuit containing neuron classes represented by the learning proteome.

Neuron classes represented by the learning proteome were identified using the gene enrichment tool from WormBase (Angeles-Albores et al., 2016) and the CeNGEN database (threshold = 2) (Taylor et al., 2021). Neurons are represented by pink triangles (sensory), orange pentagons (interneurons), and purple circles (motor neurons). Chemical synapse (black arrows) and gap junction (dotted arrows: grey for gap junctions only or yellow for synapses and gap junctions) information is provided using the software WormWeb (Bhatla, 2009). Learning regulators validated in this study are also represented: ACC-1 (brown rectangles), ACC-3 (pink squares) and LGC-46 (purple diamonds) are annotated above based on single neuron expression profiles from CeNGEN (Taylor et al., 2021). Notably, KIN-2 and F46H5.3, discussed in detail below, are expressed in all neurons shown except for DD.

general, we observed no significant difference in CIs between naïve groups for all genotypes, reflecting no gross locomotor or chemotaxis defects in the strains tested (**Figures 5**, **6**, **S7**, & **S8**).

We tested 26 candidates in total for this study. Although this represents a small subset of the 706 proteins identified in the learning proteome, several proteins in the full list are unsuitable for functional testing due to key constraints: (1) having essential roles, with corresponding single-gene mutants being lethal; (2) involvement in neurodevelopment rather than mature neuronal function; and (3) being required for locomotion, with severe locomotion defects precluding assessment using chemotaxis assays.

Candidates tested were classified as either strong (detected in biological replicates ≥ 3) or weak (replicates < 3) based on the number of mass spectrometry replicates in which they were uniquely identified in TurboID-trained *C. elegans* (shown in brackets). We determined these numbers by considering both assigned and unassigned protein lists, which contained mostly neuron-expressed proteins (**Table 1**) including known learning regulators (**Figure 2** & **Table S8**). The list of 26 candidates for further testing includes both weak and strong hits. In addition, although candidates tested were mostly detected in more replicates of trained versus control groups, we also assayed seven candidates for which this was not the case. **Table 2** summarises the potential learning regulators explored in this study, including strong/weak classifications and replicate numbers between experimental groups.

We first tested the regulatory subunit of PKA, *kin-2* (1 replicate), since it is a known regulator of memory and was detected as a weak candidate by TurboID. Adenylyl cyclase is a key signalling effector for $G\alpha_s$ and $G\alpha_i$ proteins and regulates levels of the secondary messenger cyclic AMP (cAMP) within the cell. cAMP binding to PKA regulates its activity, and therefore its downstream effects (Sassone-Corsi, 2012). We tested worms with a mutation in *kin-2* (R92C) in which a conserved residue in the inhibitory domain (which normally functions to keep PKA turned off in the absence of cAMP) is mutated, resulting in increased PKA activity (Schade et al., 2005). *kin-2* has previously been shown to be required for intermediate-term memory in *C. elegans* (Stein and Murphy, 2014), with cAMP/PKA signalling previously shown to be involved in memory in multiple systems (Kandel, 2012). We found that these *kin-2* mutant animals showed enhanced learning compared with wild-type (i.e., Non-Tg worms) (**Figure 5A**). We next re-expressed the *kin-2(R92C)* allele in wild-type worms using a pan-neuronal promoter, these worms showed a similar phenotype to *kin-2(ce179)* worms, with enhanced learning compared with non-transgenic siblings (**Figure 5B**). These data suggest that increased PKA activity in the nervous system drives salt associative learning.

We next assessed two strong candidates not previously assessed for their role in learning: putative arginine kinase F46H5.3 (4 replicates) or armadillo-domain containing protein C30G12.6 (3 replicates). Unlike *kin-2(ce179)* worms, neither single gene mutant obtained from the *Caenorhabditis* Genetics Center had been backcrossed. We backcrossed mutant strains four times to N2, and tested both non-backcrossed and backcrossed versions. An improved learning phenotype was displayed by both non-backcrossed and backcrossed F46H5.3(-) worms (**Figures 6A** & **6B**). In contrast, we found that non-backcrossed C30G12.6(-) animals displayed an enhanced learning phenotype, whereas backcrossed C30G12.6(-) mutants behaved like wild-type (**Figures 6C** & **6D**). This suggests that the non-backcrossed C30G12.6(-) strain contains a background mutation that impacts learning capacity, a potential avenue for future work.

F46H5.3 is a homolog for creatine kinase B (cytoplasmic) and mitochondrial creatine kinase 1B, but is considered an arginine kinase since *C. elegans* use arginine instead of creatine. Both creatine kinases are essential for energy metabolism via ATP modulation. Given notable representation of metabolic pathways from the learning proteome identified here (e.g., **Figure 3**), this provided additional rationale for testing F46H5.3. Moreover, the depletion of creatine kinase B reportedly

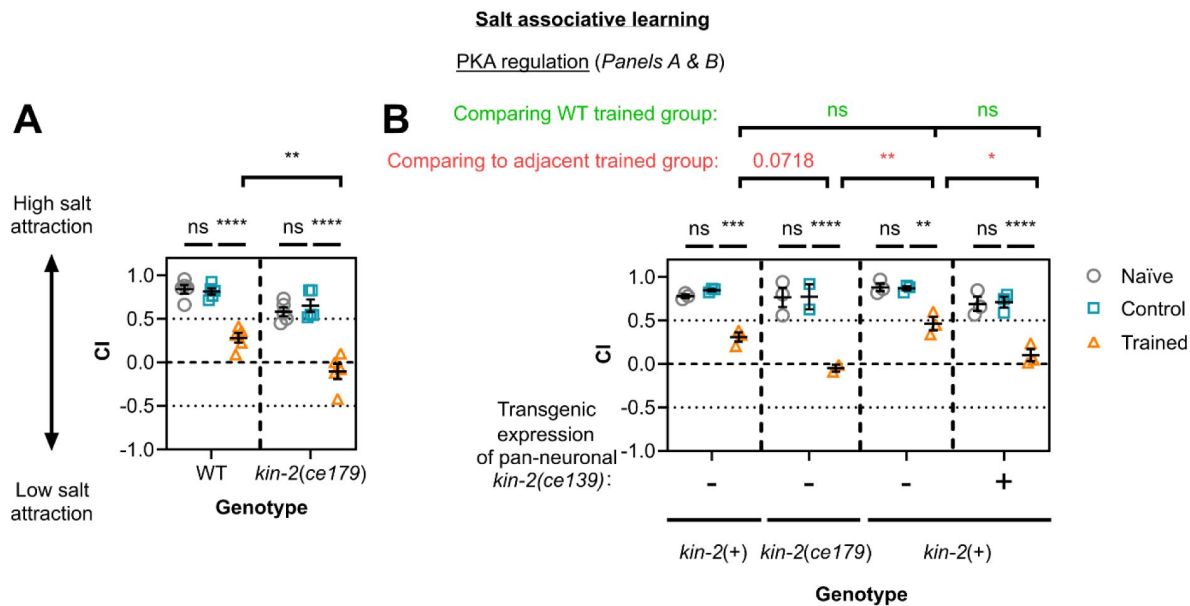


Figure 5

C. elegans PKA regulatory subunit KIN-2 acts in neurons to regulate salt associative learning.

Salt chemotaxis behaviour was measured in the form of chemotaxis indices (CI) for naïve/untrained worms (grey circles), high-salt control (blue squares), and trained worms (orange triangles). This was done for **(A & B)** wild-type (WT) animals, **(A & B)** *kin-2(ce179)* mutants, and **(B)** transgenic worms with a WT background engineered to overexpress KIN-2 from the *ce179* allele in all neurons (10-60% transgenic worms per technical replicate, both non-transgenic (-) and transgenic (+) siblings are plotted above). Each data point represents one biological replicate where **(A)** $n = 5$ and **(B)** $n = 3$ (one biological replicate was excluded from high-salt control and trained *kin-2(ce179)* groups due to insufficient sample size). **(A)** 32-487 worms and **(B)** 5-184 worms per technical replicate. Error bars = mean $\pm$ SEM. Annotations above graphs represent P-values from Two-way ANOVA and Tukey's multiple comparison tests (**** ≤ 0.0001 ; *** ≤ 0.001 ; ** ≤ 0.01 ; * ≤ 0.05 ; ns = non-significant). **(B)** Statistical comparisons between WT trained and siblings in transgenic lines are in red (top row), between adjacent trained groups are in green (middle row), and between groups within each line in black (bottom row).

Table 2

Summary of candidates assessed for their effect in learning.

The number (#) of biological replicates (total n = 5) in which each candidate was detected as an assigned hit (by the MASCOT software) or in assigned + unassigned hits (identified by bulk BLAST search) is provided under ‘# Biological replicates in TurboID trained’ and ‘# Biological replicates in TurboID high-salt control’ columns. These values exclude proteins from non-transgenic trained and non-transgenic high-salt control groups, respectively. Orange highlights indicate candidates detected in more replicates in the TurboID-trained group. Candidates are also defined as ‘weak’ or ‘strong’ based on the frequency of detection across biological replicates.

Candidates tested	# Biological replicates in <i>TurboID</i> trained (assigned hits)	# Biological replicates in <i>TurboID</i> high-salt control (assigned hits)	# Biological replicates in <i>TurboID</i> trained (assigned + unassigned hits)	Classification for candidate
IFT-139	4	1	5	Strong
ACR-2	1	0	4	Strong
F46H5.3	3	2	4	Strong
SAEG-1	2	0	4	Strong
UEV-3	4	1	4	Strong
AEX-3	0	0	3	Strong
C30G12.6	0	0	3	Strong
ELO-6	3	0	3	Strong
ELP-1	2	1	3	Strong
FSN-1	0	0	3	Strong
GAP-2	2	0	3	Strong
RIG-4	0	1	3	Strong
TAG-52	1	0	3	Strong
TAP-1	2	0	3	Strong
VER-3	3	0	3	Strong
ACC-3	1	0	2	Weak
DLK-1	1	0	2	Weak
GBB-2	2	0	2	Weak
GPA-2	2	0	2	Weak
RHO-1	2	1	2	Weak
ACC-1	1	1	1	Weak
GAP-1	1	0	1	Weak
GLR-1	0	1	1	Weak
KIN-2	1	0	1	Weak
LGC-46	1	1	1	Weak
MACO-1	1	0	1	Weak

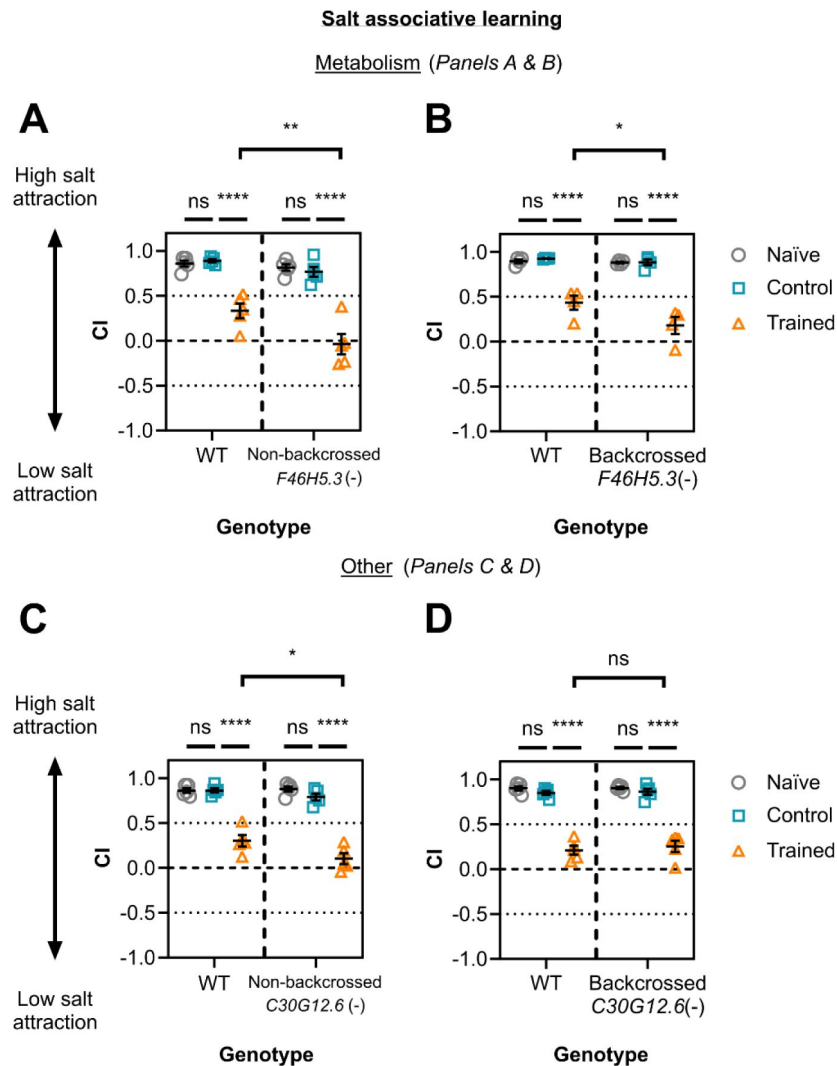


Figure 6

Salt associative learning is dependent on arginine kinase F46H5.3 and not armadillo-domain containing protein C30G12.6.

Chemotaxis indices (CI) are shown for wild-type/WT animals versus mutants for **(A)** *F46H5.3* (non-backcrossed with WT, $n = 5$), **(B)** *F46H5.3* backcrossed with WT ($n = 4$), **(C)** *C30G12.6* (non-backcrossed with WT, $n = 5$), and **(D)** *C30G12.6* backcrossed with WT ($n = 5$). These animals were assessed for salt associative learning by preparing three groups for each line: naïve/untrained (grey circles), high-salt control (blue squares), and trained (orange triangles) (27-395 worms per technical replicate). Each data point is for one biological replicate each comprising of three technical replicates. Error bars = mean $\pm$ SEM. Statistical analyses were done by Two-way ANOVA and Tukey's multiple comparison test (**** ≤ 0.0001 ; ** ≤ 0.01 ; * ≤ 0.05 ; ns = non-significant).

increases the latency needed for memory encoding of a spatial learning task in mice (Jost et al., 2002). Since F46H5.3 is expressed in most neurons in the worm (Taylor et al., 2021), this putative arginine kinase may affect learning capacity through modulating ATP levels in neurons.

We also tested genetic mutants for candidates involved in neurotransmission, a key function for all neurons and a requirement for learning (reviewed in Myhrer (2003) and Rahmani and Chew (2021)). Two strong candidates, cationic acetylcholine-gated calcium ion channel ACR-2 (4 replicates) and cholinergic receptor interactor ELP-1 (3 replicates) were not observed to regulate learning (Figures S7A & S7B). Interestingly, several acetylcholine receptors that were weak candidates showed differences in learning compared with wild-type controls. Those tested include ACC-1 (1 replicate), ACC-3 (2 replicates), and LGC-46 (1 replicate) in the subfamily of anionic “ACC” acetylcholine-gated ligand-gated ion channels (Morud et al., 2021). Interestingly, *lgc-46* mutants demonstrate a reduced capacity for learning compared with wild-type, whereas *acc-1* and *acc-3* mutants appear to display better learning (Figures S8A, S8B, & S8C). Although these ACC receptors are activated by acetylcholine (Park et al., 2000, Putrenko et al., 2005), or have been reported to possess the same protein domain/s as known acetylcholine receptors (Takayanagi-Kiya et al., 2016), they vary substantially in expression pattern in the nervous system (Taylor et al., 2021). We postulate that acetylcholine signalling in specific neurons may therefore contribute to learning in different directions, an interesting avenue for future research. We also tested other genes involved in neurotransmission, such as *gbb-2* (for GABAergic signalling, 2 replicates) and *glr-1* (for glutamatergic signalling, 1 replicate), neither showed a detectable change in learning capacity compared with wild-type (Figures S8D & S8E). *maco-1* (1 replicate) encodes a macoilin family protein that functions broadly in neurotransmission (Arellano-Carbajal et al., 2011), and has previously been shown to regulate memory of olfactory adaptation in worms (Kitazono et al., 2017). The *maco-1* mutant tested in our study (*nj21*) did not show a locomotor defect, but also did not show any obvious learning phenotypes (Figure S8F). Our data therefore indicate that specific components of cholinergic signalling are required for salt associative learning.

We tested several components of GPCR signalling that were identified as learning proteome hits, including *gap-1* (1 replicate), *gap-2* (3 replicates), *gpa-2* (2 replicates), and *rho-1* (2 replicates) (Figures S7C, S8G, S8H, & S8I), but found that worms with single mutations for these genes, when compared to wild-type controls, did not show statistically significant differences in learning capacity. We postulate that although GPCR signalling is broadly important for learning (reviewed in Jong et al. (2018) and Rahmani and Chew (2021)), there may be high levels of redundancy built within this pathway such that single pathway components can be compensated for by other functionally similar genes.

Several components of MAPK signalling have been shown to be involved in different forms of learning (reviewed in (Peng et al., 2010, Ryu and Lee, 2016)), including NSY-1/MAPKKK, MEK-2/MAPKK, and MPK-1/ERK that were identified as part of the learning proteome in our study (Ohno et al., 2014). We tested the dual leucine zipper MAPKKK-encoding gene *dlk-1* (2 replicates), a mutant of this gene showed no difference in learning capacity compared to wild-type (Figure S8J). The E2 ubiquitin-conjugating enzyme variant UEV-3 (4 replicates) has been shown to be a member of the DLK-1 pathway and a potential interactor of p38/MAPK PMK-3 (Trujillo et al., 2010). *uev-3* mutants showed a lower average CI after training compared with wild-type, but this did not reach statistical significance (p value = 0.1806) (Figure S7D). We also tested worms with a mutation in *fsn-1* (3 replicates), proposed to attenuate synapse growth in a DLK-1-dependent manner (Hung et al., 2013). These animals displayed learning capacity similar to wild-type animals (Figure S7E). In summary, these data indicate that mutating single components of the MAPK pathway does not generally perturb salt associative learning.

Finally, we assessed additional hits that do not fit in the pathways above, but were considered strong candidates. These include neuronal adhesion/IGCAM gene *rig-4* (3 replicates) or putative guanyl-nucleotide exchange factor genes (*aex-3* or *tag-52*, both 3 replicates). Aversive associative learning in worms and mice both rely on IGCAM gene *ncam-1/NCAM1* (Cremer et al., 1994 [DOI](#), Vukojevic et al., 2020 [DOI](#), Doyle et al., 1992 [DOI](#)). Additionally, guanine nucleotide exchange factors ‘ArhGEF4’ and ‘RapGEF2’ in mice (Jiang et al., 2024 [DOI](#), Yoo et al., 2020 [DOI](#)), as well as *unc-73* in *C. elegans* (Arey et al., 2018 [DOI](#)), have been linked to learning and memory previously. We found that *aex-3(-)*, *rig-4(-)*, and *tag-52(-)* single mutants did not show significant differences in salt associative learning compared to wild-type controls (**Figures S7F**, **S7G** [DOI](#), & **S7H** [DOI](#)). We also tested *elo-6* (3 replicates), which encodes a long chain fatty acid elongase that potentially functions together with another elongase encoded by *elo-5*, although only *elo-6* is expressed in neurons (Kniazeva et al., 2004 [DOI](#)). Fatty acid composition has previously been demonstrated to be important for learning and memory (Wallis et al., 2021 [DOI](#), Pershina et al., 2022 [DOI](#), Akefe et al., 2024 [DOI](#)). Our data showed no significant learning defect or improvement in *elo-6* mutant animals (**Figure S7I** [DOI](#)), although as mentioned, its role may be masked by functional redundancy.

Other strong candidates tested that did not show a learning phenotype include: (1) *ift-139* (5 replicates), a ciliogenesis gene that was explored since hippocampal cilia structures have been seen to be important for memory in mice (Niwa, 2016 [DOI](#), Jovasevic et al., 2021 [DOI](#), Berbari et al., 2014 [DOI](#)). (2) *tap-1* (3 replicates), which encodes an ortholog for TGF- β activated kinase 1 (Meneghini et al., 1999 [DOI](#)), (3) SAEG-1 (4 replicates), a suppressor for protein kinase G (PKG) ortholog EGL-4 activity, which has been implicated in behavioural changes induced by odour-sensory fatigue (L'Etoile et al., 2002 [DOI](#)); cGMP-dependent kinase PKG also promotes long-term memory in rodents (Ota et al., 2008 [DOI](#), Paul et al., 2008 [DOI](#)), and (4) VER-3 (3 replicates), which encodes a predicted vascular endothelial growth factor (VEGF) receptor-like protein (Popovici et al., 2002 [DOI](#)). VEGF/VEGFR was reported to be upregulated in rats following spatial learning (Cao et al., 2004 [DOI](#)). These data are shown in **Figure S7** [DOI](#).

There are several potential reasons why many mutants tested did not display a learning phenotype. Firstly, as mentioned above, effects may be masked by redundant or compensatory pathways. For example, IGCAM genes in the worm have been reported to act redundantly in axon navigation, including *rig-4* (Schwarz et al., 2009 [DOI](#)). It is also possible that these mutations do not fully knockout protein function. We generally assessed animals with deletion mutations predicted to disrupt protein function, but we did not confirm this through qualitative or quantitative means. Additionally, many mutants used here were not backcrossed as it was beyond the scope of this study, so these lines may have background mutations masking learning phenotypes of the mutations of interest. This was seen for *C30G12.6(-)* animals in this study, where an enhanced learning phenotype in non-backcrossed worms was lost after backcrossing (**Figure 6** [DOI](#)). Finally, the population-wide chemotaxis assays we perform here to validate candidates may not be sensitive enough to capture subtle potential behavioural differences caused by these mutations. Pirouette and weathervane behaviours in *C. elegans* change based on previously experienced salt concentrations in the presence of food (Kunitomo et al., 2013 [DOI](#)). These behaviours can be measured through more in-depth investigation of locomotor behaviour through live tracking and analysis, providing a more sensitive measure for learning responses compared to the chemotaxis assays used here. These factors are important considerations for future experiments utilising the learning proteome as rationale to assess novel mechanisms in learning.

Arginine kinase F46H5.3 regulates both appetitive and aversive gustatory learning

Next, we explored whether the learning regulators identified from our learning proteome functioned more broadly in other types of learning. To do this, we assayed salt aversive learning capacity (training with aversive cue i.e. starvation) for candidates seen to affect salt associative learning (training with an appetitive cue i.e. presence of food): PKA regulatory protein KIN-2

(**Figure 5**), arginine kinase F46H5.3 (**Figure 6**), and acetylcholine receptor subunits ACC-1, ACC3, and LGC-46 (**Figure S7**). In the salt associative learning assay used thus far, ‘trained’ worms are exposed to a pairing of no salt + food (‘control’ worms with high salt + food), whereas in salt aversive learning, ‘conditioned’ worms are exposed to high salt + no food (and control ‘mock-conditioned’ worms with no salt + no food). In the salt aversive learning assay, ‘conditioned’ worms therefore learn to avoid high salt (as it is associated with starvation, a strongly negative cue), whereas ‘mock-conditioned’ worms and naïve worms retain a preference for high salt (Nagashima et al., 2019, Hiroki et al., 2022). We found that only *F46H5.3(-)* mutant worms showed a significant change in learning capacity for salt aversive learning compared with wild-type (**Figure S9**). Specifically, *F46H5.3(-)* mutants displayed a larger decrease in CI in trained animals compared to wild-type trained worms, demonstrating a potential learning improvement (**Figure S9A**). *F46H5.3(-)* mutant phenotype also showed enhanced learning for the salt associative learning paradigm (**Figures 6A** & **6B**). Although *kin-2(ce179)* mutants were not shown to impact salt aversive learning, they have been reported previously to display impaired intermediate-term memory (but intact learning and short-term memory) for butanone appetitive learning (Stein and Murphy, 2014). These findings therefore suggest (1) a generalised effect for F46H5.3 in gustatory learning paradigms involving salt, (2) a specific role for KIN-2 in appetitive learning paradigms, and (3) a unique effect for ACC receptors ACC-1/2 and LGC-46 in salt associative learning only.

Using TurboID to predict potential molecular and cellular pathways for learning

Our TurboID approach provides a unique benefit as a systems-based tool, in that it can be used to map individual candidates onto broader molecular networks. F46H5.3 is mostly uncharacterised beyond its predicted homology to creatine kinase B and mitochondrial creatine kinase 1B. As its role within learning pathways is unknown, we tested for protein-protein interactions between F46H5.3 and other candidates in the learning proteome using the software STRING (version 12.0) (Szklarczyk et al., 2023), aiming to infer its potential function within a molecular network. Mitochondrial creatine kinases regulate phosphocreatine synthesis using ATP, which requires calcium influx into mitochondria to induce ATP synthesis (reviewed in (Schlattner et al., 2001, Schlattner et al., 2006)). Voltage-dependent anion channel VDAC-1 (identified in 3 replicates of learning proteome data) plays a critical role in calcium homeostasis in *C. elegans* mitochondria (Shoshan-Barmatz et al., 2017), and is predicted interactor of F46H5.3. Moreover, calcium influx into the mitochondria is regulated by calcium/calmodulin kinase II (Nguyen et al., 2018), an established and highly-conserved regulator of learning (reviewed in (Ataei et al., 2015, Zalcman et al., 2018)). The sole calcium/calmodulin kinase II in *C. elegans* (UNC-43, 2 replicates) is predicted to interact with VDAC-1. Our learning proteome also includes proteins involved in calcium/calmodulin complex formation (e.g., calmodulin/CMD-1 in 4 replicates and cyclic nucleotide-gated ion channel TAX-4 in 1 replicate) (Karabinos et al., 2003, Komatsu et al., 1999). Calcium/calmodulin complexes can also modulate cAMP levels (reviewed in (Sharma and Kalra, 1994)), which influences PKA/KIN-1 activity regulation by KIN-2 and A-kinase anchoring protein AKAP-1 (2 replicates) (reviewed in (Sadeghian et al., 2022)). Therefore, cytoplasmic calcium homeostasis is one potential pathway through which both KIN-2 and F46H5.3, validated in our study as learning regulators, modulate learning and memory (**Figure 7**).

Finally, we can combine the protein network analysis with analysis of neuron representation within the learning proteome, as described above and shown in **Figure 4**. There are two neuron classes that express all five candidates shown here to affect salt associative learning (KIN-2, F46H5.3, ACC-1, ACC-3, LGC-46): RIM interneurons and DB motor neurons (Taylor et al., 2021) (threshold = 2). These neurons are involved in reversals and forward locomotion, respectively (Guo et al., 2009, Chalfie et al., 1985). It is possible that these learning regulators influence experience-based behaviour through modulating the function of these neurons to alter chemotaxis responses in the presence of gustatory cues.

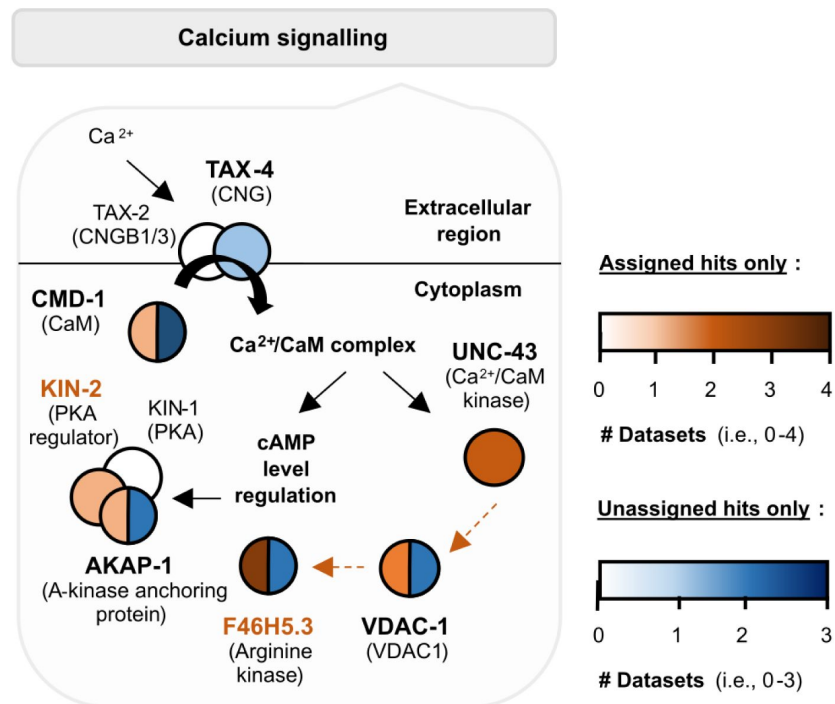


Figure 7

Learning regulators KIN-2 and F46H5.3 may modulate learning through calcium signalling pathways.

Pathway components present within the learning proteome are shown with protein names in bold. Darker colours mean the protein has been detected in more than one biological replicate (see legend). Orange and/or blue circles represent candidates that are 'assigned hits' and/or 'unassigned hits' respectively. Orange dotted arrows denote protein-protein interactions predicted by STRING (version 12.0), whereas black arrows are based on known interactions.

KIN-2 and F46H5.3 share the same expression pattern in many neuron classes, whereas neurons expressing the three ACC receptors are more diverse (Taylor et al., 2021 [↗](#)) (**Figure 4** [↗](#)). ACC-3 is expressed in salt-sensing ASE, mechanosensory and dopaminergic CEP neurons, polymodal sensory neuron IL1, and two interneuron classes (ADA and RIM) (Taylor et al., 2021 [↗](#)). ADA's function is not well-characterised but it is predicted to be involved in chemosensation (Sohn et al., 2011 [↗](#)). In contrast, ACC-1 and LGC-46 are expressed in several interneurons and motor neurons including those implicated in gustatory or olfactory learning paradigms (AIB, AVK, NSM, RIG, & RIS) (Beets et al., 2012 [↗](#), Fadda et al., 2020 [↗](#), Wang et al., 2025 [↗](#), Zhou et al., 2023 [↗](#), Sato et al., 2021 [↗](#)) and important for backward or forward locomotion (AVE, DA, DB, & VB) (Chalfie et al., 1985 [↗](#)). There are also highly represented neuron classes which are not as well defined (ADA, I4, M1, M2, & M5), which may present interesting directions for future research. Cholinergic signalling may therefore regulate gustatory learning through integration of sensory signals as well as direct modulation of the motor circuit. In contrast, KIN-2 and F46H5.3 may play more general functions within the nervous system, such as through modulate calcium homeostasis (**Figure 7** [↗](#)). Overall, our TurboID dataset offers a valuable foundation for future investigations into individual proteins involved in learning, and provides a resource for systemic analyses at the level of tissues, neuron types, subcellular localisations, and molecular networks.

Discussion

Our study demonstrates the effectiveness of using the protein labelling-technique TurboID to explore *C. elegans* learning and memory. The expression of TurboID in the whole nervous system of the worm, and addition of biotin only during the training step of salt associative learning, allowed us to label proteins in neurons during this critical stage of memory encoding. We identified these proteins by mass spectrometry and revealed a putative 'learning proteome' including known learning regulators (**Figure 2** [↗](#) & Table S8). Moreover, we identified five novel regulators of appetitive gustatory learning, namely three acetylcholine receptors (ACC-1, ACC-3, & LGC-46), PKA regulatory subunit KIN-2, and the arginine kinase F46H5.3 (**Figures 5** [↗](#), **6** & **S8**). Finally, F46H5.3 was observed to modulate an aversive gustatory learning paradigm in **Figure S9** [↗](#). These findings highlight that proximity labelling can be used in *C. elegans* to elucidate novel learning regulators, which may function across learning paradigms characterised by different modalities or valences.

Learning and memory are key functions of the nervous system and are critical for survival. Forms of associative learning have been studied in invertebrate and vertebrate animals for decades, revealing many important insights on the behavioural, neuroanatomical, and molecular requirements for learning and memory formation (reviewed in Hawkins and Byrne (2015) [↗](#), Jong et al. (2018) [↗](#), Kandel (2012) [↗](#), Matsumoto et al. (2018) [↗](#), Peng et al. (2010) [↗](#), Peng et al. (2011) [↗](#), Rahmani and Chew (2021) [↗](#), Ryu and Lee (2016) [↗](#)). However, many studies focus on single genes of interest and are therefore unable to reveal the entire network of molecular players that drive (or inhibit) learning. We used a complementary approach to take a snapshot of the proteins present in the brain during learning, using the protein-labelling tool TurboID. Previously, RNA sequencing has been used to characterise transcriptomic changes during long-term memory formation in the *C. elegans* nervous system (Lakhina et al., 2015 [↗](#)). While this provides useful insights into the mechanisms of memory formation, it does not capture proteomic information, which may differ from RNA levels. Separately, Hiroki et al. (2022) [↗](#) applied TurboID to map protein-protein interactions of PKC/PKC-1 in untrained *C. elegans*, providing insight into how PKC-1 affects gustatory appetitive learning. Our work builds on this by directly comparing trained and high-salt control conditions, offering new insights into the proteomic landscape of learning. Our strategy identified several components of molecular pathways previously shown to be generally required for learning, including neurotransmitter signalling, MAPK signalling, insulin signalling, synaptic vesicle exocytosis, and GPCR signalling (**Figure 2** [↗](#), **Figure S5** [↗](#), & Table S8). We identified regulators of learning that may not have been obvious choices for a candidate screen,

and that may have had phenotypes too subtle to be highlighted through a random mutagenesis screen. We therefore conclude that our approach is a useful and scalable method that can be used in multiple systems to delineate the molecular requirements for different forms of learning.

There are several interesting unanswered questions: Firstly, why did some candidates only seem to affect gustatory appetitive learning, as opposed to showing effects in both gustatory appetitive and aversive paradigms? Mutants of ACC-1, ACC-3, LGC-46, and KIN-2 show significantly different learning capacities for salt associative learning compared with wild-type (**Figures 5** & **S8**) but did not show differences when tested for salt aversive learning (**Figure S9**). In contrast, only *F46H5.3(-)* mutants showed a significant learning difference in both salt associative learning and salt aversive learning (**Figures 6** & **S9A**). One possibility is that our method for selecting hits from our mass spectrometry data for downstream validation introduced an unintended bias: to identify the learning proteome, we subtracted proteins in the list for ‘high-salt control’ worms from the protein list for ‘trained’ worms, for each biological replicate. The main difference between high-salt control and trained worms was whether they were exposed to a pairing of ‘salt + food’ (control) or ‘no salt + food’ (trained). It is possible that proteins present in trained worms, but not high-salt control worms, during conditioning are those that are strongly regulated by changes in salt concentration (and therefore impact mainly appetitive gustatory learning with ‘no salt’, versus salt aversive learning induced in the presence of salt). Indeed, our ‘simple subtraction’ approach may be an overly conservative method for selecting learning regulators, as it is likely that many neuronal proteins generally important for learning (i.e., across multiple learning paradigms) are present in both groups, but at higher (or lower) levels in trained worms.

Secondly, why do the different acetylcholine receptors that we identified in our study impact learning differently? We showed that loss-of-function mutants of ACC-1 and ACC-3 show improved learning, whereas *lgc-46(-)* mutants displayed a learning defect compared with wild-type (**Figure S8**). One study suggested that ACC-1 and ACC-3 may function together: ACC-3 homomers do not respond robustly to acetylcholine, but ACC-1 and ACC-3 can form a functional heteromer, albeit with lower sensitivity to acetylcholine than ACC-1 homomers (Putrenko et al., 2005). ACC-1 and ACC-3 functioning as heteromers may be why these proteins impact learning in the same direction. On the other hand, LGC-46 is also a member of the acetylcholine-gated chloride channel (ACC) family but impacts salt associative learning in the opposite direction to ACC-1 and ACC-3. LGC-46 (in 81 neurons) has a much broader expression pattern than ACC-1 (in 32 neurons) or ACC-3 (in 14 neurons), and is expressed in many more interneurons and motor neurons (Taylor et al., 2021). It is therefore possible that some of the LGC-46-expressing neurons function to regulate learning in a different manner to ACC-1- or ACC-3-expressing cells.

Our approach demonstrates a powerful method to uncover regulatory networks for a variety of behaviours; however, one factor we aim to improve in future is the amount of background ‘noise’ observed in the learning proteome. While the five biological replicates of learning proteome data did reveal genes and molecular pathways implicated in learning and memory, there was also potential background. For example, when assessing the overlap between all proteins (i.e., both assigned and unassigned proteins (**Figure S6**)), a large proportion of genes identified were categorised as ribosomal (4-12%), mitochondrial (8-12%), or involved in reproduction (16-19%) (STRING database, accessed Feb 2024). Mass spectrometers detect peptides by abundance (Bakalarski et al., 2008), so many of these highly abundant proteins may have preferentially been detected over less abundant neuronal proteins. There are several ways to reduce background and improve the signal-to-noise ratio: (1) using an integrated transgenic line that expresses TurboID with 100% transmission, as the line used in our study had a 70-80% transmission rate, (2) using a cell-sorting strategy (e.g., by flow cytometry) to isolate a tissue of interest (Beets et al., 2020); this is particularly useful if the tissue of interest is only a low proportion of the worm biomass (e.g., neurons are 1% (Froehlich et al. (2021)), as proteins from larger tissues, such the

gut and germline, may interfere with detection, and (3) removing highly abundant background proteins during mass spectrometry sample preparation, such as mitochondrial carboxylases that are endogenously biotinylated, as in Artan et al. (2022) [\[link\]](#).

There are also inherent limitations to using a qualitative approach. While our approach includes weak candidates and does not include a statistical framework for comparing protein abundance between experimental groups, this flexibility allows for the identification of potentially novel regulators that might otherwise be overlooked in more stringent analyses. Notably, we did observe relationships between weak candidates and learning. For example, ACC-1, which modulates salt associative learning in *C. elegans*, was detected in one replicate of mass spectrometry as a potential learning regulator (**Figure S8A** [\[link\]](#)). To address the lack of quantitative comparison, we categorised each candidate with their occurrence per replicate of mass spectrometry data for TurboID trained versus high-salt control datasets (summary of this data for candidates tested shown in **Table 2** [\[link\]](#)). In addition, the raw mass spectrometry data is provided for each biological replicate and experimental group via an open-access server (**link**), enabling transparency and further analysis by the research community. Future studies could benefit from implementing a quantitative approach to directly measure protein abundance differences between trained and control groups. While integrating such approaches with TurboID is challenging due to the requirement for biotinylated protein enrichment, overcoming these limitations, or using an alternative proteomic strategy, could uncover additional learning regulators.

Finally, while the learning regulators identified in this study support the validity of our proteomic approach, further functional validation is important. Testing for rescue of learning phenotypes in transgenic lines re-expressing learning regulators pan-neuronally, endogenously, or in single neurons would provide valuable insight into their functions within the nervous system. Our attempts to generate such rescue lines using standard microinjection techniques encountered several technical challenges, including difficulties with low transmission rate, potentially due to plasmid toxicity, and culturing issues potentially caused by transgene-dependent reproductive defects. To overcome these challenges, future work may utilise single-copy integration methods to reduce transgene dosage or use tissue- or cell-specific RNA interference to achieve targeted knockdown. These approaches could provide more precise insights into the roles of learning regulators within specific neuronal contexts.

In conclusion, we present an effective and scalable approach to identify the network of molecular processes that drive learning and memory formation, using the compact *C. elegans* nervous system. Our data reveal proteins from established biological pathways linked to associative memory, and through which we have identified novel regulators of gustatory associative learning. Future studies using this approach to identify learning regulators in other contexts will advance our understanding of the complex spatiotemporal regulation of learning and memory. This may help to elucidate the principles through which different memory types arise from the combination of specific neuronal signals, individual brain regions/cells, and different sensory modalities, relevant to brains of many sizes.

Experimental Procedures

C. elegans strain maintenance

Young adult (day 1) hermaphrodite *C. elegans* were grown using standard conditions on nematode growth medium (NGM) agar in petri dishes at 22 °C for all experiments (Brenner, 1974 [\[link\]](#)). This was done for at least two generations for salt associative learning assays involving TurboID, as well as assays involving salt aversive learning involving *kin-2(ce179)* mutants, and all butanone appetitive learning assays (assays defined below). Animals were otherwise cultured at 22 °C for one generation, and at 15 °C prior to this. For TurboID-based labelling experiments, worms were

cultured for at least two generations with the biotin-auxotrophic strain as their food source (Branon et al., 2018 [DOI](#)). For all other experiments, animals were fed *Escherichia coli* (*E. coli*) strain OP50. *C. elegans* lines used in this paper are listed in Table S1. **Biotin treatment**

The biotin treatment strategy was adapted from Artan et al. (2021) [DOI](#). Briefly, a solution of 100 mM biotin, 250 mM KOH, 5 mM K₃PO₄ (pH 6.0), 1 mM CaCl₂, and 1 mM MgSO₄ was diluted 1:100 in *E. coli* MG1655 *bioB::kan* washed with modified Luria Broth (LB; 25 mM NaCl, 5 mM K₃PO₄ (pH 6.0), 1 mM CaCl₂, 1 mM MgSO₄, 1.0% (w/v) Bacto™ Tryptone, 0.5% (w/v) yeast extract, 0.05 mg/mL Kanamycin). Biotin-depleted worms were fed *E. coli* MG1655 on NGM agar during a 6 h conditioning period (see the ‘Salt associative learning’ section). The bacterial pH was increased by a negligible amount (i.e., 0.1) from the addition of KOH, thus it is expected it will not significantly impact worm physiology (Khanna et al., 1997 [DOI](#), Cong et al., 2020 [DOI](#)). At least 3,000 worms per group were utilised for downstream proteomic experiments.

Protein extraction and quantification

The following protocol was adapted from Liang et al. (2014) [DOI](#) and Artan et al. (2021) [DOI](#): After biotin treatment (end of learning assay), worms were washed twice using ‘washing buffer’ (50 mM NaCl, 5 mM K₃PO₄ (pH 6.0), 1 mM CaCl₂, 1 mM MgSO₄) and then stored as pellets in a -80 °C freezer. Each pellet was suspended in 200 µL Radioimmunoprecipitation assay (RIPA) buffer containing 2 M urea (prepared as in Sanchez and Feldman (2021) [DOI](#)), as well as 150 mM NaCl, 50 mM Tris-Cl (pH 8.0), 5 mM EDTA, 10 mM NaF, 2 mM Na₃VO₄, 1 mM NaPP, 1% (v/v) Nonidet-P40, 1% (w/v) SDS, 0.5% (w/v) sodium deoxycholate, 0.1% (w/v) β-glycerophosphate, and 1× cOmplete Mini Protease Inhibitor (Merck). Worm pellets were sonicated (10× 4 sec total sonication time, 2 sec ‘on’ pulse, 3 sec ‘off’ pulse’, and 25% amplitude) with the Q125 Sonicator (Q Sonica) in a temperature-controlled room (~2-8 °C) and allowed to rest on ice for >20 sec between each sonication. All samples were vortexed for ~5 sec at room temperature, and then centrifuged (14,000 rcf, 4 °C, 10 min) to separate carcasses/debris from supernatant containing proteins. The supernatant was then used for protein quantification by BCA assay (ThermoFisher Scientific, #23225).

SDS-PAGE, protein transfer, and western blotting

20-40 µL protein samples, containing 1× sample buffer (0.25 M Tris (pH 6.8), 10% (v/v) β-mercaptoethanol, 10% (v/v) glycerol, 10% (w/v) SDS, 0.25% (w/v) bromophenol blue), were boiled at 95 °C for 3 min. Proteins were electrophoresed through 8% (v/v) polyacrylamide gels for 40-60 min at 80 V, before the voltage was increased to 100 V for an additional 40-60 min, and then increased to 120 V until completion of electrophoresis (Jeong et al., 2018 [DOI](#), Stefanoska et al., 2022 [DOI](#)). The standard protocol was utilised for semi-dry protein transfer (constant 25 V, 40-45 min) onto nitrocellulose membranes and western blotting, using the following antibodies/probes (dilution in brackets, all made in 5% (w/v) bovine serum albumin): rabbit anti-α tubulin (1:1000) (Abcam, #ab4074), mouse anti-V5 (1:1000) (Cell Signalling Technology, #80076), goat anti-rabbit HRP (1:20,000) (Cell Signalling Technology, #7074P2), goat anti-mouse HRP (1:20,000) (ThermoFisher Scientific, #G-21040), and streptavidin-HRP (1:5000) (Cell Signalling Technology, #3999) (Jeong et al., 2018 [DOI](#), Stefanoska et al., 2022 [DOI](#)). Probed proteins were visualised by chemiluminescence using Clarity Western ECL Substrate (Bio-Rad, #1705060) according to manufacturer’s instructions.

Mass spectrometry

1. Sample preparation

Our protocol was adapted from Artan et al. (2022) [DOI](#), Sanchez et al. (2021) [DOI](#), and Prikas et al. (2020) [DOI](#): ~1 mg of protein was desalted using 7 kDa molecular weight cut-off desalting spin column (ThermoFisher Scientific, #89883) by buffer exchange with RIPA buffer containing a lower SDS content (i.e., 0.1% (w/v) SDS) and no urea (Artan et al., 2022 [DOI](#)). Desalted protein samples were

quantified by BCA assay so that ~0.6-1.0 mg of total protein per sample could be used in subsequent pull-down experiments to enrich for biotinylated peptides using Streptavidin Magnetic Beads (NEB, #S1420S) equilibrated with TBS-T (150 mM NaCl, 10 mM Tris (pH 7.4), 0.1% (v/v) Tween-20). Total protein was gently agitated in a tube rotator in 1 mL of total volume (4.5 μ L bead:8.0 μ g total protein ratio) for 18 h at 4 °C. Magnetic beads were sequentially washed with the following solutions (number of washes in brackets): TBST (x3), 1 M KCl (1x), 0.1 M Na₂CO₃ (1x), and PBS (5x; ThermoFisher Scientific, #10010023) (Sanchez et al., 2021 [DOI](#)).

The beads were then incubated in a ThermoMixer (Eppendorf # 5384000063) at 800 rpm, 55 °C for 1 h in 200 μ L per sample of reducing solution containing 4 M urea, 50 mM NH₄HCO₃, 5 mM dithiothreitol, and 0.1% (w/v) Protease-Max Surfactant (Promega, #V2071) in 50 mM NH₄HCO₃. Alkylation was promoted by adding 4 μ L of 0.5 M iodoacetamide to each solution and re-incubating each sample at 800 rpm, 55 °C for 20 min in the dark. Finally, samples were incubated at 800 rpm, 37 °C, for 18 h) with the addition of 162 μ L of digesting solution per sample (50 mM NH₄HCO₃, 0.1% (w/v) Protease-Max Surfactant (Promega, #V2071) in 50 mM NH₄HCO₃, 0.01 μ g/ μ L of Sequencing Grade Modified Trypsin (Promega #V5111)) to facilitate an on-bead protein digest.

This digest was stopped using a protocol modified from Prikas et al. (2020) [DOI](#). Beads were placed on a magnetic rack (NEB, #S1506S) so the supernatant could be transferred to new 1.5 mL tubes without the beads; peptides from digested proteins in the supernatant will henceforth be referred to as ‘unbound’ samples, whereas peptides attached to beads will be called ‘bound’ samples. For ‘unbound samples’, trifluoroacetic acid (TFA) was added to each sample to a final concentration of 0.1% (w/v), centrifuged (20,000 rcf, 22 °C, 20 min), and the supernatant transferred to new 1.5 mL tubes. To ensure as much protein as possible was recovered, 0.1% (w/v) of TFA was then added to the previously centrifuged 1.5 mL tubes, re-centrifuged (20,000 rcf, 22 °C, 10 min), and the resulting supernatant pooled with the first supernatant sample. For ‘bound’ samples, magnetic beads were first gently agitated on a tube rotator in 250 μ L of elution buffer (EB) solution (80% (v/v) acetonitrile, 0.2% (w/v) TFA, 0.1% (v/v) formic acid), magnetised so the supernatant could be transferred, and then boiled in 200 μ L of EB solution at 800 rpm, 95 °C for 5 min. These beads were then magnetised again so that supernatant could be transferred for TFA treatment.

Peptides for both ‘unbound’ and ‘bound’ samples were desalted using tC18 cartridges (Waters, #WAT036810), vacuum-dried at ambient temperature for ~3 h, and then resuspended in a compatible solution (0.2% (v/v) heptafluorobutyric acid in 1% (v/v) formic acid) for liquid chromatography with tandem mass spectrometry (LC-MS/MS).

2. Mass spectrometry

For technical reasons, we used two mass spectrometers as outlined in Prikas et al. (2020) [DOI](#) – the ThermoFisher Scientific Q-Exactive Orbitrap (QE) and ThermoScientific Orbitrap Exploris (Exploris). Samples from biological replicates 1 and 2 were run on the QE, replicates 4 and 5 were run on the Exploris, and replicate 3 was run on both machines. We treated these as 6 separate experiments, although there were only 5 biological replicates, as considerably more proteins were identified using the Exploris compared with the QE – for this reason, the Exploris was used for subsequent experiments. The overlap between learning proteomes for each biological replicate (i.e., proteins unique to ‘TurboID, trained’) has been summarised in **Figure S9** [DOI](#), based on the mass spectrometer used. The resulting data was processed using the MASCOT search engine (Matrix Science) and *C. elegans* Swiss-Prot database (downloaded on 01/02/2021). An MS/MS ion search was performed with the following settings: ‘semi-trypsin’ enzyme, ‘monoisotopic’ mass values, ‘unrestricted’ protein mass, ‘ $\pm$ 5 ppm’ peptide tolerance, ‘ $\pm$ 0.05 Da’ fragment mass tolerance, and ‘3’ maximum missed cleavages. Biotin (K), Carbamidomethyl (C), Oxidation (M), Phospho (ST), and Phospho (Y) were selected as variable modifications. Data from the MASCOT search engine will be made accessible on the Dryad platform upon publication.

3. Data analysis

We designated protein identities from all mass spectrometry experiments as either ‘assigned’ or ‘unassigned’ hits (see **Tables S3 & S4** for full lists). ‘Assigned proteins/hits’ are defined as proteins identified by MASCOT, based on peptide sequences detected during mass spectrometry, with at least one unique peptide detected for that protein. This threshold was chosen based on [Prikas et al. \(2020\)](#) to ensure sensitivity in detecting low-abundance neuronal proteins. We did not restrict our definition of ‘assigned hits’ to any peptide or protein score threshold. In contrast, ‘unassigned hits’ were determined using peptide sequences with a peptide score ≥ 15 , but that were not assigned a protein identity by MASCOT, as the peptide was not detected as unique for a specific protein by MASCOT. Our criteria for ‘unassigned hits’ was the protein identity required (1) at least one peptide for the protein with a peptide score ≥ 15 calculated by MASCOT, (2) an 100% identity match between peptide sequences with a peptide score ≥ 15 and the protein determined by BLAST, and (3) an e value < 0.05 for the identity match calculated by BLAST, such that a smaller number represents an increased probability that the identity is true and not given by random chance. To collect a list of unassigned hits, we used a custom Python script to perform bulk BLAST-p searches for these sequences using the ‘Reference proteins (refseq_protein)’ database and ‘*Caenorhabditis elegans* (taxid:6239)’ organism. This Python code uses Anaconda 2023.03 ([Anaconda Software Distribution, 2016](#)), BioPython 1.78 ([Cock et al., 2009](#)), and pandas software 1.5.3 ([The pandas development team, 2020](#)). Our pipeline then assigned protein accession numbers to searched peptide sequences where the percent identity = 100% and e value < 0.05 . Finally, the Batch Entrez online software available on the NCBI website was used to convert accession numbers to protein identities. This generated lists of unassigned hits for samples in each biological replicate.

Proteins detected in TurboID, trained worms only were calculated as follows: (1) proteins detected from ‘non-transgenic, control’ and ‘non-transgenic, trained’ worms were subtracted from corresponding protein lists generated from TurboID worms, and then (2) proteins that overlap between lists for ‘TurboID, control’ and ‘TurboID, trained’ worms were subtracted from each other. Venn diagrams were generated using an online tool (<https://bioinformatics.psb.ugent.be/webtools/Venn/>).

GO term analyses were achieved using STRING (version 12.0), Cytoscape (version 3.10.0), and the Cytoscape App ClueGO (version 2.5.10) ([Bindea et al., 2009](#)). All 1010 proteins from ‘TurboID, trained’ lists were entered into STRING as a single list, so k-means clustering could be utilised to separate proteins into 10 clusters, to parse data into more accessible clusters with enough proteins to output enriched GO terms. The ‘tabular text’ protein-protein interaction information exported from STRING for each cluster was then uploaded onto Cytoscape. GO term analyses were performed with ClueGO for the following categories per cluster using default settings (network specificity in brackets): (i) cellular component (medium), (ii) biological process (detailed), and (iii) molecular function (medium). The ClueGO results were exported to spreadsheets for each cluster, as Table S5 (cellular component) and Table S6 (biological process/molecular function), such that each row corresponds to a GO term based on gene/s within a specific cluster. Each list of genes in each row was consolidated with other gene lists with a matching or similar corresponding GO term, to generate the data shown in **Tables S5–S6, Figure 2**, & **Figure S5**. Nodes that did not show protein-protein interactions with other nodes and/or were not categorised into any GO term by ClueGO were manually categorised through a literature search and added onto these figures through Cytoscape.

We assessed overlap between biological replicates for individual candidates (**Figure S6**) using two mass spectrometry systems: Thermo-Fisher Q-Exactive Orbitrap (‘QE’) and Orbitrap Exploris (‘Exploris’). Candidates detected in multiple replicates comprised 17% of assigned hits in QE runs, 13% in Exploris, and 21–23% when including unassigned hits (**Figure S6A–D**). Of the 1,010 assigned QE hits, 17% were also identified with Exploris, increasing to 29% when including all

2,065 protein identities (**Figures S6E–F**). Despite modest overlap (<25%), key learning-related pathways (**Figure 2**, Table S6) and other biological processes, including metabolic pathways (**Figure 3**), were consistently represented, supporting the biological relevance of the identified learning proteome.

Behavioural assays

We adapted previously established methods to perform three behavioural paradigms that model associative learning: (1) salt associative learning (Tang et al., 2023, Hiroki et al., 2022, Nagashima et al., 2019), (2) salt aversive learning (Lim et al., 2018), and (3) butanone appetitive learning (Kauffman et al., 2010).

1. Salt associative learning

This experiment had three groups: naïve worms that did not undergo training, ‘control’ worms that were paired with 100 mM NaCl and food, and ‘trained’ worms that were paired with no NaCl and food. Worms from all groups were washed off agar plates using washing buffer as in the **‘Protein extraction and quantification’** section above. The naïve group could be then immediately tested for their innate response to salt using the salt chemotaxis assay (Rahmani et al., 2024). For experimental groups, worms were washed a third time with washing buffer (containing 50 mM NaCl; for ‘control’) or no-salt buffer (washing buffer without NaCl; for ‘trained’). These washing steps were completed within 10 min per group. Trained animals were placed on 9 cm ‘conditioning plates’ containing salt-deficient agar (5 mM K₃PO₄ (pH 6.0), 1 mM CaCl₂, 1 mM MgSO₄, 2.0% (w/v) agar) and their bacterial food source *E. coli* MG1655 *bioB::kan*. Control worms were placed on conditioning plates containing salt-deficient agar supplemented with 100 mM NaCl. Worms were left on these plates for 6 h at 22 °C in the dark (Nagashima et al., 2019). Following the training step, worms were washed twice with washing buffer and then transferred within 2 min to test their learning capacity using a salt chemotaxis assay (described below in **Section 3**).

The food source (*E. coli* MG1655 *bioB::kan*) was prepared 3–4 d before each experiment, which involved (1) pelleting 1.5 mL of bacteria by centrifugation at 11,000 rpm for 30 sec in a 2 mL tube, (2) discarding the supernatant, and then (3) vortexing cells in 750 µL of modified LB. 750 µL of washed bacteria was transferred onto each conditioning plate, left to dry at room temperature overnight, and then left at 22 °C for 3–4 d before use. Notably, the salt-deficient agar in conditioning plates for training contains 0.728 mM of NaCl due to the use of modified LB, meaning that it is not completely lacking NaCl but has only a very small amount.

2. Salt aversive learning

This assay involves three groups: naïve (did not undergo training), mock-conditioned worms (paired with no salt and no food), and conditioned worms (paired with 50 mM NaCl and no food). All groups were first washed as described in the **Salt associative learning** section, except the third wash was performed with no-salt buffer (for ‘mock-conditioned’) or washing buffer (for ‘conditioned’). Naïve worms were placed on chemotaxis assay plates after two washes with washing buffer. Worms undergoing conditioning were incubated at room temperature for 3 h in 1.5 mL tubes, placed on a shaker at 175 rpm. These tubes contained no-salt buffer for ‘mock-conditioned’ groups or washing buffer (containing 50 mM NaCl) for ‘conditioned’ groups. Worms were pelleted by sedimentation for 1–2 min before use in salt chemotaxis assays as described below.

3. Salt chemotaxis assay

Salt chemotaxis assay (CTX) plates contain a salt gradient prepared by placing 5 mm cubes of salt-deficient agar on top of salt-deficient agar, with one cube containing 0 mM salt on one side and another cube containing 200 mM salt on the other side (Jang et al., 2019) (see [Figure S2A](#) for a schematic). Worms were allowed to crawl freely on chemotaxis assay plates for 45 min in the dark at 22 °C (Nagashima et al., 2019), becoming immobilised when they encountered the paralytic agent sodium azide at the extremes of the salt gradient. These animals were then counted within the regions outlined in [Figure S2B](#) to calculate the salt chemotaxis index (CI) based on the below equation:

$$\frac{\# \text{ worms on high salt} - \# \text{ worms on low salt}}{\# \text{ total population} - \# \text{ worms on origin}}$$

CI values range from -1.0 (strong preference for low salt concentrations) to +1.0 (strong preference for high salt concentrations). We note several differences in our chemotaxis assay compared with other studies: in our study, (1) CTX plates contained 0 mM salt prior to the addition of salt cubes (containing 0 or 200 mM salt) (Jang et al., 2019), while other studies use CTX plates containing 50 mM NaCl (Kunitomo et al., 2013), and (2) the food source used to induce learning is biotin-auxotrophic strain (*E. coli bioB::kan* MG1655), which is grown with 50 mM Kanamycin antibiotic, differing from other studies that used *E. coli* NA22 without antibiotic (Kunitomo et al., 2013, Nagashima et al., 2019).

TurboID worm populations exposed to biotin during training by salt associative learning were scored for the total percentage of transgenic animals on chemotaxis assay plates ([Figure 1B](#)). This is because TurboID worms expressed the enzyme from an extrachromosomal array, and this scoring confirmed the presence of TurboID-positive animals in worm pellets to be used for downstream proteomics. To do this, fluorescing worms only (identified by co-injection marker *Punc-122::rfp*) were counted in all zones excluding the origin and compared to the total number of worms (fluorescing and non-fluorescing worms) in these zones. The average percentage for TurboID-positive worms was 27-57% per biological replicate.

Neuron class analysis for learning proteome data

Analyses utilised protein lists (containing assigned hits only) from mass spectrometry experiments for the following groups: (1) non-Tg high-salt control, (2) non-Tg trained, (3) TurboID high-salt control, and (4) TurboID high-salt trained ([Figure 1A](#)). Briefly, protein identities from non-Tg high-salt control and non-Tg trained were subtracted from TurboID high-salt control and TurboID trained, respectively. Two lists containing proteins unique to TurboID high-salt control (388 proteins) and TurboID trained (706 proteins) were separately input into the CeNGEN database (threshold = 2) (Taylor et al., 2021), to identify gene expression profiles for each list. CeNGEN output a neuron class list for each protein list; the number of proteins expressed in each neuron class was calculated for TurboID high-salt control and TurboID trained. For TurboID trained, these numbers were normalised by a factor of ~1.8. Neuron-specific fold-differences in the number of proteins expressed in TurboID trained versus TurboID high-salt control was used to rank each neuron class. We interpreted a higher fold-difference value as a relatively greater enrichment (fold-change) of training-associated genes compared to control.

Statistical analyses

Statistica analyses were performed in GraphPad Prism (version 8.0). The Shapiro-Wilk normality test was used to assess chemotaxis assay data. Following confirmation of normality, an ordinary two-way ANOVA with Tukey's multiple comparisons post-test ($\alpha = 0.05$) was performed to compare differences between mean CI values for each group. This statistical analysis was chosen given that

it aligns with recent publications that employ similar experimental designs and data structures (Beets et al., 2020 [DOI](#), Jang et al., 2019 [DOI](#), Kitazono et al., 2017 [DOI](#), Lim et al., 2018 [DOI](#), Lin et al., 2010 [DOI](#)). Exact p-values for each statistical comparison are reported in Table S9.

Supplementary Figures

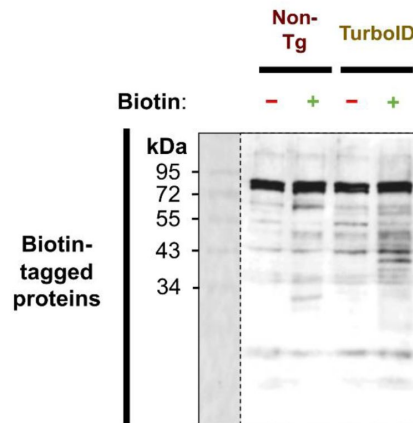


Figure S1

Western blot to quantify biotin-tagged protein levels following biotin exposure.

C. elegans used to generate this blot were either wild-type/non-transgenic/Non-Tg animals or transgenic worms expressing TurboID in all neurons. Animals were grown for at least two generations via their diet (i.e., *E. coli* bioB::kan MG1655), and then their biotin-depleted progeny (young hermaphrodite adults) were lysed by sonication for protein extraction. Whole worms were lysed immediately (-) or fed MG1655 supplemented with 1 mM biotin for 6 h (+) before lysis (+). Biotin-tagged proteins were probed with streptavidin-horseradish peroxidase (40 µg of total protein /lane). Here, the sum of areas containing biotin-tagged protein in each lane is interpreted as signal intensity for these proteins. By comparing this signal for untreated and treated lanes within each worm line, treatment was seen to increase signal (1.3-fold for non-Tg and 1.7-fold for transgenic *C. elegans*). The leftmost lane contains protein standards (kDa, sizes annotated on the left).

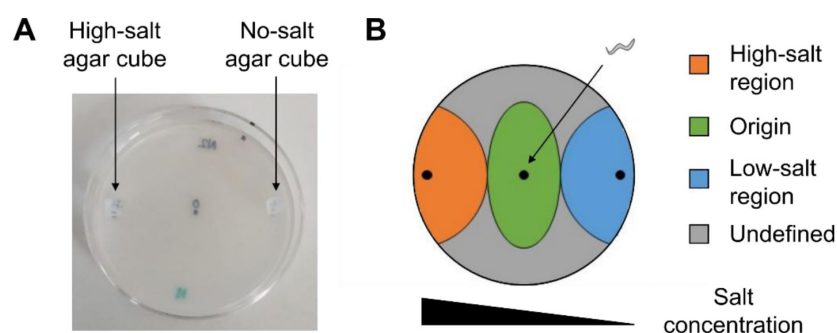


Figure S2

Design of chemotaxis assay plates to quantify salt chemotaxis behaviours.

(A) Method used to generate a salt concentration gradient (0-200 mM NaCl) on chemotaxis assay plates: 5 mm cubes of no-salt/salt-deficient agar with or without high-salt (200 mM NaCl) were placed on top of salt-deficient agar, 5 mm from the plate edge (6.0 cm diameter). Cubes were incubated overnight at 22 °C, or incubated sequentially for (1) 5.5 h at 37 °C, (2) 15 min at 15 °C, and then (3) 0.5-2 h at 22 °C. Each cube was replaced with a 1 µL spot of 0.1 M sodium azide <2 min before use. **(B)** Template used to quantify salt chemotaxis on chemotaxis assay plates: regions of high salt (orange) and low salt (blue) were defined as a circle (diameter = 2.4 cm). These regions were cut-off on their longitudinal axis since they were centred 0.5 cm from the edge of each plate. Worms were transferred onto the region of origin (green; 2 cm short axis and 2.4 cm long axis). Points where agar cubes or worms were transferred onto are represented by a small black circle.

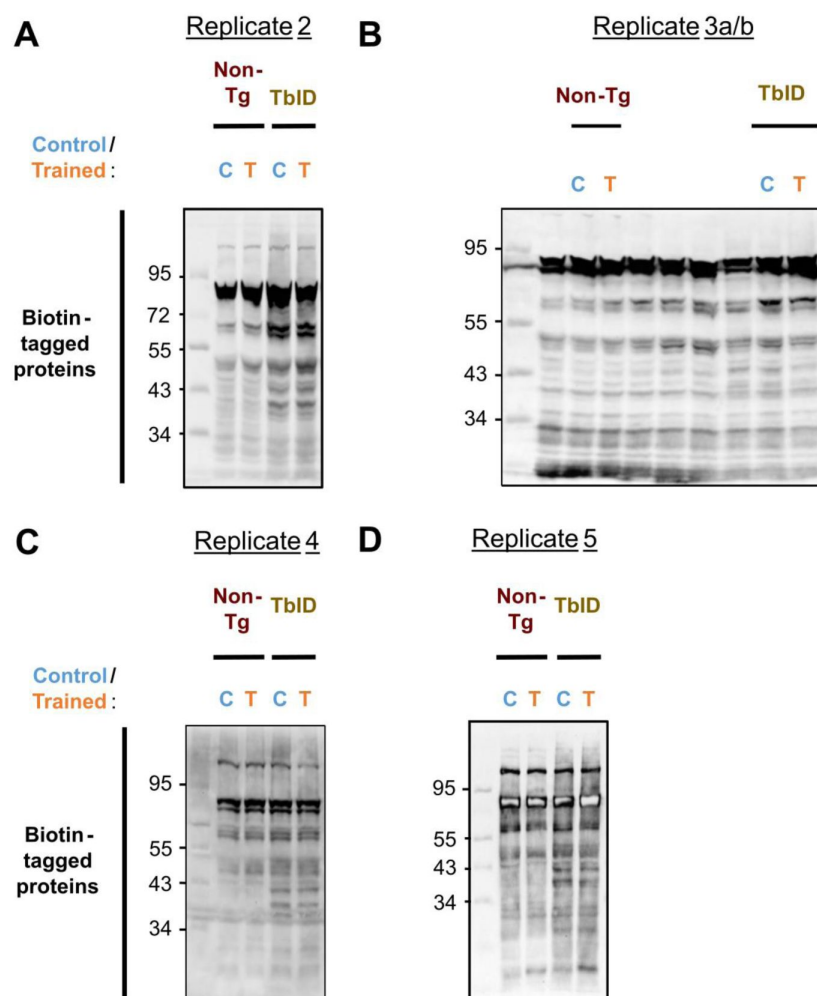


Figure S3

Western blots to assess biotinylation in *C. elegans* by TurboID during memory encoding of salt associative learning.

Each lane contains whole worm lysate from wild-type/non-transgenic animals (Non-Tg, red) or TurboID-positive transgenic *C. elegans* (TbID, yellow). Worms were biotin-depleted for at least two generations via their diet (i.e., *E. coli* bioB::kan MG1655), and then treated with 1 mM of exogenous biotin for 6 h. This biotin treatment was paired with salt associative learning to generate high-salt control/C (food + salt, blue) and trained/T worms (food + no salt, orange) for protein extraction. Each panel corresponds to a biological replicate of mass spectrometry data with TurboID as annotated above (underlined in black). Each lane contains (A) 50 µg, (B, D) 40 µg, or (C) 20 µg of total protein, with biotin-tagged proteins visualised using streptavidin-horseradish peroxidase. For panel (C), unannotated lanes correspond to protein samples not relevant to this experiment. The leftmost lane for each image contains protein standards (kDa, sizes annotated on the left). Comparisons regarding signal intensity between each experimental group are in Table S2.

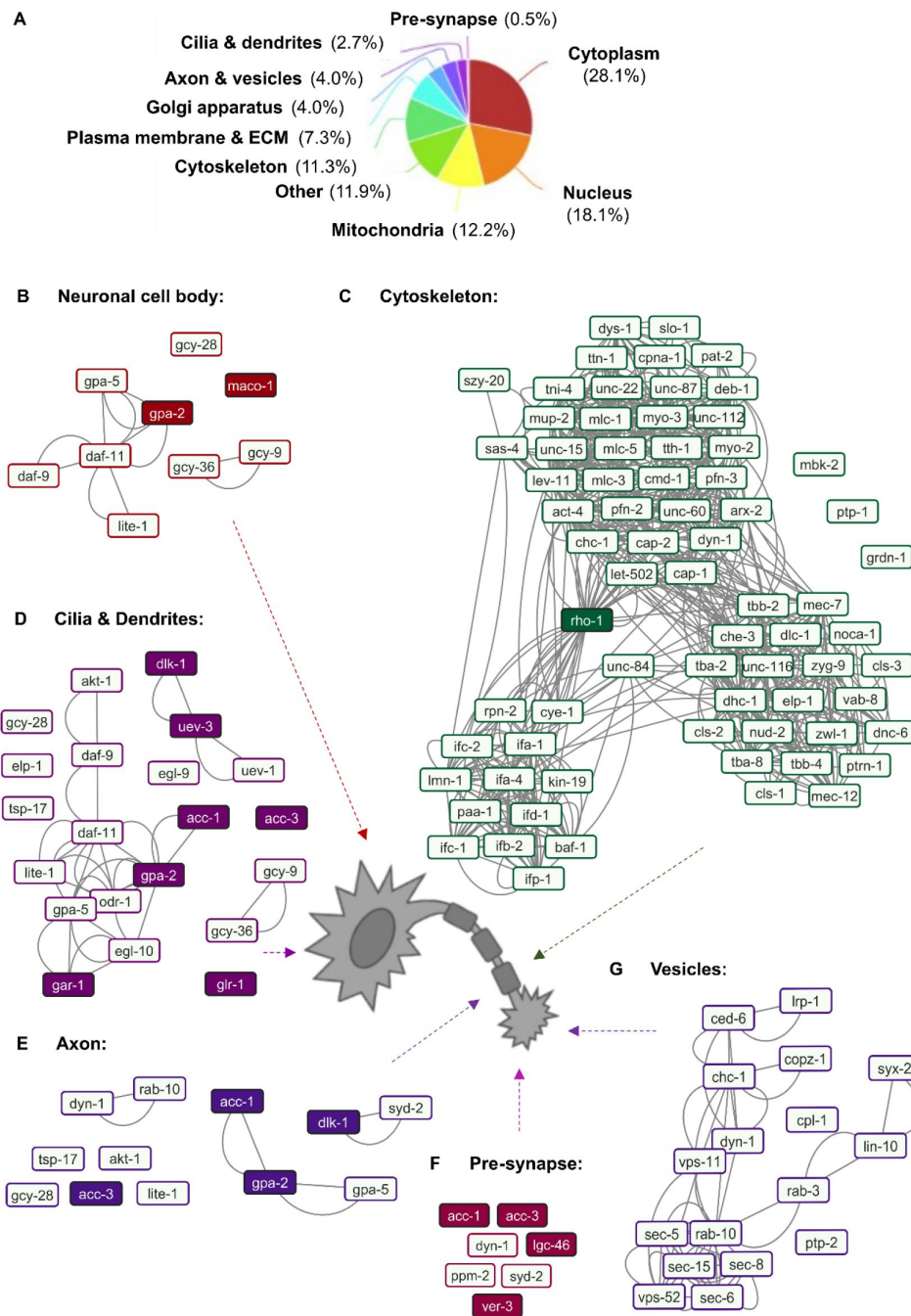
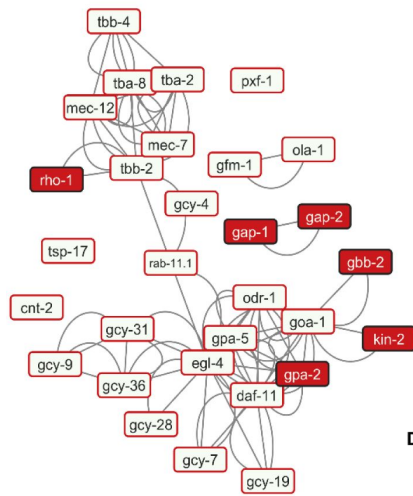


Figure S4

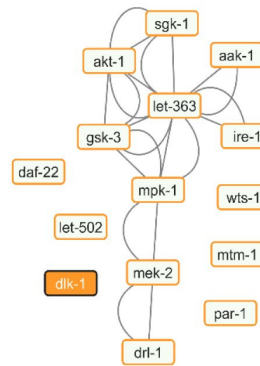
Analysis of subcellular localisations of assigned protein hits in the learning proteome.

Analyses based on gene ontology (GO) terms were performed with ClueGO (version 2.5.10) using Cytoscape (version 3.10.0). **(A)** Pie chart summarising the percentage of all proteins in 'TurboID, trained' protein lists assigned GO terms within different 'cellular components' (from Table S3). **Abbreviations:** ECM = extracellular matrix. **(B-G)** Protein-protein interaction (PPI) networks of proteins in the learning proteome assigned to cellular components most relevant to the nervous system: **(B)** Neuronal cell body/cytoplasm, **(C)** Cytoskeleton, **(D)** Cilia and dendrites, **(E)** Axon, **(F)** Pre-synapse, and **(G)** Vesicles. PPI interactions were exported from STRING (version 12.0) and visualised with Cytoscape. Arrows from networks in **B-G** annotate the relevant region on a neuron schematic. Highlighted proteins were further tested in this study. Known information on subcellular localisation and predictions based on homology were also used for several nodes (Gaudet et al., 2011 [Takayanagi-Kiya et al., 2016](#) [Lans et al., 2004](#) [Arellano-Carbajal et al., 2011](#) [Popovici et al., 2002](#)). The consolidated data from GO term analyses can be accessed on Table S5.

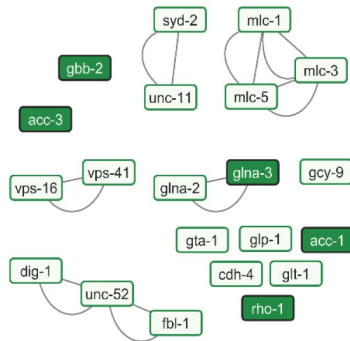
A G protein signalling:



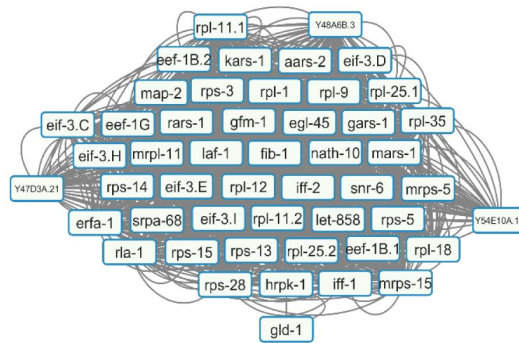
B Insulin & protein kinases:



C Neurotransmission:



D Protein synthesis:



E Protein degradation:

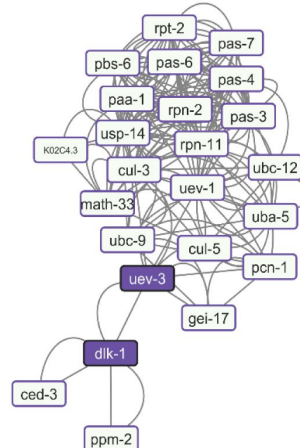


Figure S5

Analysis of molecular functions of assigned hits in the learning proteome.

Analyses using GO terms for molecular function were performed with ClueGO (version 2.5.10) using Cytoscape (version 3.10.0). Protein-protein interaction (PPI) networks generated for proteins assigned the GO term **(A)** G protein signalling, **(B)** Insulin and protein kinases, **(C)** Neurotransmission, **(D)** Protein synthesis, and **(E)** Protein degradation. PPI interactions were exported from STRING (version 12.0) to be visualised with Cytoscape using the protein lists in Table S3. Highlighted proteins were tested in this study using appropriate genetic mutants. Known information on molecular function and predictions based on homology were also used for several nodes (Putrenko et al., 2005 [\[1\]](#), Park et al., 2000 [\[2\]](#), Gottschling et al., 2017 [\[3\]](#), Gaudet et al., 2011 [\[4\]](#), Hayashizaki et al., 1998 [\[5\]](#), Lans et al., 2004 [\[6\]](#)). The consolidated data from GO term analyses can be accessed on Table S6.

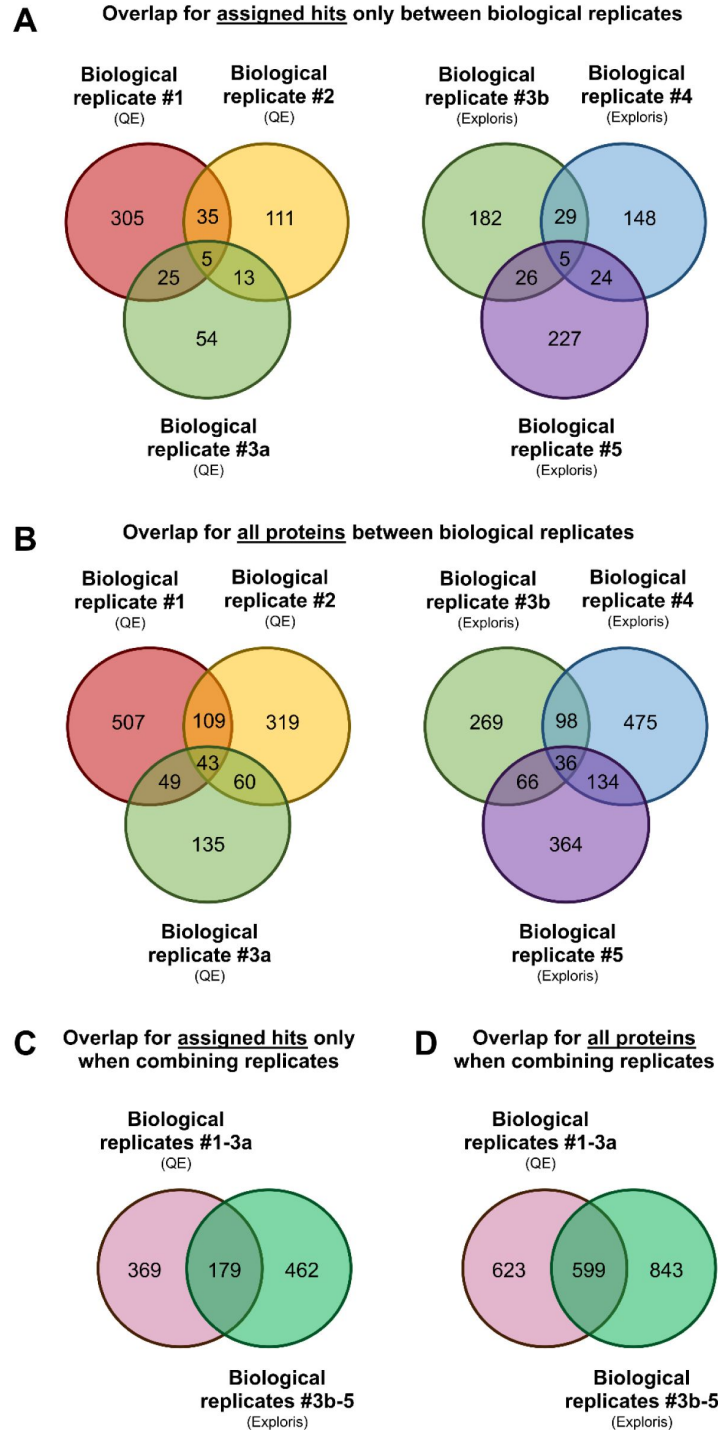


Figure S6

Assessing overlap for proteins detected in trained worms across all biological replicates.

(A-B) Numbers represent (A) assigned proteins only or (B) all (assigned and unassigned) proteins detected with the Q Exactive (QE) (left) or the Exploris (right) in ‘TurboID, trained’ animals only; these proteins were identified by subtracting proteins also in ‘non-transgenic, trained’ and/or in ‘TurboID, control’ worms. Six protein lists are compared here since one replicate was run in both mass spectrometers. (C-D) Lists containing proteins specific to trained animals for each biological replicate were compared based on the mass spectrometer used for (C) assigned proteins only and (D) all proteins.

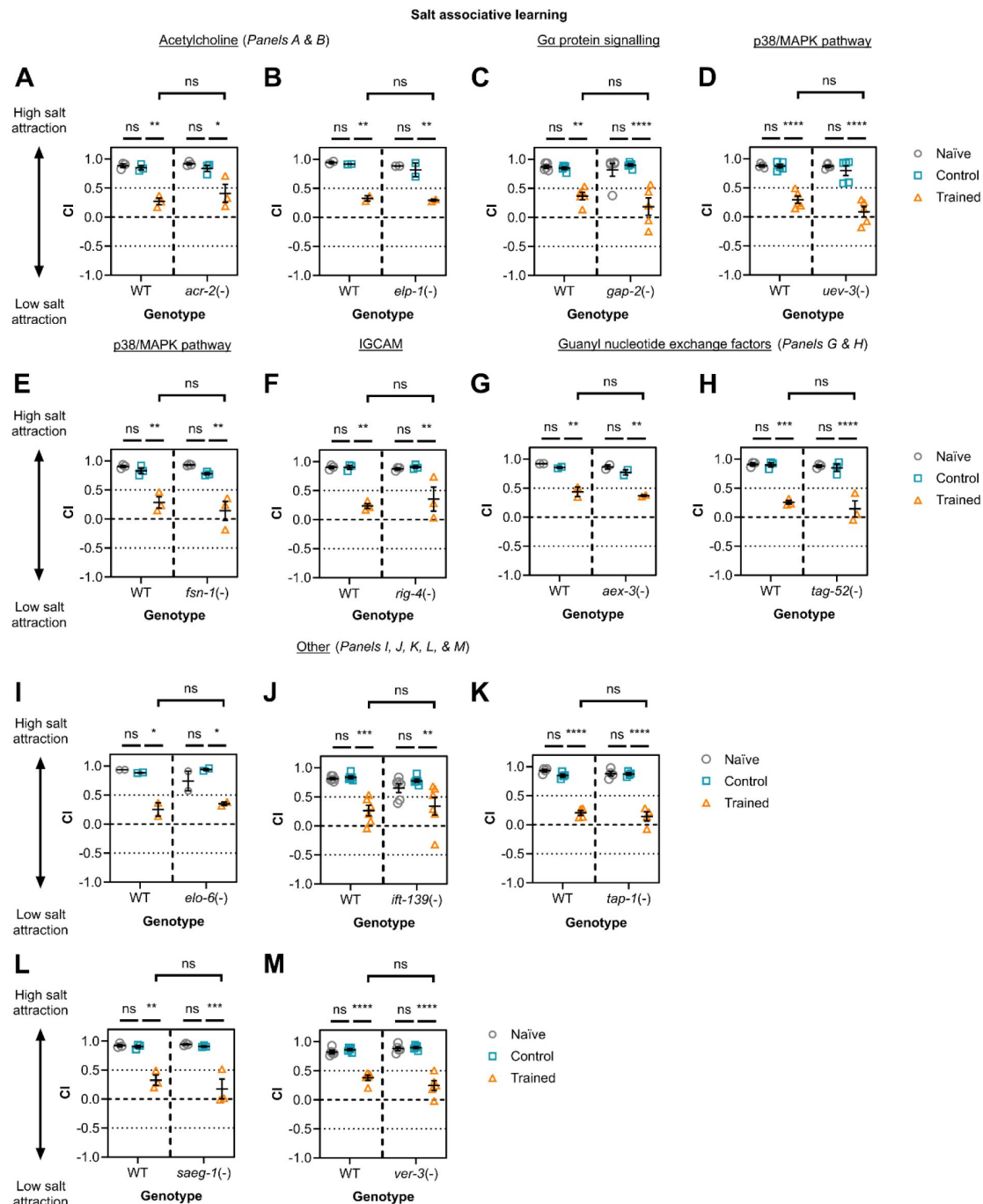


Figure S7

Behavioural testing for strong candidates identified by TurboID-based mass spectrometry experiments.

Worms with single mutations targeting molecular components in the following pathways were tested for their salt associative learning capacity, compared to wild-type (WT) worms: **(A & B)** Acetylcholine signalling: *acr-1* (n = 3) and *elp-1* (n = 2), **(C)** G protein signalling (*gap-2*, n = 5), **(D & E)** p38/MAPK signalling: *uev-3* (n = 5) and *fsn-1* (n = 3), **(F)** IGCAM (*rig-4*, n = 3), **(G & H)** Guanylyl nucleotide exchange factors: *aex-3* (n = 2) and *tag-52* (n = 3), and **(I-M)** Other proteins of interest: *elo-6* (n = 2), *ift-139* (n = 6), *tap-1* (n = 4), *saeg-1* (n = 3), and *ver-3* (n = 5). Each data point represents the average chemotaxis index (CI) per biological replicate (three technical replicates per biological replicate). Statistical analysis: Two-way ANOVA & Tukey's multiple comparisons test (**** ≤ 0.0001; *** ≤ 0.001; ** ≤ 0.01; * ≤ 0.05; ns = non-significant). Error bars = mean ± SEM.

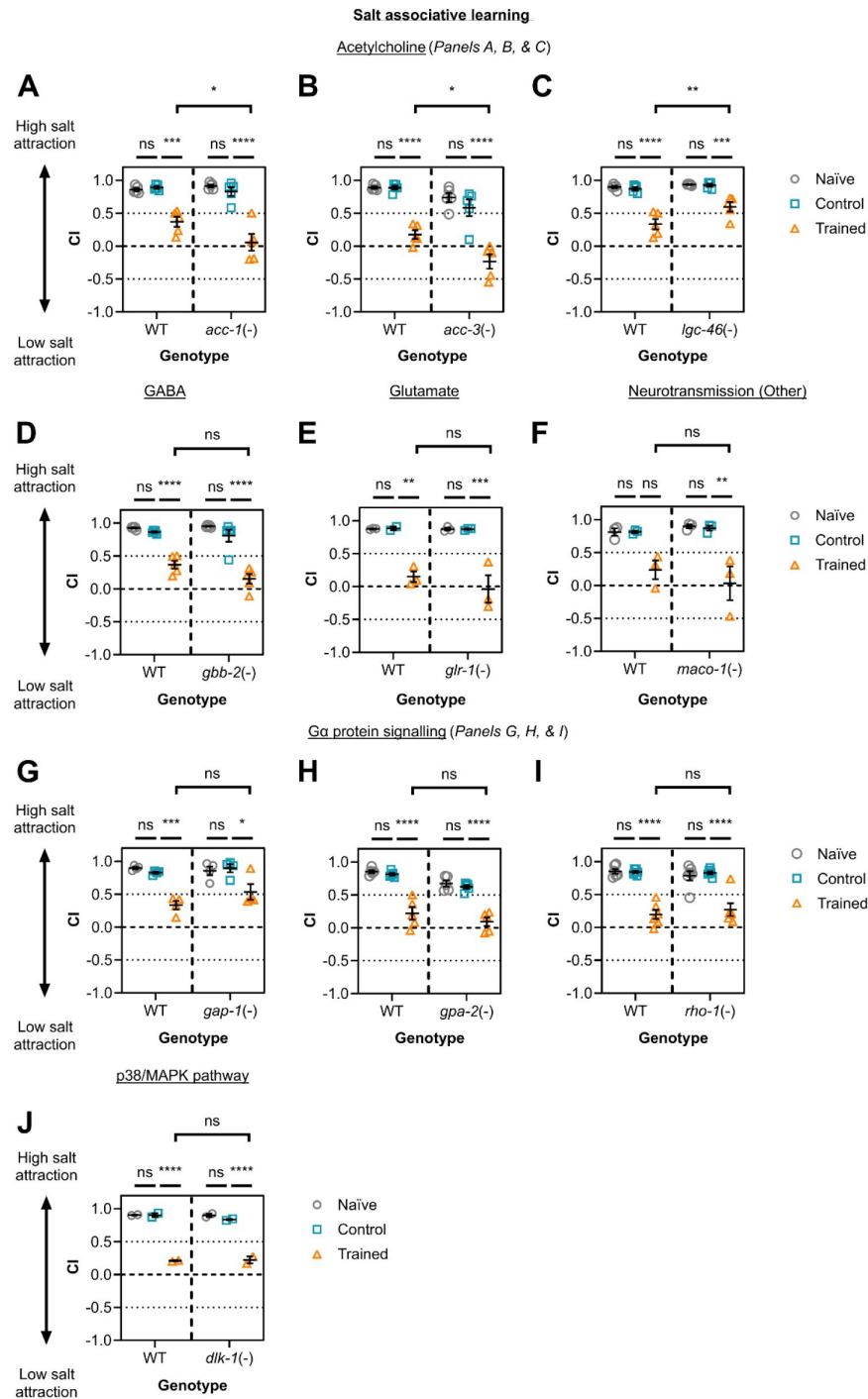


Figure S8

Behavioural testing for weak candidates in the learning proteome.

Weak candidates from TurboID-based mass spectrometry data were tested by assessing learning capacity for single mutant *C. elegans*, compared to wild-type (WT) worms. Each mutation targets a molecular component in one of the following pathways: (**A, B, & C**) Acetylcholine signalling: *acc-1*, *acc-3*, or *lgc-46* (n = 5), (**D, E, & F**) Neurotransmission: *gbb-2* (n = 5), *glr-1* (n = 3), or *maco-1* (n = 3), (**G, H, & I**) G protein signalling: *gap-1*, (n = 4), *gpa-2* (n = 5), or *rho-1* (n = 6), or (**J**) p38/MAPK signalling (*dlk-1*, n = 2). One biological replicate (n) constitutes each data point, with three technical replicates per biological replicate (23-372 worms per technical replicate). Error bars = mean ± SEM. Statistical analysis: Two-way ANOVA & Tukey's multiple comparisons test (**** ≤ 0.0001; *** ≤ 0.001; ** ≤ 0.01; * ≤ 0.05; ns = non-significant).

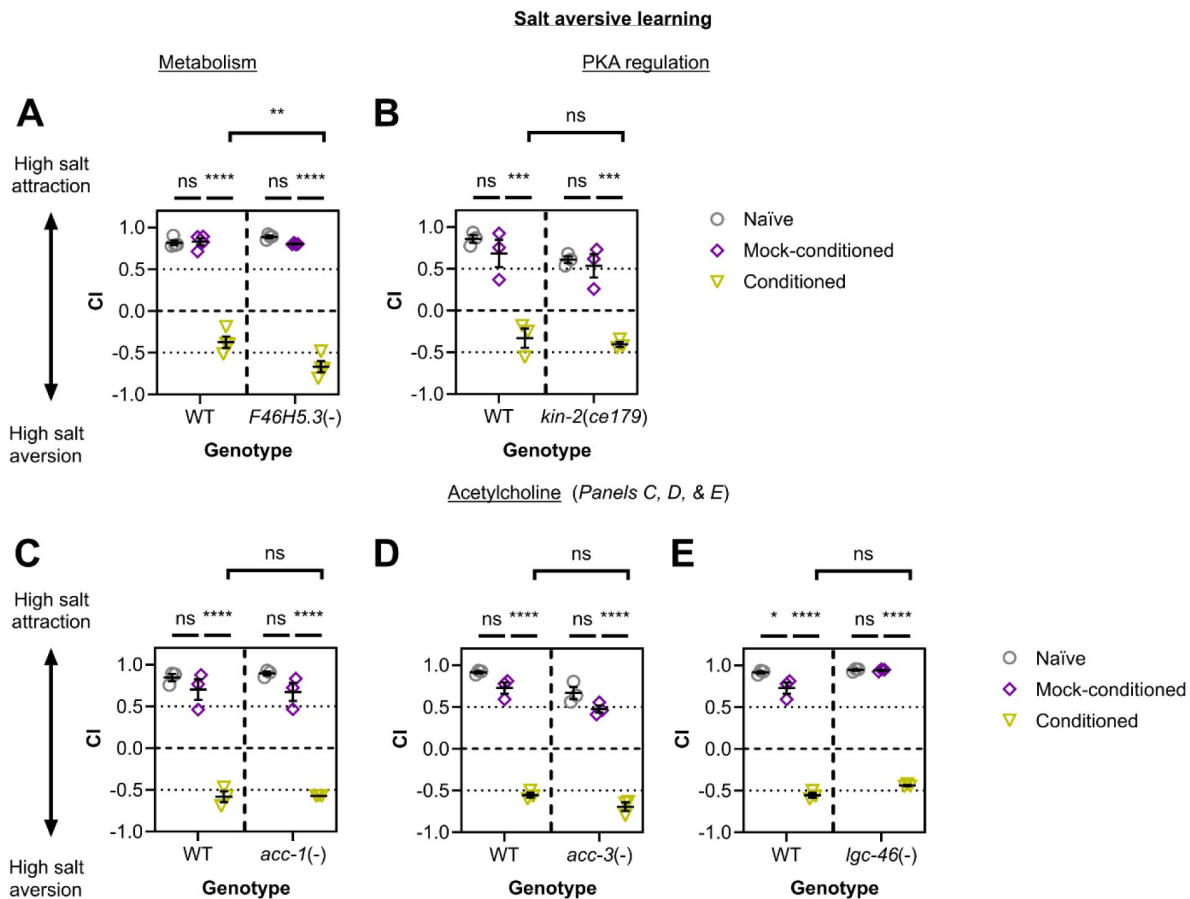


Figure S9

Salt aversive learning is modulated by arginine kinase F46H5.3.

Worms were naïve, mock-conditioned with no salt + no food, or conditioned with salt + no food. Three technical replicates were generated per biological replicate (26-481 worms per technical replicate). **(A)** $n = 4$ and **(B-E)** $n = 3$ biological replicates. Each data point represents a chemotaxis index (CI) for an individual biological replicate. Statistical analysis: Two-way ANOVA & Tukey's multiple comparisons test (**** ≤ 0.0001 ; *** ≤ 0.001 ; ** ≤ 0.01 ; * ≤ 0.05 ; ns = non-significant). Error bars = mean $\pm$ SEM.

Data Availability

The source data from this publication, including mass spectrometry data (files from MASCOT search), raw data from learning assays, videos from behavioural tracking, have been uploaded to the **Dryad database**: this will be made available publicly following publication. In the interim, please contact the corresponding author for requests for raw data. Custom Python code is available through **GitHub** (<https://github.com/ChewWormLab/Chew-Worm-Lab-Post-Mass-Spectrometry-Peptide-processing> ). *C. elegans* strains used in this study are available upon request.

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Author contributions

Conceptualisation/Methodology/Project administration/Supervision – AP, YLC. Data curation/Formal analysis/Investigation – AR, AM, EA, RA, RG, MJ, AP, YLC. Funding acquisition – YLC. Visualisation/Writing – original draft/Writing – review & editing – AR, MJ, YLC.

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Additional files

Tables S1-S9. Table S1: List of *C. elegans* lines used in this study. Table S2: Quantifying biotinylated protein levels in western blots for high-salt control and trained TurboID *C. elegans*. Biotinylated protein signal is compared between experimental groups based on western blots in Figure S3. Each row corresponds to a unique biological replicate of western blot and mass spectrometry data as annotated under the 'Replicate' column. For each sample, the sum of areas containing bands within an entire lane is used as a readout for biotin-tagged protein signal intensity from whole worm lysates. Non-transgenic/Non-Tg *C. elegans* and TurboID/TbID were treated with biotin during salt associative learning, to generate high-salt control and trained groups for each line. **Table S3: Protein lists for all assigned hits detected in TurboID, trained worms.** *C. elegans* peptides were assigned corresponding protein identities by MASCOT and then assigned protein lists were generated for each biological replicate by subtracting proteins also present in 'non-transgenic, trained' worms and/or in 'TurboID, control' worms within the same replicate. Each protein list in rows 9-378 are defined in this table by its corresponding biological replicate, the mass spectrometer used, and the inclusion of data from unbound and bound peptide samples. Peptide samples generated for biological replicate 3 were run in both mass spectrometers used in this study, the Q Exactive or the Exploris, and the corresponding protein list is named '3a' and '3b' respectively. **Table S4: Protein lists for all unassigned hits detected in TurboID, trained worms.** Protein identities were determined through bulk BLAST searching *C. elegans* peptides detected by mass spectrometry, rather than MASCOT, and then unassigned protein lists were generated for each experimental group in each biological replicate. Proteins also in 'non-transgenic, trained' and/or 'TurboID, control' lists for the same biological replicate were subtracted from the corresponding 'TurboID, trained' list. Each protein list in rows 9-587 are defined in this table by its corresponding biological replicate, the mass spectrometer used, and the inclusion of data from unbound and bound peptide samples. Peptide samples generated for biological replicate 3 were run in both mass spectrometers used in this study, the Q Exactive or the Exploris, and the corresponding protein list is named '3a' and '3b' respectively. **Table S5: Cellular component GO terms associated with assigned hits in the learning proteome.** All (1010) assigned protein hits detected in TurboID, trained worms were separated into 10 clusters by k-means clustering on STRING (version 12.0) and GO term analysis was conducted for each cluster using ClueGO (version 2.5.10) on Cytoscape (version 3.10.0). GO terms were then sorted into 'plasma membrane & extracellular matrix (ECM)', 'cilia & dendrites', 'cell body & cytoplasm', 'nucleus', 'Golgi apparatus', 'cytoskeleton', 'mitochondria', 'axon & vesicles', 'pre-synapse', and 'other' (i.e., non-informative terms) categories due to recurring and/or similar terms. Colour-coding is based on the GO Group. Each category is separated by underlined and bolded titles corresponding to the cellular component in rows 1-285. Consolidated lists of genes/proteins are detailed from rows 287-465. **Table S6: Biological process and molecular function GO terms associated with assigned hits in the learning proteome.** All (1010) assigned protein hits detected in TurboID, trained worms were separated into 10 clusters by k-means clustering on STRING (version 12.0) and GO term analysis was conducted for each cluster using ClueGO (version 2.5.10) on Cytoscape (version 3.10.0). GO terms were then sorted into 'protein synthesis', 'protein degradation', 'insulin & protein kinases', 'G protein signalling', and 'neurotransmission' categories due to recurring and/or similar terms. Colour-coding is based on the GO Group. Each category is separated by underlined and bolded titles corresponding to the biological processes listed above in rows 1-109. Consolidated lists of genes/proteins are detailed from rows 111-162. **Table S7: Identifying neuron classes represented within the learning proteome in this study.** This table presents neurons that express ≥ 1 gene encoding proteins found in either the trained or control lists. For neurons present in both datasets, fold-change values were calculated as the ratio of the number of genes expressed in the trained condition to those in the control. Neurons are ranked in descending order of fold-change. Values under the '#Proteins from trained (normalised)' column are normalised based on the fold-difference between the number of proteins in the trained vs control protein lists (normalisation factor = ~ 1.8). Classifications for neuron type are assigned to each neuron class based on Pereira et al. (2015). The CeNGEN database was used to identify gene expression patterns Taylor et al. (2021) and includes all 302 neurons in the *C. elegans* hermaphrodite nervous system. Only 128 neuron classes are

listed here since neurons that do not express any proteins from trained or control groups are not shown. **Table S8: Molecular pathways for proteins detected in TurboID, trained *C. elegans* only.** Proteins unique to trained animals that express TurboID were identified by subtracting those also in 'TurboID, control' and/or 'non-transgenic, trained' protein lists. Gene names are listed with assigned hits in bold. Each gene/protein has been grouped based on 'Biological Process' GO term annotations provided by STRING (version 12.0) and information available on WormBase (version WS290) based on data accessed during September–November 2023. **Table S9: Statistical analyses for learning assays performed in this study.** Data from statistical analyses are separated by figure with appropriate underlined and bolded titles, including Figure 1B (red, rows 1–7), Figure 5 (yellow, rows 9–31), Figure 6 (green, rows 33–63), Figure S7 (blue, rows 65–167), Figure S8 (purple, rows 169–247), and Figure S9 (pink, rows 249–287). Statistical analyses were performed on GraphPad PRISM 8 using two-way ANOVA and Tukey's multiple comparisons tests (alpha value = $p < 0.05$). Columns left-to-right list the groups being compared, the mean difference (Diff.), the 95% confidence interval of diff., whether the difference is significant, the degree of significance (**** ≤ 0.0001 ; *** ≤ 0.001 ; ** ≤ 0.01 ; * ≤ 0.05 ; ns = non-significant), and the adjusted p value. [↗](#)

Note

This reviewed preprint has been updated to add the middle initial to one of the authors.

References

- Akefe I. O., Saber S. H., Matthews B., Venkatesh B. G., Gormal R. S., Blackmore D. G., Alexander S., Sieriecki E., Gambin Y., Bertran-Gonzalez J., Vitale N., Humeau Y., Gaudin A., Ellis S. A., Michaels A. A., Xue M., Cravatt B., Joensuu M., Wallis T. P., Meunier F. A. (2024) **The DDHD2-STXBP1 interaction mediates long-term memory via generation of saturated free fatty acids** *EMBO J.* **43**:533–567 [Google Scholar](#)
- Anaconda Software Distribution (2016) **Anaconda: Computer software Vers. 2-2.4.0** <https://anaconda.com>
- Angeles-Albores D., Ry N. L., Chan J., Sternberg P. W (2016) **Tissue enrichment analysis for *C. elegans* genomics** *BMC Bioinformatics* **17**:366 [Google Scholar](#)
- Appleby P. A (2012) **A model of chemotaxis and associative learning in *C. elegans*** *Biological Cybernetics* **106**:373–387 [Google Scholar](#)
- Ardiel E. L., Rankin C. H (2015) **Cross-referencing online activity with the connectome to identify a neglected but well-connected neuron** *Current Biology* **25**:R405–R406 [Google Scholar](#)
- Arellano-Carbajal F., Briseño-Roa L., Couto A., Cheung B. H., Labouesse M., De Bono M. (2011) **Macoilin, a conserved nervous system-specific ER membrane protein that regulates neuronal excitability** *PLoS Genet* **7**:e1001341 [Google Scholar](#)
- Arey R. N., Stein G. M., Kaletsky R., Kauffman A., Murphy C. T (2018) **Activation of G(aq) Signaling Enhances Memory Consolidation and Slows Cognitive Decline** *Neuron* **98**:562–574 [Google Scholar](#)
- Artan M., Barratt S., Flynn S. M., Begum F., Skehel M., Nicolas A., De Bono M. (2021) **Interactome analysis of *Caenorhabditis elegans* synapses by TurboID-based proximity**

labeling *J Biol Chem* **297**:101094 [Google Scholar](#)

Artan M., Hartl M., Chen W., De Bono M. (2022) **Depletion of endogenously biotinylated carboxylases enhances the sensitivity of TurboID-mediated proximity labeling in *Caenorhabditis elegans*** *J Biol Chem* **298**:102343 [Google Scholar](#)

Ataei N., Sabzghabae A. M., Movahedian A (2015) **Calcium/Calmodulin-dependent Protein Kinase II is a Ubiquitous Molecule in Human Long-term Memory Synaptic Plasticity: A Systematic Review** *Int J Prev Med* **6**:88 [Google Scholar](#)

Bailey C. H., Kandel E. R., Harris K. M (2015) **Structural Components of Synaptic Plasticity and Memory Consolidation** *Cold Spring Harb Perspect Biol* **7**:a021758 [Google Scholar](#)

Bakalarski C. E., Elias J. E., Villén J., Haas W., Gerber S. A., Everley P. A., Gygi S. P (2008) **The impact of peptide abundance and dynamic range on stable-isotope-based quantitative proteomic analyses** *J Proteome Res* **7**:4756–65 [Google Scholar](#)

Barzilai A., Kennedy T. E., Sweatt J. D., Kandel E. R (1989) **5-HT modulates protein synthesis and the expression of specific proteins during long-term facilitation in *Aplysia* sensory neurons** *Neuron* **2**:1577–1586 [Google Scholar](#)

Beets I., Janssen T., Meelkop E., Temmerman L., Suetens N., Rademakers S., Jansen G., Schoofs L (2012) **Vasopressin/oxytocin-related signaling regulates gustatory associative learning in *C. elegans*** *Science* **338**:543–5 [Google Scholar](#)

Beets I., Zhang G., Fenk L. A., Chen C., Nelson G. M., Félix M. A., De Bono M. (2020) **Natural Variation in a Dendritic Scaffold Protein Remodels Experience-Dependent Plasticity by Altering Neuropeptide Expression** *Neuron* **105**:106–121 [Google Scholar](#)

Berbari N. F., Malarkey E. B., Yazdi S. M., McNair A. D., Kippe J. M., Croyle M. J., Kraft T. W., Yoder B. K (2014) **Hippocampal and cortical primary cilia are required for aversive memory in mice** *PLoS One* **9**:e106576 [Google Scholar](#)

Bhatla N. (2009) **WormWeb: *C. elegans* Interactive Neural Network [Online]** <http://www.wormweb.org/>

Bindea G., Mlecnik B., Hackl H., Charoentong P., Tosolini M., Kirilovsky A., Fridman W. H., Pagès F., Trajanoski Z., Galon J (2009) **ClueGO: a Cytoscape plug-in to decipher functionally grouped gene ontology and pathway annotation networks** *Bioinformatics* **25**:1091–3 [Google Scholar](#)

Bradshaw K. D., Emptage N. J., Bliss T. V (2003) **A role for dendritic protein synthesis in hippocampal late LTP** *Eur J Neurosci* **18**:3150–2 [Google Scholar](#)

Branon T. C., Bosch J. A., Sanchez A. D., Udeshi N. D., Svinkina T., Carr S. A., Feldman J. L., Perrimon N., Ting A. Y (2018) **Efficient proximity labeling in living cells and organisms with TurboID** *Nature biotechnology* **36**:880–887 [Google Scholar](#)

Brenner S (1974) **The genetics of *Caenorhabditis elegans*** *Genetics* **77**:71–94 [Google Scholar](#)

Cao L., Jiao X., Zuzga D. S., Liu Y., Fong D. M., Young D., During M. J (2004) **VEGF links hippocampal activity with neurogenesis, learning and memory** *Nature Genetics* **36**:827–835 [Google Scholar](#)

- Chalfie M., Sulston J. E., White J. G., Southgate E., Thomson J. N., Brenner S (1985) **The neural circuit for touch sensitivity in *Caenorhabditis elegans*** *J Neurosci* **5**:956–64 [Google Scholar](#)
- Chen C.-C., Wu J.-K., Lin H.-W., Pai T.-P., Fu T.-F., Wu C.-L., Tully T., Chiang A.-S (2012) **Visualizing Long-Term Memory Formation in Two Neurons of the *Drosophila* Brain** *Science* **335**:678–685 [Google Scholar](#)
- Cock P. J. A., Antao T., Chang J. T., Chapman B. A., Cox C. J., Dalke A., Friedberg I., Hamelryck T., Kauff F., Wilczynski B., De Hoon M. J. L. (2009) **Biopython: freely available Python tools for computational molecular biology and bioinformatics** *Bioinformatics* **25**:1422–1423 [Google Scholar](#)
- Cong Y., Yang H., Zhang P., Xie Y., Cao X., Zhang L (2020) **Transcriptome Analysis of the Nematode *Caenorhabditis elegans* in Acidic Stress Environments** *Front Physiol* **11**:1107 [Google Scholar](#)
- Cremer H., Lange R., Christoph A., Plomann M., Vopper G., Roes J., Brown R., Baldwin S., Kraemer P., Scheff S., et al. (1994) **Inactivation of the N-CAM gene in mice results in size reduction of the olfactory bulb and deficits in spatial learning** *Nature* **367**:455–9 [Google Scholar](#)
- Das S., Lituma P. J., Castillo P. E., Singer R. H (2023) **Maintenance of a short-lived protein required for long-term memory involves cycles of transcription and local translation** *Neuron* **111**:2051–2064 [Google Scholar](#)
- Doyle E., Nolan P. M., Bell R., Regan C. M (1992) **Intraventricular infusions of anti-neural cell adhesion molecules in a discrete posttraining period impair consolidation of a passive avoidance response in the rat** *J Neurochem* **59**:1570–3 [Google Scholar](#)
- Fadda M., De Fruyt N., Borghgraef C., Watteyne J., Peymen K., Vandeweyer E., Naranjo Galindo F. J., Kieswetter A., Mirabeau O., Chew Y. L., Beets I., Schoofs L. (2020) **NPY/NPF-Related Neuropeptide FLP-34 Signals from Serotonergic Neurons to Modulate Aversive Olfactory Learning in *Caenorhabditis elegans*** *J Neurosci* **40**:6018–6034 [Google Scholar](#)
- Fazeli M. S., Corbet J., Dunn M. J., Dolphin A. C., Bliss T. V (1993) **Changes in protein synthesis accompanying long-term potentiation in the dentate gyrus in vivo** *J Neurosci* **13**:1346–53 [Google Scholar](#)
- Feng J., Zhou Y., Campbell S. L., Le T., Li E., Sweatt J. D., Silva A. J., Fan G (2010) **Dnmt1 and Dnmt3a maintain DNA methylation and regulate synaptic function in adult forebrain neurons** *Nat Neurosci* **13**:423–30 [Google Scholar](#)
- Froehlich J. J., Rajewsky N., Ewald C. Y (2021) **Estimation of *C. elegans* cell- and tissue volumes** *MicroPubl Biol* **2021** [Google Scholar](#)
- Gaudet P., Livstone M. S., Lewis S. E., Thomas P. D (2011) **Phylogenetic-based propagation of functional annotations within the Gene Ontology consortium** *Briefings in Bioinformatics* **12**:449–462 [Google Scholar](#)
- Gottschling D. C., Döring F., Lüersen K (2017) **Locomotion Behavior Is Affected by the Gα(S) Pathway and the Two-Pore-Domain K(+) Channel TWK-7 Interacting in GABAergic Motor Neurons in *Caenorhabditis elegans*** *Genetics* **206**:283–297 [Google Scholar](#)

- Guo Z. V., Hart A. C., Ramanathan S (2009) **Optical interrogation of neural circuits in *Caenorhabditis elegans*** *Nature Methods* **6**:891–896 [Google Scholar](#)
- Gyurkó M. D., Csermely P., Sóti C., Steták A (2015) **Distinct roles of the RasGAP family proteins in *C. elegans* associative learning and memory** *Sci Rep* **5**:15084 [Google Scholar](#)
- Hart A. C., Sims S., Kaplan J. M (1995) **Synaptic code for sensory modalities revealed by *C. elegans* GLR-1 glutamate receptor** *Nature* **378**:82–85 [Google Scholar](#)
- Hawkins R. D., Byrne J. H (2015) **Associative learning in invertebrates** *Cold Spring Harb Perspect Biol* **7** [Google Scholar](#)
- Hayashizaki S., Iino Y., Yamamoto M (1998) **Characterization of the *C. elegans* gap-2 gene encoding a novel Ras-GTPase activating protein and its possible role in larval development** *Genes to Cells* **3**:189–202 [Google Scholar](#)
- Hernandez P. J., Abel T (2008) **The role of protein synthesis in memory consolidation: Progress amid decades of debate** *Neurobiology of Learning and Memory* **89**:293–311 [Google Scholar](#)
- Hiroki S., Iino Y (2022) **The redundancy and diversity between two novel PKC isoforms that regulate learning in *Caenorhabditis elegans*** *Proc Natl Acad Sci U S A* **119** [Google Scholar](#)
- Hiroki S., Yoshitane H., Mitsui H., Sato H., Umatani C., Kanda S., Fukada Y., Iino Y (2022) **Molecular encoding and synaptic decoding of context during salt chemotaxis in *C. elegans*** *Nature Communications* **13**:2928 [Google Scholar](#)
- Holzer E., Rumpf-Kienzl C., Falk S., Dammermann A. (2021) **A Modified TurboID Approach Identifies Tissue-Specific Centriolar Components In *C. elegans*** *bioRxiv* <https://doi.org/10.1101/2021.12.20.473533> | [Google Scholar](#)
- Huang Q., Liao C., Ge F., Ao J., Liu T (2022) **Acetylcholine bidirectionally regulates learning and memory** *Journal of Neurorestoration* **10**:100002 [Google Scholar](#)
- Huckleberry K. A., Ferguson L. B., Drew M. R (2016) **Behavioral mechanisms of context fear generalization in mice** *Learn Mem* **23**:703–709 [Google Scholar](#)
- Hukema R. (2006) **Gustatory behaviour in *C. elegans*** *Erasmus MC* [Google Scholar](#)
- Hung W. L., Hwang C., Gao S., Liao E. H., Chitturi J., Wang Y., Li H., Stigloher C., Bessereau J. L., Zhen M (2013) **Attenuation of insulin signalling contributes to FSN-1-mediated regulation of synapse development** *Embo j* **32**:1745–60 [Google Scholar](#)
- Inberg S., Elkobi A., Edri E., Rosenblum K (2013) **Taste Familiarity Is Inversely Correlated with Arc/Arg3.1 Hemispheric Lateralization** *The Journal of Neuroscience* **33**:11734 [Google Scholar](#)
- Jang M. S., Toyoshima Y., Tomioka M., Kunitomo H., Iino Y (2019) **Multiple sensory neurons mediate starvation-dependent aversive navigation in *Caenorhabditis elegans*** *Proceedings of the National Academy of Sciences* **116**:18673–18683 [Google Scholar](#)
- Jeong D.-E., Lee Y., Lee S.-J. V (2018) **Western Blot Analysis of *C. elegans* Proteins** In: Huang L. E., editors. *Hypoxia: Methods and Protocols* New York, NY: Springer New York [Google Scholar](#)

- Jiang S. Z., Shahoha M., Zhang H. Y., Brancaleone W., Elkahloun A., Tejeda H. A., Ashery U., Eiden L. E (2024) **The guanine nucleotide exchange factor RapGEF2 is required for ERK-dependent immediate-early gene (Egr1) activation during fear memory formation** *Cell Mol Life Sci* **81**:48 [Google Scholar](#)
- Jong Y. I., Harmon S. K., O'malley K. L (2018) **Intracellular GPCRs Play Key Roles in Synaptic Plasticity** *ACS Chem Neurosci* **9**:2162–2172 [Google Scholar](#)
- Jost C. R., Van Der Zee C. E. E. M., 'T Zandt H. J. A., Oerlemans F., Verheij M., Streijger F., Fransen J., Heerschap A., Cools A. R., Wieringa B. (2002) **Creatine kinase B-driven energy transfer in the brain is important for habituation and spatial learning behaviour, mossy fibre field size and determination of seizure susceptibility** *European Journal of Neuroscience* **15**:1692–1706 [Google Scholar](#)
- Jovasevic V., Zhang H., Sananbenesi F., Guedea A. L., Soman K. V., Wiktorowicz J. E., Fischer A., Radulovic J (2021) **Primary cilia are required for the persistence of memory and stabilization of perineuronal nets** *iScience* **24**:102617 [Google Scholar](#)
- Kandel E. R (2012) **The molecular biology of memory: cAMP, PKA, CRE, CREB-1, CREB-2, and CPEB** *Molecular Brain* **5**:14 [Google Scholar](#)
- Kano T., Brockie P. J., Sassa T., Fujimoto H., Kawahara Y., Iino Y., Mellem J. E., Madsen D. M., Hosono R., Maricq A. V (2008) **Memory in *Caenorhabditis elegans* is mediated by NMDA-type ionotropic glutamate receptors** *Curr Biol* **18**:1010–5 [Google Scholar](#)
- Karabinos A., Büssing I., Schulze E., Wang J., Weber K., Schnabel R (2003) **Functional analysis of the single calmodulin gene in the nematode *Caenorhabditis elegans* by RNA interference and 4-D microscopy** *European Journal of Cell Biology* **82**:557–563 [Google Scholar](#)
- Kauffman A. L., Ashraf J. M., Corces-Zimmerman M. R., Landis J. N., Murphy C. T (2010) **Insulin signaling and dietary restriction differentially influence the decline of learning and memory with age** *PLoS Biol* **8**:e1000372 [Google Scholar](#)
- Khanna N., Cressman C. P., Tata C. P., Williams P. L. (1997) **Tolerance of the Nematode *Caenorhabditis elegans* to pH, Salinity, and Hardness in Aquatic Media** *Archives of Environmental Contamination and Toxicology* **32**:110–114 [Google Scholar](#)
- Kitazono T., Hara-Kuge S., Matsuda O., Inoue A., Fujiwara M., Ishihara T (2017) **Multiple Signaling Pathways Coordinately Regulate Forgetting of Olfactory Adaptation through Control of Sensory Responses in *Caenorhabditis elegans*** *J Neurosci* **37**:10240–10251 [Google Scholar](#)
- Kniazeva M., Crawford Q. T., Seiber M., Wang C. Y., Han M (2004) **Monomethyl branched-chain fatty acids play an essential role in *Caenorhabditis elegans* development** *PLoS Biol* **2**:E257 [Google Scholar](#)
- Komatsu H., Jin Y. H., L'etoile N., Mori I., Bargmann C. I., Akaike N., Ohshima Y (1999) **Functional reconstitution of a heteromeric cyclic nucleotide-gated channel of *Caenorhabditis elegans* in cultured cells** *Brain Res* **821**:160–8 [Google Scholar](#)
- Krebs H. A., Johnson W. A (1937) **Metabolism of ketonic acids in animal tissues** *Biochem J* **31**:645–60 [Google Scholar](#)

- Kunitomo H., Sato H., Iwata R., Satoh Y., Ohno H., Yamada K., Iino Y (2013) **Concentration memory-dependent synaptic plasticity of a taste circuit regulates salt concentration chemotaxis in *Caenorhabditis elegans*** *Nature Communications* **4**:2210 [Google Scholar](#)
- L'etoile N. D., Coburn C. M., Eastham J., Kistler A., Gallegos G., Bargmann C. I (2002) **The cyclic GMP-dependent protein kinase -4 regulates olfactory adaptation in *C. elegans*** *Neuron* **36**:1079–89 [Google Scholar](#)
- Lakhina V., Arey Rachel N., Kaletsky R., Kauffman A., Stein G., Keyes W., Xu D., Murphy Coleen T (2015) **Genome-wide Functional Analysis of CREB/Long-Term Memory-Dependent Transcription Reveals Distinct Basal and Memory Gene Expression Programs** *Neuron* **85**:330–345 [Google Scholar](#)
- Lamprecht R (2014) **The actin cytoskeleton in memory formation** *Progress in Neurobiology* **117**:1–19 [Google Scholar](#)
- Lans H., Rademakers S., Jansen G. (2004) **A Network of Stimulatory and Inhibitory Gα-Subunits Regulates Olfaction in *Caenorhabditis elegans*** *Genetics* **167**:1677–1687 [Google Scholar](#)
- Lee S. H., Choi J. H., Lee N., Lee H. R., Kim J. I., Yu N. K., Choi S. L., Lee S. H., Kim H., Kaang B. K (2008) **Synaptic protein degradation underlies destabilization of retrieved fear memory** *Science* **319**:1253–6 [Google Scholar](#)
- Liang V., Ullrich M., Lam H., Chew Y. L., Banister S., Song X., Zaw T., Kassiou M., Götz J., Nicholas H. R (2014) **Altered proteostasis in aging and heat shock response in *C. elegans* revealed by analysis of the global and de novo synthesized proteome** *Cell Mol Life Sci* **71**:3339–61 [Google Scholar](#)
- Liao C.-P., Chiang Y.-C., Tam W. H., Chen Y.-J., Chou S.-H., Pan C.-L (2022) **Neurophysiological basis of stress-induced aversive memory in the nematode *Caenorhabditis elegans*** *Current Biology* **32**:5309–5322 [Google Scholar](#)
- Lim J. P., Fehlauer H., Das A., Saro G., Glauser D. A., Brunet A., Goodman M. B (2018) **Loss of CaMKI Function Disrupts Salt Aversive Learning in *C. elegans*** *The Journal of neuroscience : the official journal of the Society for Neuroscience* **38**:6114–6129 [Google Scholar](#)
- Lin C. H. A., Tomioka M., Pereira S., Sellings L., Iino Y., Kooy D. V. D (2010) **Insulin Signaling Plays a Dual Role in *Caenorhabditis elegans* Memory Acquisition and Memory Retrieval** *The Journal of Neuroscience* **30**:8001 [Google Scholar](#)
- Lindsay J. H., Mathies L. D., Davies A. G., Bettinger J. C (2022) **A neuropeptide signal confers ethanol state dependency during olfactory learning in *Caenorhabditis elegans*** *Proc Natl Acad Sci U S A* **119**:e2210462119 [Google Scholar](#)
- Lu N. C., Goetsch K. M (1993) **Carbohydrate Requirement of *Caenorhabditis Elegans* and the Final Development of a Chemically Defined Medium** *Nematologica* **39**:303–311 [Google Scholar](#)
- Luo L., Wen Q., Ren J., Hendricks M., Gershow M., Qin Y., Greenwood J., Soucy E. R., Klein M., Smith-Parker H. K., Calvo A. C., Colón-Ramos D. A., Samuel A. D., Zhang Y (2014) **Dynamic encoding of perception, memory, and movement in a *C. elegans* chemotaxis circuit** *Neuron* **82**:1115–28 [Google Scholar](#)

- Matsumoto Y., Matsumoto C. S., Mizunami M (2018) **Signaling Pathways for Long-Term Memory Formation in the Cricket** *Front Psychol* **9**:1014 [Google Scholar](#)
- Meneghini M. D., Ishitani T., Carter J. C., Hisamoto N., Ninomiya-Tsuji J., Thorpe C. J., Hamill D. R., Matsumoto K., Bowerman B (1999) **MAP kinase and Wnt pathways converge to downregulate an HMG-domain repressor in *Caenorhabditis elegans*** *Nature* **399**:793–7 [Google Scholar](#)
- Morud J., Hardege I., Liu H., Wu T., Choi M.-K., Basu S., Zhang Y., Schafer W. R (2021) **Deorphanization of novel biogenic amine-gated ion channels identifies a new serotonin receptor for learning** *Current Biology* **31**:4282–4292 [Google Scholar](#)
- Myhrer T (2003) **Neurotransmitter systems involved in learning and memory in the rat: a meta-analysis based on studies of four behavioral tasks** *Brain Research Reviews* **41**:268–287 [Google Scholar](#)
- Nagashima T., Iino Y., Tomioka M (2019) **DAF-16/FOXO promotes taste avoidance learning independently of axonal insulin-like signaling** *PLoS genetics* **15**:e1008297–e1008297 [Google Scholar](#)
- Nguyen E. K., Koval O. M., Noble P., Broadhurst K., Allamargot C., Wu M., Strack S., Thiel W. H., Grumbach I. M (2018) **CaMKII (Ca(2+)/Calmodulin-Dependent Kinase II) in Mitochondria of Smooth Muscle Cells Controls Mitochondrial Mobility, Migration, and Neointima Formation** *Arterioscler Thromb Vasc Biol* **38**:1333–1345 [Google Scholar](#)
- Niwa S (2016) **The nephronophthisis-related gene *ift-139* is required for ciliogenesis in *Caenorhabditis elegans*** *Sci Rep* **6**:31544 [Google Scholar](#)
- Ohno H., Kato S., Naito Y., Kunitomo H., Tomioka M., Iino Y (2014) **Role of synaptic phosphatidylinositol 3-kinase in a behavioral learning response in *C. elegans*** *Science* **345**:313–317 [Google Scholar](#)
- Ota K. T., Pierre V. J., Ploski J. E., Queen K., Schafe G. E (2008) **The No-cGMP-PKG signaling pathway regulates synaptic plasticity and fear memory consolidation in the lateral amygdala via activation of ERK/MAP kinase** *Learn Mem* **15**:792–805 [Google Scholar](#)
- Park H., Kaang B. K (2019) **Balanced actions of protein synthesis and degradation in memory formation** *Learn Mem* **26**:299–306 [Google Scholar](#)
- Park Y. S., Lee Y. S., Cho N. J., Kaang B. K (2000) **Alternative splicing of *gar-1*, a *Caenorhabditis elegans* G-protein-linked acetylcholine receptor gene** *Biochem Biophys Res Commun* **268**:354–8 [Google Scholar](#)
- Paul C., Schöberl F., Weinmeister P., Micale V., Wotjak C. T., Hofmann F., Kleppisch T (2008) **Signaling through cGMP-dependent protein kinase I in the amygdala is critical for auditory-cued fear memory and long-term potentiation** *J Neurosci* **28**:14202–12 [Google Scholar](#)
- Pedreira M. E., Dimant B., Tomsic D., Quesada-Allue L. A., Maldonado H (1995) **Cycloheximide inhibits context memory and long-term habituation in the crab *Chasmagnathus*** *Pharmacology Biochemistry and Behavior* **52**:385–395 [Google Scholar](#)
- Peixoto L. L., Wimmer M. E., Poplawski S. G., Tudor J. C., Kenworthy C. A., Liu S., Mizuno K., Garcia B. A., Zhang N. R., Giese K. P., Abel T (2015) **Memory acquisition and retrieval impact**

different epigenetic processes that regulate gene expression *BMC Genomics* **16**:S5 [Google Scholar](#)

Peng S., Zhang Y., Zhang J., Wang H., Ren B (2010) **ERK in learning and memory: a review of recent research** *Int J Mol Sci* **11**:222–232 [Google Scholar](#)

Peng S., Zhang Y., Zhang J., Wang H., Ren B (2011) **Glutamate receptors and signal transduction in learning and memory** *Molecular Biology Reports* **38**:453–460 [Google Scholar](#)

Pereira L., Kratsios P., Serrano-Saiz E., Sheftel H., Mayo A. E., Hall D. H., White J. G., Leboeuf B., Garcia L. R., Alon U., Hobert O (2015) **A cellular and regulatory map of the cholinergic nervous system of *C. elegans*** *eLife* **4**:e12432 <https://doi.org/10.7554/eLife.12432> | [Google Scholar](#)

Pershina E. V., Kulagina T. P., Savina T. A., Aripovsky A. V., Levin S. G., Arkhipov V. I (2022) **Changes in the level of fatty acids in the brain of rats during memory acquisition** *Behav Brain Res* **417**:113599 [Google Scholar](#)

Popovici C., Isnardon D., Birnbaum D., Roubin R (2002) **Caenorhabditis elegans receptors related to mammalian vascular endothelial growth factor receptors are expressed in neural cells** *Neuroscience Letters* **329**:116–120 [Google Scholar](#)

Prikas E., Poljak A., Ittner A (2020) **Mapping p38 α mitogen-activated protein kinase signaling by proximity-dependent labeling** *Protein Science* **29**:1196–1210 [Google Scholar](#)

Putrenko I., Zakikhani M., Dent J. A (2005) **A Family of Acetylcholine-gated Chloride Channel Subunits in *Caenorhabditis elegans*** *Journal of Biological Chemistry* **280**:6392–6398 [Google Scholar](#)

Rahmani A., Chew Y. L (2021) **Investigating the molecular mechanisms of learning and memory using *Caenorhabditis elegans*** *Journal of Neurochemistry* **159**:417–451 [Google Scholar](#)

Rahmani A., Mcmillen A., Allen E., Minervini C., Chew Y. L (2024) **Behavioral Tests for Associative Learning in *Caenorhabditis elegans*** In: Dworkin S., editors. *Neurobiology: Methods and Protocols* New York, NY: Springer US [Google Scholar](#)

Rose J. K., Kaun K. R., Chen S. H., Rankin C. H (2003) **GLR-1, a non-NMDA glutamate receptor homolog, is critical for long-term memory in *Caenorhabditis elegans*** *J Neurosci* **23**:9595–9 [Google Scholar](#)

Rosenberg T., Gal-Ben-Ari S., Dieterich D. C., Kreutz M. R., Ziv N. E., Gundelfinger E. D., Rosenblum K (2014) **The roles of protein expression in synaptic plasticity and memory consolidation** *Frontiers in Molecular Neuroscience* **7** [Google Scholar](#)

Roux K. J., Kim D. I., Raida M., Burke B (2012) **A promiscuous biotin ligase fusion protein identifies proximal and interacting proteins in mammalian cells** *J Cell Biol* **196**:801–10 [Google Scholar](#)

Ryu H.-H., Lee Y.-S (2016) **Cell type-specific roles of RAS-MAPK signaling in learning and memory: Implications in neurodevelopmental disorders** *Neurobiology of Learning and Memory* **135**:13–21 [Google Scholar](#)

- Sadeghian F., Castaneda P. G., Amin M. R., Cram E. J (2022) **Functional Insights into Protein Kinase A (PKA) Signaling from *C. elegans*** *Life* **12** [Google Scholar](#)
- Sakai N., Ohno H., Tomioka M., Iino Y (2017) **The intestinal TORC2 signaling pathway contributes to associative learning in *Caenorhabditis elegans*** *PLOS One* **12**:e0177900 [Google Scholar](#)
- Sanchez A. D., Branon T. C., Cote L. E., Papagiannakis A., Liang X., Pickett M. A., Shen K., Jacobs-Wagner C., Ting A. Y., Feldman J. L (2021) **Proximity labeling reveals non-centrosomal microtubule-organizing center components required for microtubule growth and localization** *Current Biology* **31**:3586–3600 [Google Scholar](#)
- Sanchez A. D., Feldman J. L (2021) **A proximity labeling protocol to probe proximity interactions in *C. elegans*** *STAR Protoc* **2**:100986 [Google Scholar](#)
- Sassone-Corsi P (2012) **The cyclic AMP pathway** *Cold Spring Harb Perspect Biol* **4** [Google Scholar](#)
- Sato H., Kunitomo H., Fei X., Hashimoto K., Iino Y (2021) **Glutamate signaling from a single sensory neuron mediates experience-dependent bidirectional behavior in *Caenorhabditis elegans*** *Cell Reports* **35**:109177 [Google Scholar](#)
- Schade M. A., Reynolds N. K., Dollins C. M., Miller K. G (2005) **Mutations that rescue the paralysis of *Caenorhabditis elegans* *ric-8* (synembryn) mutants activate the G alpha(s) pathway and define a third major branch of the synaptic signaling network** *Genetics* **169**:631–49 [Google Scholar](#)
- Schlattner U., Dolder M., Wallimann T., Tokarska-Schlattner M (2001) **Mitochondrial Creatine Kinase and Mitochondrial Outer Membrane Porin Show a Direct Interaction That Is Modulated by Calcium*** *Journal of Biological Chemistry* **276**:48027–48030 [Google Scholar](#)
- Schlattner U., Tokarska-Schlattner M., Wallimann T (2006) **Mitochondrial creatine kinase in human health and disease** *Biochimica et Biophysica Acta (BBA) - Molecular Basis of Disease* **1762**:164–180 [Google Scholar](#)
- Schwarz V., Pan J., Voltmer-Irsch S., Hutter H (2009) **IgCAMs redundantly control axon navigation in *Caenorhabditis elegans*** *Neural Development* **4**:13 [Google Scholar](#)
- Shahmoradi A (2015) **Analyses of improved long term memory in SHARP1 and SHARP2 double knockout mice** University of Göttingen [Google Scholar](#)
- Sharma R. K., Kalra J (1994) **Molecular interaction between cAMP and calcium in calmodulin-dependent cyclic nucleotide phosphodiesterase system** *Clin Invest Med* **17**:374–82 [Google Scholar](#)
- Shoshan-Barmatz V., De S., Meir A (2017) **The Mitochondrial Voltage-Dependent Anion Channel 1, Ca(2+) Transport, Apoptosis, and Their Regulation** *Front Oncol* **7**:60 [Google Scholar](#)
- Smith W. B., Starck S. R., Roberts R. W., Schuman E. M (2005) **Dopaminergic stimulation of local protein synthesis enhances surface expression of GluR1 and synaptic transmission in hippocampal neurons** *Neuron* **45**:765–79 [Google Scholar](#)

Sohn Y., Choi M.-K., Ahn Y.-Y., Lee J., Jeong J (2011) **Topological Cluster Analysis Reveals the Systemic Organization of the *Caenorhabditis elegans* Connectome** *PLOS Computational Biology* **7**:e1001139 [Google Scholar](#)

Stefanoska K., Gajwani M., Tan A. R. P., Ahel H. I., Asih P. R., Volkerling A., Poljak A., Ittner A (2022) **Alzheimer's disease: Ablating single master site abolishes tau hyperphosphorylation** *Sci Adv* **8**:eabl8809 [Google Scholar](#)

Stefanoska K., Prikas E., Lin Y., Kosonen R., Ittner A (2023) **Remote memory engrams are controlled by encoding-specific tau phosphorylation** *bioRxiv* <https://doi.org/10.1101/2023.12.01.569663> | [Google Scholar](#)

Stein G. M., Murphy C. T (2014) **C. elegans positive olfactory associative memory is a molecularly conserved behavioral paradigm** *Neurobiology of Learning and Memory* **115**:86–94 [Google Scholar](#)

Sutton M. A., Schuman E. M (2006) **Dendritic Protein Synthesis, Synaptic Plasticity, and Memory** *Cell* **127**:49–58 [Google Scholar](#)

Suzuki H., Thiele T. R., Faumont S., Ezcurra M., Lockery S. R., Schafer W. R (2008) **Functional asymmetry in *Caenorhabditis elegans* taste neurons and its computational role in chemotaxis** *Nature* **454**:114–117 [Google Scholar](#)

Szklarczyk D., Kirsch R., Koutrouli M., Nastou K., Mehryary F., Hachilif R., Gable A. L., Fang T., Doncheva N. T., Pyysalo S., Bork P., Jensen L. J., Von Mering C. (2023) **The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest** *Nucleic Acids Res* **51**:D638–d646 [Google Scholar](#)

Takayanagi-Kiya S., Zhou K., Jin Y (2016) **Release-dependent feedback inhibition by a presynaptically localized ligand-gated anion channel** *eLife* **5**:e21734 <https://doi.org/10.7554/eLife.21734> | [Google Scholar](#)

Tang L. T. H., Lee G. A., Cook S. J., Ho J., Potter C. C., Bülow H. E (2023) **Restructuring of an asymmetric neural circuit during associative learning** *bioRxiv* [Google Scholar](#)

Taylor S. R., Santpere G., Weinreb A., Barrett A., Reilly M. B., Xu C., Varol E., Oikonomou P., Glenwinkel L., Mcwhirter R., Poff A., Basavaraju M., Rafi I., Yemini E., Cook S. J., Abrams A., Vidal B., Cros C., Tavazoie S., Sestan N., Hammarlund M., Hobert O., Miller D. M (2021) **Molecular topography of an entire nervous system** *Cell* **184**:4329–4347 [Google Scholar](#)

The Pandas Development Team (2020) **pandas-dev/pandas: Pandas. Latest ver** .Zenodo <https://doi.org/10.5281/zenodo.3509134>

Tomioka M., Adachi T., Suzuki H., Kunitomo H., Schafer W. R., Iino Y (2006) **The insulin/PI 3-kinase pathway regulates salt chemotaxis learning in *Caenorhabditis elegans*** *Neuron* **51**:613–25 [Google Scholar](#)

Trujillo G., Nakata K., Yan D., Maruyama I. N., Jin Y (2010) **A ubiquitin E2 variant protein acts in axon termination and synaptogenesis in *Caenorhabditis elegans*** *Genetics* **186**:135–45 [Google Scholar](#)

Voglis G., Tavernarakis N (2008) **A synaptic DEG/ENaC ion channel mediates learning in C. elegans by facilitating dopamine signalling** *Embo j* **27**:3288–99 [Google Scholar](#)

Vukojevic V., Gschwind L., Vogler C., Demougin P., De Quervain D. J., Papassotiropoulos A., Stetak A. (2012) **A role for α -adducin (ADD-1) in nematode and human memory** *Embo j* **31**:1453–66 [Google Scholar](#)

Vukojevic V., Mastrandreas P., Arnold A., Peter F., Kolassa I. T., Wilker S., Elbert T., De Quervain D. J., Papassotiropoulos A., Stetak A. (2020) **Evolutionary conserved role of neural cell adhesion molecule-1 in memory** *Transl Psychiatry* **10**:217 [Google Scholar](#)

Wallis T. P., Venkatesh B. G., Narayana V. K., Kvaskoff D., Ho A., Sullivan R. K., Windels F., Sah P., Meunier F. A (2021) **Saturated free fatty acids and association with memory formation** *Nat Commun* **12**:3443 [Google Scholar](#)

Wang P.-Z., Ge M.-H., Su P., Wu P.-P., Wang L., Zhu W., Li R., Liu H., Wu J.-J., Xu Y., Zhao J.-L., Li S.-J., Wang Y., Chen L.-M., Wu T.-H., Wu Z.-X (2025) **Sensory plasticity caused by up-down regulation encodes the information of short-term learning and memory** *iScience* **28**:112215 [Google Scholar](#)

Watteyne J., Peymen K., Van Der Auwera P., Borghgraef C., Vandeweyer E., Van Damme S., Rutten I., Lammertyn J., Jelier R., Schoofs L., Beets I. (2020) **Neuromedin U signaling regulates retrieval of learned salt avoidance in a C. elegans gustatory circuit** *Nat Commun* **11**:2076 [Google Scholar](#)

Yoo K. S., Lee K., Lee Y. S., Oh W. J., Kim H. K (2020) **Rho Guanine Nucleotide Exchange Factor 4 (Arhgef4) Deficiency Enhances Spatial and Object Recognition Memory** *Exp Neurol* **29**:334–343 [Google Scholar](#)

Zalcman G., Federman N., Romano A (2018) **CaMKII Isoforms in Learning and Memory: Localization and Function** *Front Mol Neurosci* **11**:445 [Google Scholar](#)

Zhou S., Zhang Y., Kaletsky R., Toraason E., Zhang W., Dong M.-Q., Murphy C. T (2023) **Signaling from the C. elegans Hypodermis Non-autonomously Facilitates Short-term Associative Memory** *bioRxiv* [Google Scholar](#)

Author information

Aelon Rahmani

Flinders Health and Medical Research Institute, College of Medicine and Public Health, Flinders University, Bedford Park, Australia

Anna McMillen

Flinders Health and Medical Research Institute, College of Medicine and Public Health, Flinders University, Bedford Park, Australia

Ericka Allen

Flinders Health and Medical Research Institute, College of Medicine and Public Health, Flinders University, Bedford Park, Australia

Radwan Ansaar

Flinders Health and Medical Research Institute, College of Medicine and Public Health, Flinders University, Bedford Park, Australia

Renee Green

Flinders Health and Medical Research Institute, College of Medicine and Public Health,
Flinders University, Bedford Park, Australia

Michaela E Johnson

Flinders Health and Medical Research Institute, College of Medicine and Public Health,
Flinders University, Bedford Park, Australia

Anne Poljak

Bioanalytical Mass Spectrometry Facility, Mark Wainwright Analytical Centre, University of
New South Wales, Sydney, Australia

Yee Lian Chew

Flinders Health and Medical Research Institute, College of Medicine and Public Health,
Flinders University, Bedford Park, Australia
ORCID iD: [0000-0001-6078-9312](https://orcid.org/0000-0001-6078-9312)

For correspondence: yeelian.chew@flinders.edu.au

Editors

Reviewing Editor

Mei Zhen

Lunenfeld-Tanenbaum Research Institute, Toronto, Canada

Senior Editor

Albert Cardona

University of Cambridge, Cambridge, United Kingdom

Reviewer #1 (Public review):

Summary:

Rahmani et al. utilize the TurboID method to characterize global proteome changes in the worm's nervous system induced by a salt-based associative learning paradigm. Altogether, they uncover 706 proteins tagged by the TurboID method in worms that underwent the memory-inducing protocol. Next, the authors conduct a gene enrichment analysis that implicates specific molecular pathways in salt-associative learning, such as MAP kinase and cAMP-mediated pathways, as well as specific neuronal classes including pharyngeal neurons, and specific sensory neurons, interneurons, and motor neurons. The authors then screen a representative group of hits from the proteome analysis. They find that mutants of candidate genes from the MAP kinase pathway, namely *dlk-1* and *uev-3*, do not affect performance in the learning paradigm. Instead, multiple acetylcholine signaling mutants, as well as a protein-kinase-A mutant, significantly affected performance in the associative memory assay (e.g., *acc-1*, *acc-3*, *lgc-46*, and *kin-2*). Finally, the authors demonstrate that protein-kinase-A mutants, as well as acetylcholine signaling mutants, do not exhibit a phenotype in a related but distinct conditioning paradigm-aversive salt conditioning-suggesting their effect is specific to appetitive salt conditioning.

Overall, the authors addressed the concerns raised in the previous review round, including the statistics of the chemotaxis experiments and the systems-level analysis of the neuron class expression patterns of their hits. I also appreciate the further attempt to equalize the sample size of the chemotaxis experiments and the transparent reporting of the sample size

and statistics in the figure captions and Table S9. The new results from the panneuronal overexpression of the kin-2 gain-of-function allele also contribute to the manuscript. Together, these make the paper more compelling. The additional tested hits provide a comprehensive analysis of the main molecular pathways that could have affected learning. However, the revised manuscript includes more information and analysis, raising additional concerns.

Major comments:

As reviewer 4 noted, and as also shown to be relevant for C30G12.6 presented in Figure 6, the backcrossing of the mutants is important, as background mutations may lead to the observed effects. Could the authors add to Table 1, sheet 1, the outcrossing status of the tested mutants? It is important to validate that the results of the positive hits (where learning was affected), such as *acc-1*, *acc-3*, and *lgc-46*, do not stem from background mutations.

The fold change in the number of hits for different neurons in the CENGEN-based rank analysis requires a statistical test (discussed on pages 17-19 and summarized in Table S7). Similar to the other gene enrichment analyses presented in the manuscript, the new rank analysis also requires a statistical test. Since the authors extensively elaborate on the results from this analysis, I think a statistical analysis is especially important for its interpretation. For example, if considering the IL1 neurons, which ranked highest, and assuming random groups of genes-each having the same size as those of the ranked neurons (209 genes in total for IL1 in Table S7)-how common would it be to get the calculated fold change of 1.38 or higher? Such bootstrapping analysis is common for enrichment analysis. Perhaps the authors could consult with an institutional expert (Dr. Pawel Skuza, Flinders University) for the statistical aspects of this analysis.

The learning phenotypes from Figure S8, concerning *acc-1*, *acc-3*, and *lgc-46* mutants, are summarized in a scheme in Figure 4; however, the chemotaxis results are found in the supplemental Figure S8. Perhaps I missed the reasoning, but for transparency, I think the relevant Figure S8 results should be shown together with their summary scheme in Figure 4.

<https://doi.org/10.7554/eLife.108438.1.sa3>

Reviewer #2 (Public review):

Summary:

In this study by Rahmani in colleagues, the authors sought to define the "learning proteome" for a gustatory associative learning paradigm in *C. elegans*. Using a cytoplasmic TurboID expressed under the control of a pan-neuronal promoter, the authors labeled proteins during the training portion of the paradigm, followed by proteomics analysis. This approach revealed hundreds of proteins potentially involved in learning, which the authors describe using gene ontology and pathway analysis. The authors performed functional characterization of over two dozen of these genes for their requirement in learning using the same paradigm. They also compared the requirement for these genes across various learning paradigms and found that most hits they characterized appear to be specifically required for the training paradigm used for generating the "learning proteome".

Strengths:

- The authors have thoughtfully and transparently designed and reported the results of their study. Controls are carefully thought-out, and hits are ranked as strong and weak. By combining their proteomics with behavioral analysis, the authors also highlight the biological significance of their proteomics findings, and support that even weak hits are meaningful.

- The authors display a high degree of statistical rigor, incorporating normality tests into their behavioral data which is beyond the field standard.
- The authors include pathway analysis that generates interesting hypotheses about processes involved learning and memory
- The authors generally provide thoughtful interpretations for all of their results, both positive and negative, as well as any unexpected outcomes.

Weaknesses:

- The authors use the Cengen single cell-transcriptomic atlas to predict where the proteins in the "learning proteome" are likely to be expressed and use this data to identify neurons that are likely significant to learning, and building hypothetical circuit. This is an excellent idea; however, the Cengen dataset only contains transcriptomic data from juvenile L4 animals, while the authors performed their proteome experiments in Day 1 Adult animals. It is well documented that the *C. elegans* nervous system transcriptome is significant different between these two stages (Kaletsky et al., 2016, St. Ange et al., 2024), so the authors might be missing important expression data, resulting in inaccurate or incomplete networks. The adult neuronal single-cell atlas data (<https://cestaan.princeton.edu/>) would be better suited to incorporate into neuronal expression analysis.
- The authors offer many interpretations for why mutants in "learning proteome" hits have no detectable phenotype, which is commendable. They are however overlooking another important interpretation, it is possible that these changes to the proteome are important for memory, which is dependent upon translation and protein level changes, and is molecularly distinct from learning. It is well established in the field mutating or knocking down memory regulators in other paradigms will often have no detectable effect on learning. Incorporating this interpretation into the discussion and highlighting it as an area for future exploration would strengthen the manuscript.
- A minor weakness - In the discussion, the authors state that the Lakhina, et al 2015 used RNA-seq to assess memory transcriptome changes. This study used microarray analysis.

Significance:

The approach used in this study is interesting and has the potential to further our knowledge about the molecular mechanisms of associative behaviors. There have been multiple transcriptomic studies in the worm looking at gene expression changes in the context of behavioral training. This study compliments and extends those studies, by examining how the proteome changes in a different training paradigm. This approach here could be employed for multiple different training paradigms, presenting a new technical advance for the field. This paper would be of interest to the broader field of behavioral and molecular neuroscience. Though it uses an invertebrate system, many findings in the worm regarding learning and memory translate to higher organisms, making this paper of interest and significant to the broader field of behavioral neuroscience.

<https://doi.org/10.7554/eLife.108438.1.sa2>

Reviewer #4 (Public review):

Summary:

In this manuscript, authors used a learning paradigm in *C. elegans*; when worms were fed in a saltless plate, its chemotaxis to salt is greatly reduced. To identify learning-related proteins, authors employed nervous system-specific transcriptome analysis to compare whole proteins

in neurons between high-salt-fed animals and saltless-fed animals. Authors identified "learning-specific proteins" which are observed only after saltless feeding. They categorized these proteins by GO analyses, pathway analyses and expression site analyses, and further stepped forward to test mutants in selected genes identified by the proteome analysis. They find several mutants that are defective or hyper-proficient for learning, including *acc-1/3* and *lgc-46* acetylcholine receptors, *F46H5.3* putative arginine kinase, and *kin-2*, a cAMP pathway gene. These mutants were not previously reported to have abnormality in the learning paradigm.

Concerns:

Upon revision, authors addressed all concerns of this reviewer, and the results are now presented in a way that facilitates objective evaluation. Authors' conclusions are supported by the results presented, and the strength of the proteomics approach is persuasively demonstrated.

Significance:

(1) Total neural proteome analysis has not been conducted before for learning-induced changes, though transcriptome analysis has been performed for odor learning (Lakhina et al., <http://dx.doi.org/10.1016/j.neuron.2014.12.029>). This warrants the novelty of this manuscript, because for some genes, protein levels may change even though mRNA levels remain the same. Although in a few reports TurboID has been used in *C. elegans*, this is the first report of a systematic analysis of tissue-specific differential proteomics.

(2) Authors found five mutants that have abnormality in the salt learning. These genes have not been described to have the abnormality, providing novel knowledge to the readers, especially those who work on *C. elegans* behavioural plasticity. Especially, involvement of acetylcholine neurotransmission has not been addressed before. Although transgenic rescue experiments have not been performed except *kin-2*, and the site of action (neurons involved) has not been tested in this manuscript, it will open the venue to further determine the way in which acetylcholine receptors, cAMP pathway etc. influences the learning process.

<https://doi.org/10.7554/eLife.108438.1.sa1>

Author response:

General Statements

We thank the reviewers for providing us the opportunity to revise our manuscript titled "Identifying regulators of associative learning using a protein-labelling approach in *C. elegans*." We appreciate the insightful feedback that we received to improve this work. In response, we have extensively revised the manuscript with the following changes: we have (1) clarified the criteria used for selecting candidate genes for behavioural testing, presenting additional data from 'strong' hits identified in multiple biological replicates (now testing 26 candidates, previously 17), (2) expanded our discussion of the functional relevance of validated hits, including providing new tissue-specific and neuron class-specific analyses, and (3) improved the presentation of our data, including visualising networks identified in the 'learning proteome', to better highlight the significance of our findings. We also substantially revised the text to indicate our attempts to address limitations related to background noise in the proteomic data and outlined potential refinements for future studies. All revisions are clearly marked in the manuscript in red font. A detailed, point-by-point response to each comment is provided below.

Point-by-point description of the revisions:

Reviewer #1 (Evidence, reproducibility and clarity):

Summary:

*Rahmani et al., utilize the TurboID method to characterize the global proteome changes in the worm's nervous system induced by a salt-based associative learning paradigm. Altogether, Rahmani et al., uncover 706 proteins that are tagged by the TurboID method specifically in samples extracted from worms that underwent the memory inducing protocol. Next, the authors conduct a gene enrichment analysis that implicates specific molecular pathways in salt associative learning, such as MAP-kinase and cAMP-mediated pathways. The authors then screen a representative group of the hits from the proteome analysis. The authors find that mutants of candidate genes from the MAP-kinase pathway, namely *dlk-1* and *uev-3*, do not affect the performance in the learning paradigm. Instead multiple acetylcholine signaling mutants significantly affected the performance in the associative memory assay, e.g., *acc-1*, *acc-3*, *gar-1*, and *lgc-46*. Finally, the authors demonstrate that the acetylcholine signaling mutants did not exhibit a phenotype in similar but different conditioning paradigms, such as aversive salt-conditioning or appetitive odor conditioning, suggesting their effect is specific to appetitive salt conditioning.*

Major comments:

(1) The statistical approach and analysis of the behavior assay:

The authors use a 2-way ANOVA test which assumes normal distribution of the data. However, the chemotaxis index used in the study is bounded between -1 and 1, which prevents values near the boundaries to be normally distributed.

Since most of the control data in this assay in this study is very close to 1, it strongly suggests that the CI data is not normally distributed and therefore 2-way ANOVA is expected to give skewed results.

I am aware this is a common mistake and I also anticipate that most conclusions will still hold also under a more fitting statistical test.

We appreciate the point raised by Reviewer 1 and understand the importance of performing the correct statistical tests.

The statistical tests used in this study were chosen since parametric tests, particularly ANOVA tests to assess differences between multiple groups, are commonly used to assess behaviour in the *C. elegans* learning and memory field. Below is a summary of the tests used by studies that perform similar behavioural tests cited in this work, as examples:

Author response table 1.

A summary for the statistical tests performed by similar studies for chemotaxis assay data. References (listed in the leftmost column) were observed to (A) use parametric tests only or (B) performed either a parametric or non-parametric test on each chemotaxis assay dataset depending on whether the data passed a normality test. Listings for ANOVA tests are in bold to demonstrate their common use in the *C. elegans* learning and memory field.

Reference	Parametric test/s used in the reference	Non-parametric test/s used in the reference
Beets <i>et al.</i> , 2020	Two-way ANOVA	None
Hiroki & Iino 2022	One-way ANOVA	None
Hiroki <i>et al.</i> , 2022	One-way ANOVA	None
Hukema <i>et al.</i> , 2006	T-tests	None
Hukema <i>et al.</i> , Learn. Mem. 2008	T-tests	None
Jang <i>et al.</i> , 2019	ANOVA	None
Kitazono <i>et al.</i> , 2017	Two-way ANOVA and t-tests	None
Lans <i>et al.</i> , 2004	One-way ANOVA	None
Lim <i>et al.</i> , 2018	Two-way ANOVA	Wilcoxon rank sum test adjusted with the Benjamini–Hochberg method
Lin <i>et al.</i> , 2010	Two-way or three-way ANOVA	None
Nagashima <i>et al.</i> , 2019	One-way ANOVA	None
Ohno <i>et al.</i> , 2014		None
Sakai <i>et al.</i> , 2017	One-way ANOVA or t-tests	None
Stein & Murphy 2014	Two-way ANOVA and t-tests	None
Tang <i>et al.</i> , 2023	One-way ANOVA or t-tests	None
Tomioka <i>et al.</i> , 2006	T tests	None
Wattayne <i>et al.</i> , 2020	One-way ANOVA	Two-sided Kruskal–Wallis

We note Reviewer 1's concern that this may stem from a common mistake. As stated, Two-way ANOVA generally relies on normally distributed data. We used GraphPad Prism to perform the Shapiro-Wilk normality test on our chemotaxis assay data as it is generally appropriate for sample sizes < 50 ($\alpha = 0.05$), and found that most data passes this test including groups with skewed indices. For example, this is the data for Figure S8C:

Author response table 2.

Shapiro-Wilk normality test results for chemotaxis assay data in Figure S8C. Chemotaxis assay data was generated to assess salt associative learning capacity for wild-type (WT) versus *lgc-46(-)* mutant *C. elegans*. Three experimental groups were prepared for each *C. elegans* strain (naïve, high-salt control, and trained). From top-to-bottom, the data below displays the 'W' value, 'P value', a binary yes/no for whether the data passes the Shapiro-Wilk normality test, and a 'P value summary' (ns = nonsignificant). W values measure the similarity between a normal distribution and the chemotaxis assay data. Data is considered normal in the Shapiro-Wilk normality test when a W value is near 1.0 and the null hypothesis is not rejected (i.e., P value > 0.05).

	WT naïve	WT high-salt control	WT trained	<i>lgc-46</i> naïve	<i>lgc-46</i> high-salt control	<i>lgc-46</i> trained
W	0.9196	0.9114	0.8926	0.8334	0.8151	0.8769
P value	0.5272	0.4758	0.3705	0.1475	0.1070	0.2954
Passed normality test (alpha=0.05)?	Yes	Yes	Yes	Yes	Yes	Yes
P value summary	ns	ns	ns	ns	ns	ns

The manuscript now includes the use of the Shapiro-Wilk normality test to assess chemotaxis assay data before using two-way ANOVA on page 51.

Nevertheless an appropriate statistical analysis should be performed. Since I assume the authors would wish to take into consideration both the different conditions and

biological repeats, I can suggest two options:

- Using a Generalized linear mixed model, one can do with R software.
- Using a custom bootstrapping approach.

We thank Reviewer 1 for suggesting these two options. We carefully considered both approaches and consulted with the in-house statistician at our institution (Dr Pawel Skuza, Flinders University) for expert advice to guide our decision. In summary:

(1) Generalised linear mixed models: Generalised linear mixed models (GLMMs) are generally most appropriate for nested/hierarchical data. However, our chemotaxis assay data does not exhibit such nesting. Each biological replicate (N) consists of three technical replicates, which are averaged to yield a single chemotaxis index per N. Our statistical comparisons are based solely on these averaged values across experimental groups, making GLMMs less applicable in this context.

(2) Bootstrapping: Based on advice from our statistician, while bootstrapping can be a powerful tool, its effectiveness is limited when applied to datasets with a low number of biological replicates (N). Bootstrapping relies on resampling existing data to simulate additional observations, which may artificially inflate statistical power and potentially suggest significance where the biological effect size is minimal or not meaningful. Increasing the number of biological replicates to accommodate bootstrapping could introduce additional variability and compromise the interpretability of the results.

The total number of assays, especially controls, varies quite a bit between the tested mutants. For example compare the acc-1 experiment in Figure 4.A., and gap-1 or rho-1 in Figure S4.A and D. It is hard to know the exact N of the controls, but I assume that for example, lowering the wild type control of acc-1 to equivalent to gap-1 would have made it non significant. Perhaps the best approach would be to conduct a power analysis, to know what N should be acquired for all samples.

We thoroughly evaluated performing the power analysis: however, this is typically performed with the assumption that an N = 1 represents a singular individual/person. An N = 1 in this study is one biological replicate that includes hundreds of worms, which is why it is not typically employed in our field for this type of behavioural test.

Considering these factors, we have opted to continue using a two-way ANOVA for our statistical analysis. This choice aligns with recent publications that employ similar experimental designs and data structures. Crucially, we have verified that our data meet the assumptions of normality, addressing key concerns regarding the suitability of parametric testing. We believe this approach is sufficiently rigorous to support our main conclusions. This rationale is now outlined on page 51.

To be fully transparent, our aim is to present differences between wild-type and mutant strains that are clearly visible in the graphical data, such that the choice of statistical test does not become a limiting factor in interpreting biological relevance. We hope this rationale is understandable, and we sincerely appreciate the reviewer's comment and the opportunity to clarify our analytical approach.

We hope that Reviewer 1 will appreciate these considerations as sufficient justification to retain the statistical tests used in the original manuscript. Nevertheless, to constructively address this comment, we have performed the following revisions:

(1) Consistent number of biological replicates: We performed additional biological replicates of the learning assay to confirm the behavioural phenotypes for the key candidates described

(KIN-2 , F46H5.3, ACC-1, ACC-3, LGC-46). We chose $N = 5$ since most studies cited in this paper that perform similar behavioural tests do the same (see Author response table 3 below).

Author response table 3.

A summary for sample sizes generated by similar studies for chemotaxis assay data. References (listed in the leftmost column) were observed to the sample sizes (N) below corresponding to biological replicates of chemotaxis assay data. N values are in bold when the study uses $N \leq 5$.

Reference	<u>N used in the study for chemotaxis assay data</u>
Beets <i>et al.</i> , 2020	8
Hiroki & Iino 2022	5-8
Hiroki <i>et al.</i> , 2022	6-7
Hukema <i>et al.</i> , 2006	≥ 4
Hukema <i>et al.</i> , Learn. Mem. 2008	≥ 4
Jang <i>et al.</i> , 2019	≥ 4
Kitazono <i>et al.</i> , 2017	≥ 4
Kauffman <i>et al.</i> , 2010	≥ 3
Kauffman <i>et al.</i> , J. Vis. Exp. 2011	≥ 3
Lans <i>et al.</i> , 2004	2
Lim <i>et al.</i> , 2018	2-4
Lin <i>et al.</i> , 2010	≥ 4
Nagashima <i>et al.</i> , 2019	≥ 7
Ohno <i>et al.</i> , 2014	≥ 11
Sakai <i>et al.</i> , 2017	≥ 4
Stein & Murphy 2014	3-5
Tang <i>et al.</i> , 2023	≥ 9
Watteyne <i>et al.</i> , 2020	≥ 10

(1) Grouped presentation of behavioural data: We now present all behavioural data by grouping genotypes tested within the same biological replicate, including wild-type controls, rather than combining genotypes tested separately. This ensures that each graph displays data from genotypes sharing the same N, also an important consideration for performing parametric tests. Accordingly, we re-performed statistical analyses using this reduced N for relevant graphs. As anticipated, this rendered some comparisons non-significant. All statistical comparisons are clearly indicated on each graph.

(2) Improved clarity of figure legends: We revised figure legends for Figures 5, 6, S7, S8, & S9 to make clear how many biological replicates have been performed for each genotype by adding N numbers for each genotype in all figures.

The authors use the phrasing "a non-significant trend", I find such claims uninterpretable and should be avoided. Examples: Page 16. Line 7 and Page 18, line 16.

This is an important point. While we were not able to find the specific phrasing "a non-significant trend" from this comment in the original manuscript, we acknowledge that referring to a phenotype as both a trend and non-significant may confuse readers, which was originally stated in the manuscript in two locations.

The main text has been revised on pages 27 & 28 when describing comparisons between trained groups between two *C. elegans* lines, by removing mentions of trends and retaining descriptions of non-significance.

(2) Neuron-specific analysis and rescue of mutants:

Throughout the study the authors avoid focusing on specific neurons. This is understandable as the authors aim at a systems biology approach, however, in my view this limits the impact of the study. I am aware that the proteome changes analyzed in this study were extracted from a pan neuronally expressed TurboID. Yet, neuron-specific changes may nevertheless be found. For example, running the protein lists from Table S2, in the Gene enrichment tool of wormbase, I found, across several biological replicates, enrichment for the NSM, CAN and RIG neurons. A more careful analysis may uncover specific neurons that take part in this associative memory paradigm. In addition, analysis of the overlap in expression of the final gene list in different neurons, comparing them, looking for overlap and connectivity, would also help to direct towards specific circuits.

This is an important and useful suggestion. We appreciate the benefit in exploring the data from this study from a neuron class-specific lens, in addition to the systems-level analyses already presented.

The WormBase gene enrichment tool is indeed valuable for broad transcriptomic analyses (the findings from utilising this tool are now on page 16); however, its use of Anatomy Ontology (AO) terms also contains annotations from more abundant non-neuronal tissues in the worm. To strengthen our analysis and complement the Wormbase tool, we also used the CeNGEN database as suggested by Reviewer 3 Major Comment 1 (Taylor et al., 2021), which uses single cell RNA-Seq data to profile gene expression across the *C. elegans* nervous system. We input our learning proteome data into CeNGEN as a systemic analysis, identifying neurons highly represented by the learning proteome (on pages 16-20). To do this, we specifically compared genes/proteins from high-salt control worms and trained worms to identify potential neurons that may be involved in this learning paradigm. Briefly, we found:

- WormBase gene enrichment tool: Enrichment for anatomy terms corresponding to specific interneurons (ADA, RIS, RIG), ventral nerve cord neurons, pharyngeal neurons (M1, M2, M5, I4), PVD sensory neurons, DD motor neurons, serotonergic NSM neurons, and CAN.
- CeNGEN analysis: Representation of neurons previously implicated in associative learning (e.g., AVK interneurons, RIS interneurons, salt-sensing neuron ASEL, CEP & ADE dopaminergic neurons, and AIB interneurons), as well as neurons not previously studied in this context (pharyngeal neurons I3 & I6, polymodal neuron IL1, motor neuron DA9, and interneuron DVC). Methods are detailed on pages 50 & 51.

These data are summarised in the revised manuscript as Table S7 & Figure 4.

To further address the reviewer's suggestion, we examined the overlap in expression patterns of the validated learning-associated genes *acc-1*, *acc-3*, *lgc-46*, *kin-2*, and *F46H5.3* across the neuron classes above, using the CeNGEN database. This was done to explore potential neuron classes in which these regulators may act in to regulate learning. This analysis revealed both shared and distinct expression profiles, suggesting potential functional connectivity or co-regulation among subsets of neurons. To summarise, we found:

- All five learning regulators are expressed in RIM interneurons and DB motor neurons.
- KIN-2 and F46H5.3 share the same neuron expression profile and are present in many neurons, so they may play a general function within the nervous system to facilitate learning.
- ACC-3 is expressed in three sensory neuron classes (ASE, CEP, & IL1).
- In contrast, ACC-1 and LGC-46 are expressed in neuron classes (in brackets) implicated in gustatory or olfactory learning paradigms (AIB, AVK, NSM, RIG, & RIS) (Beets et al., 2012, Fadda et al., 2020, Wang et al., 2025, Zhou et al., 2023, Sato et al., 2021), neurons important for

backward or forward locomotion (AVE, DA, DB, & VB) (Chalfie et al., 1985), and neuron classes for which their function is yet detailed in the literature (ADA, I4, M1, M2, & M5).

These neurons form a potential neural circuit that may underlie this form of behavioural plasticity, which we now describe in the main text on pages 16-20 & 34-35 and summarise in Figure 4.

OPTIONAL: A rescue of the phenotype of the mutants by re-expression of the gene is missing, this makes sure to avoid false-positive results coming from background mutations. For example, a pan neuronal or endogenous promoter rescue would help the authors to substantiate their claims, this can be done for the most promising genes. The ideal experiment would be a neuron-specific rescue but this can be saved for future works.

We appreciate this suggestion and recognise its potential to strengthen our manuscript. In response, we made many attempts to generate pan-neuronal and endogenous promoter reexpression lines. However, we faced several technical issues in transgenic line generation, including poor survival following microinjection likely due to protein overexpression toxicity (e.g., C30G12.6, F46H5.3), and reduced animal viability for chemotaxis assays, potentially linked to transgene-related reproductive defects (e.g., ACC-1). As we have previously successfully generated dozens of transgenic lines in past work (e.g. Chew et al., Neuron 2018; Chew et al., Phil Trans B 2018; Gadenne/Chew et al., Life Science Alliance 2022), we believe the failure to produce most of these lines is not likely due to technical limitations. For transparency, these observations have been included in the discussion section of the manuscript on pages 39 & 40 as considerations for future troubleshooting.

Fortunately, we were able to generate a pan-neuronal promoter line for KIN-2 that has been tested and included in the revised manuscript. This new data is shown in Figure 5B and described on pages 23 & 24. Briefly, this shows that pan-neuronal expression of KIN-2 from the *ce179* mutant allele is sufficient to reproduce the enhanced learning phenotype observed in *kin2(ce179)* animals, confirming the role of KIN-2 in gustatory learning.

To address the potential involvement of background mutations (also indicated by Reviewer 4 under ‘cross-commenting’), we have also performed experiments with backcrossed versions of several mutants. These experiments aimed to confirm that salt associative learning phenotypes are due to the expected mutation. Namely, we assessed *kin-2(ce179)* mutants that had been backcrossed previously by another laboratory, as well as C30G12.6(-) and F46H5.3(-) animals backcrossed in this study. Although not all backcrossed mutants retained their original phenotype (i.e., C30G12.6) (Figure 6D, a newly added figure), we found that backcrossed versions of KIN-2 and F46H5.3 both robustly showed enhanced learning (Figures 5A & 6B).

This is described in the text on pages 23-26.

Minor comments:

(1) Lack of clarity regarding the validation of the biotin tagging of the proteome.

The authors show in Figure 1 that they validated that the combination of the transgene and biotin allows them to find more biotin-tagged proteins. However there is significant biotin background also in control samples as is common for this method. The authors mention they validated biotin tagging of all their experiments, but it was unclear in the text whether they validated it in comparison to no-biotin controls, and checked for the fold change difference.

This is an important point: We validated our biotin tagging method prior to mass spectrometry by comparing 'no biotin' and 'biotin' groups. This is shown in Figure S1 in the revised manuscript, which includes a western blot comparing untreated and biotin treated animals that are nontransgenic or expressing TurboID. As expected, by comparing biotinylated protein signal for untreated and treated lanes within each line, biotin treatment increased the signal 1.30-fold for non-transgenic and 1.70-fold for TurboID C. elegans. This is described on page 8 of the revised manuscript.

To clarify, for mass spectrometry experiments, we tested a no-TurboID (non-transgenic) control, but did not perform a no-biotin control. We included the following four groups: (1) No-TurboID 'control' (2) No-TurboID 'trained', (3) pan-neuronal TurboID 'control' and (4) pan-neuronal TurboID 'trained', where trained versus control refers to whether 'no salt' was used as the conditioned stimulus or not, respectively (illustrated in Figure 1A). Due to the complexity of the learning assay (which involves multiple washes and handling steps, including a critical step where biotin is added during the conditioning period), and the need to collect sufficient numbers of worms for protein extraction (>3,000 worms per experimental group), adding 'no-biotin' controls would have doubled the number of experimental groups, which we considered unfeasible for practical reasons. This is explained on pages 8 & 9 of the revised manuscript.

Also, it was unclear which exact samples were tested per replicate. In Page 9, Lines 17-18: "For all replicates, we determined that biotinylated proteins could be observed ...", But in Page 8, Line 24: "We then isolated proteins from ... worms per group for both 'control' and 'trained' groups,... some of which were probed via western blotting to confirm the presence of biotinylated proteins".

Could the authors specify which samples were verified and clarify how?

Thank you for pointing out these unclear statements: We have clarified the experimental groups used for mass spectrometry experiments as detailed in the response above on pages 8 & 9. In addition, western blots corresponding to each biological replicate of mass spectrometry data described in the main text on page 10 and have been added to the revised manuscript (as Figure S3). These western blots compare biotinylation signal for proteins extracted from (1) NoTurboID 'control' (2) No-TurboID 'trained', (3) pan-neuronal TurboID 'control' and (4) panneuronal TurboID 'trained'. These blots function to confirm that there were biotinylated proteins in TurboID samples, before enrichment by streptavidin-mediated pull-down for mass spectrometry.

OPTIONAL: include the fold changes of biotinylated proteins of all the ones that were tested. Similar to Figure 1.C.

This is an excellent suggestion. As recommended by the reviewer, we have included foldchanges for biotinylated protein levels between high-salt control and trained groups (on pages 9 & 10 for replicate #1 and in Table S2 for replicates #2-5). This was done by measuring protein levels in whole lanes for each experimental group per biological replicate within western blots (Figure 1C for replicate #1 and Figure S3 for replicates #2-5) of protein samples generated for mass spectrometry (N = 5).

(2) Figure 2 does not add much to the reader, it can be summarized in the text, as the fraction of proteins enriched for specific cellular compartments.

I would suggest to remove Figure 2 (originally written as figure 3) to text, or transfer it to the supplementry material.

As noted in cross-comment response to Reviewer 4, there were typos in the original figure references, we have corrected them above. Essentially, this comment is referring to Figure 2.

We appreciate this feedback from Reviewer 1. We agree that the original Figure 2 functions as a visual summary from analysis of the learning proteome at the subcellular compartment level. However, it also serves to highlight the following:

- Representation for neuron-specific GO terms is relatively low, but even this small percentage represents entire protein-protein networks that are biologically meaningful, but that are difficult to adequately describe in the main text.
- TurboID was expressed in neurons so this figure supports the relevance of the identified proteome to biological learning mechanisms.
- Many of these candidates could not be assessed by learning assay using single mutants since related mutations are lethal or substantially affect locomotion. These networks therefore highlight the benefit in using strategies like TurboID to study learning.

We have chosen to retain this figure, moving it to the supplementary material as Figure S4 in the revised manuscript, as suggested.

OPTIONAL- I would suggest the authors to mark in a pathway summary figure similar to Figure 3 (originally written as Figure 4) the results from the behavior assay of the genetic screen. This would allow the reader to better get the bigger picture and to connect to the systemic approach taken in Figures 2 and 3.

We think this is a fantastic suggestion and thank Reviewer 1 for this input. In the revised manuscript, we have added Figure 7, which summarises the tested candidates that displayed an effect on learning, mapped onto potential molecular pathways derived from networks in the learning proteome. This figure provides a visual framework linking the behavioural outcomes to the network context. This is described in the main text on pages 32-33.

(3) Typo in Figure 3: the circle of PPM1: The blue right circle half is bigger than the left one.

We thank the Reviewer for noticing this, the node size for PPM-1.A has been corrected in what is now Figure 2 in the revised work.

(4) Unclear in the discussions. In the discussion Page 24, Line 14, the authors raise this question: "why are the proteins we identified not general learning regulators?. The phrasing and logic of the argumentation of the possible answers was hard to follow. - Can you clarify?

We appreciate this feedback in terms of unclarity, as we strive to explain the data as clearly and transparently as possible. Our goal in this paragraph was to discuss why some candidates were seen to only affect salt associative learning, as opposed to showing effects in multiple learning paradigms (i.e., which we were defining as a ‘general learning regulator’). We have adjusted the wording in several places in this paragraph now on pages 36 & 37 to address this comment. We hope the rephrased paragraph provides sufficient rationalisation for the discussion regarding our selection strategy used to isolate our protein list of potential learning regulators, and its potential limitations.

Cross-Commenting

Firstly, we would like to express our appreciation for the opportunity for reviewers to crosscomment on feedback from other reviewers. We believe this is an excellent feature of the peer review process, and we are grateful to the reviewers for their thoughtful engagement and collaborative input.

I would like to thank Reviewer #4 for the great cross comment summary, I find it accurate and helpful.

I also would like to thank Reviewer #4 for spotting the typos in my minor comments, their page and figure numbers are the correct ones.

We have corrected these typos in the relevant comments, and have responded to them accordingly.

Small comment on common point 1 - My feeling is that it is challenging to do quantitative mass spectrometry, especially with TurboID. In general, the nature of MS data is that it hints towards a direction but a followup validation work is required in order to assess it. For example, I am not surprised that the fraction of repeats a hit appeared in does not predict well whether this hit would be validated behaviorally. Given these limitations, I find the authors' approach reasonable.

We thank Reviewer 1 for this positive and thoughtful feedback. We also appreciate Reviewer 4's comment regarding quantitative mass spectrometry and have addressed this in detail below (see response to Reviewer 4). However, we agree with Reviewer 1 that there are practical challenges to performing quantitative mass spectrometry with TurboID, primarily due to the enrichment for biotinylated proteins that is a key feature of the sample preparation process.

Importantly, we whole-heartedly agree with Reviewer 1's statement that "In general, the nature of MS data is that it hints towards a direction but a follow-up validation work is required in order to assess it". This is the core of our approach: however, we appreciate that there are limitations to a qualitative 'absent/present' approach. We have addressed some of these limitations by clarifying the criteria used for selecting candidate genes, based additionally on the presence of the candidate in multiple biological replicates (categorised as 'strong' hits). Based on this method, we were able to validate the role of several novel learning regulators (Figures 5, 6, & S7). We sincerely hope that this manuscript can function as a direction for future research, as suggested by this Reviewer.

I also would like to highlight this major comment from reviewer 4:

"In Experimental Procedures, authors state that they excluded data in which naive or control groups showed average CI < 0.6499, and/or trained groups showed average CI < -0.0499 or > .5499 for N2 (page 36, lines 5-7). "

This threshold seems arbitrary to me too, and it requires the clarifications requested by reviewer 4.

As detailed in our response to Reviewer 4, Major Comment 2, data were excluded only in rare cases, specifically when N2 worms failed to show strong salt attraction prior to training, or when trained N2 worms did not exhibit the expected behavioural difference compared to untrained controls – this can largely be attributed to clear contamination or over-population issues, which are visible prior to assessing CTX plates and counting chemotaxis indices.

These criteria were initially established to provide an objective threshold for excluding biological replicates, particularly when planning to assay a large number of genetic mutants.

However, after extensive testing across many replicates, we found that N2 worms (that were not starved, or not contaminated) consistently displayed the expected phenotype, rendering these thresholds unnecessary. We acknowledge that emphasizing these criteria may have been misleading, and have therefore removed them from page 50 in the revised manuscript to avoid confusion and ensure clarity.

Reviewer #1 (Significance):

This study does a great job to effectively utilize the TurboID technique to identify new pathways implicated in salt-associative learning in C. elegans. This technique was used in C. elegans before, but not in this context. The salt-associative memory induced proteome list is a valuable resource that will help future studies on associative memory in worms. Some of the implicated molecular pathways were found before to be involved in memory in worms like cAMP, as correctly referenced in the manuscript. The implication of the acetylcholine pathway is novel for C. elegans, to the best of my knowledge. The finding that the uncovered genes are specifically required for salt associative memory and not for other memory assays is also interesting.

However overall I find the impact of this study limited. The premise of this work is to use the Turbo-ID method to conduct a systems analysis of the proteomic changes. The work starts by conducting network analysis and gene enrichment which fit a systemic approach. However, since the authors find that ~30% of the tested hits affect the phenotype, and since only 17/706 proteins were assessed, it is challenging to draw conclusive broad systemic claims.

Alternatively, the authors could have focused on the positive hits, and understand them better, find the specific circuits where these genes act. This could have increased the impact of the work. Since neither of these two options are satisfied, I view this work as solid, but not wide in its impact and therefore estimate the audience of this study would be more specialized.

My expertise is in C. elegans behavior, genetics, and neuronal activity, programming and machine learning.

We thank the Reviewer for these comments and appreciate the recognition of the value of the proteomic dataset and the identification of novel molecular pathways, including the acetylcholine pathway, as well as the specificity of the uncovered genes to salt-associative memory. Regarding the reviewer's concern about the overall impact and scope of the study, we respectfully offer the following clarification. Our aim was to establish a systems-level approach for investigating learning-related proteomic changes using TurboID, and we acknowledge that only a subset of the identified proteins was experimentally tested (now 26/706 proteins in the revised manuscript). Although only five of the tested single gene mutants showed a robust learning phenotype in the revised work (after backcrossing, more stringent candidate selection, improved statistical analysis in addressing reviewer comments), our proteomic data provides us a unique opportunity to define these candidates within protein-protein networks (as illustrated in Figure 7). Importantly, our functional testing focused on single-gene mutants, which may not reveal phenotypes for genes that act redundantly (now mentioned on pages 28-30). This limitation is inherent to many genetic screens and highlights the value of our proteomic dataset, which enables the identification of broader protein-protein interaction networks and molecular pathways potentially involved in learning.

To support this systems-level perspective, we have added Figure 7, which visually integrates the tested candidates into molecular pathways derived from the learning proteome for learning regulators KIN-2 and F46H5.3. We also emphasise more explicitly in the text (on

pages 32-33) the value of our approach by highlighting the functional protein networks that can be derived from our proteomics dataset.

We fully acknowledge that the use of TurboID across all neurons limits the resolution needed to pinpoint individual neuron contributions, and understand the benefit in further experiments to explore specific circuits. Many circuits required for salt sensing and salt-based learning are highly explored in the literature and defined explicitly (see Rahmani & Chew, 2021), so our intention was to complement the existing literature by exploring the protein-protein networks involved in learning, rather than on neuron-neuron connectivity. However, we recognise the benefit in integrating circuit-level analyses, given that our proteomic data suggests hundreds of candidates potentially involved in learning. While validating each of these candidates is beyond the scope of the current study, we have taken steps to suggest candidate neurons/circuits by incorporating tissue enrichment analyses and single-cell transcriptomic data (Table S7 & Figure 4). These additions highlight neuron classes of interest and suggest possible circuits relevant to learning.

We hope this clarification helps convey the intended scope and contribution of our study. We also believe that the revisions made in response to Reviewer 1's feedback have strengthened the manuscript and enhanced its significance within the field.

Reviewer #2 (Evidence, reproducibility and clarity):

Summary:

*In this study by Rahmani in colleagues, the authors sought to define the "learning proteome" for a gustatory associative learning paradigm in *C. elegans*. Using a cytoplasmic TurboID expressed under the control of a pan-neuronal promoter, the authors labeled proteins during the training portion of the paradigm, followed by proteomics analysis. This approach revealed hundreds of proteins potentially involved in learning, which the authors describe using gene ontology and pathways analysis. The authors performed functional characterization of some of these genes for their requirement in learning using the same paradigm. They also compared the requirement for these genes across various learning paradigms, and found that most hits they characterized appear to be specifically required for the training paradigm used for generating the "learning proteome".*

Major Comments:

(1) The definition of a "hit" from the TurboID approach is does not appear stringent enough. According to the manuscript, a hit was defined as one unique peptide detected in a single biological replicate (out of 5), which could give rise to false positives. In figure S2, it is clear that there relatively little overlap between samples with regards to proteins detected between replicates, and while perhaps unintentional, presenting a single unique peptide appears to be an attempt to inflate the number of hits. Defining hits as present in more than one sample would be more rigorous. Changing the definition of hits would only require the time to re-list genes and change data presented in the manuscript accordingly.

We thank Reviewer 2 for this valuable comment, and the following related suggestion. We agree with the statement that "Defining hits as present in more than one sample would be more rigorous". Therefore, to address this comment, we have now separated candidates into two categories in Table 2 in the revised manuscript: 'strong' (present in 3 or more biological replicates) and 'weak' candidates (present in 2 or fewer biological replicates). However, we think these weaker candidates should still be included in the manuscript, considering we did observe relationships between these proteins and learning. For example, ACC-1, which influences salt associative learning in *C. elegans*, was detected in one replicate of mass

spectrometry as a potential learning regulator (Figure S8A). We describe this classification in the main text on pages 21-22.

We also agree with Reviewer 2 that the overlap between individual candidate hits is low between biological replicates; the inclusion of Figure S2 in the original manuscript serves to highlight this limitation. However, it is also important to consider that there is notable overlap for whole molecular pathways between biological replicates of mass spectrometry data as shown in Figure 2 in the revised manuscript (this consideration is now mentioned on pages 13-14). We have included Figure 3 to illustrate representation for two metabolic processes across several biological replicates normally indispensable to animal health, as an example to provide additional visual aid for the overlap between replicates of mass spectrometry. We provide this figure (described on pages 13 & 15) to demonstrate the strength of our approach in that it can detect candidates not easily assessable by conventional forward or reverse genetic screens.

We also appreciate the opportunity to explain our approach. The criteria of “at least one unique peptide” was chosen based on a previous work for which we adapted for this manuscript (Prikas et al., 2020). It was not intended to inflate the number of hits but rather to ensure sensitivity in detecting low-abundance neuronal proteins. We have clarified this in our Methods (page 46).

(2) The "hits" that the authors chose to functionally characterize do not seem like strong candidate hits based on the proteomics data that they generated. Indeed, most of the hits are present in a single, or at most 2, biological replicate. It is unclear as to why the strongest hits were not characterized, which if mutant strains are publicly available, would not be a difficult experiment to perform.

We thank the reviewer for this important suggestion. To address this, we have described two molecular pathways with multiple components that appear in more than one biological replicate of mass spectrometry data in Figure 3 (main text on page 13). In addition, we have included Figures 6 & S7 where 9 additional single mutants corresponding to candidates in three or more biological replicates of mass spectrometry were tested for salt associative learning. Briefly, we found the following (number of replicates that a protein was unique to TurboID trained animals is in brackets):

- Novel arginine kinase F46H5.3 (4 replicates) displays an effect in both salt associative learning and salt aversive learning in the same direction (Figures 6A, 6B, & S9A, pages 31-32 & 37-38).
- Worms with a mutation for armadillo-domain protein C30G12.6 (3 replicates) only displayed an enhanced learning phenotype when non-backcrossed, not backcrossed. This suggests the enhanced learning phenotype was caused by a background mutation (Figure 6, pages 24-25).
- We did not observe an effect on salt associative learning when assessing mutations for the ciliogenesis protein IFT-139 (5 replicates), guanyl nucleotide factors AEX-3 or TAG52 (3 replicates), p38/MAPK pathway interactor FSN-1 (3 replicates), IGCAM/RIG-4 (3 replicates), and acetylcholine components ACR-2 (4 replicates) and ELP-1 (3 replicates) (Figure S7, on pages 27-30). However, we note throughout the section for which these candidates are described that only single gene mutants were tested, meaning that genes that function in redundant or compensatory pathways may not exhibit a detectable phenotype.

Because of the lack of strong evidence that these are indeed proteins regulated in the context of learning based on proteomics, including evidence of changes in the proteins (by imaging expression changes of fluorescent reporters or a biochemical approach), would increase confidence that these hits are genuine.

We thank Reviewer 2 for this suggestion – we agree that it would have been ideal to have additional evidence suggesting that changes in candidate protein levels are associated directly with learning. Ideally, we would have explored this aspect further; however, as outlined in response to Reviewer 1 Major Comment 2 (OPTIONAL), this was not feasible within the scope of the current study due to several practical challenges. Specifically, we attempted to generate pan-neuronal and endogenous promoter rescue lines for several candidates, but encountered significant challenges, including poor survival post-microinjection (likely due to protein overexpression toxicity) and reduced viability for behavioural assays, potentially linked to transgene-related reproductive defects. This information is now described on pages 39 & 40 of the revised work.

To address these limitations, we performed additional behavioural experiments where possible. We successfully generated a pan-neuronal promoter line for kin-2, which was tested and included in the revised manuscript (Figure 5B, pages 30 & 31). In addition, to confirm that observed learning phenotypes were due to the expected mutations and not background effects, we conducted experiments using backcrossed versions of several mutant lines as suggested by Reviewer 4 Cross Comment 3 (Figure 6, pages 23-24 & 24-26). Briefly, this shows that panneuronal expression of KIN-2 from the ce179 mutant allele is sufficient to repeat the enhanced learning phenotype observed in backcrossed kin-2(ce179) animals, providing additional evidence that the identified hits are required for learning. We also confirmed that F46H5.3 modulates salt associative learning, given both non-backcrossed and backcrossed F46H5.3(-) mutants display a learning enhancement phenotype. The revised text now describes this data on the page numbers mentioned above.

Minor Comments:

(1) The authors highlight that the proteins they discover seem to function uniquely in their gustatory associative paradigm, but this is not completely accurate. kin-2, which they characterize in figure 4, is required for positive butanone association (the authors even say as much in the manuscript) in Stein and Murphy, 2014.

We appreciate this correction and thank the Reviewer for pointing this out. We have amended the wording appropriately on page 31 to clarify our meaning.

“Although kin-2(ce179) mutants were not shown to impact salt aversive learning, they have been reported previously to display impaired intermediate-term memory (but intact learning and short-term memory) for butanone appetitive learning (Stein and Murphy, 2014).”

Reviewer #2 (Significance):

General Assessment:

The approach used in this study is interesting and has the potential to further our knowledge about the molecular mechanisms of associative behaviors. Strengths of the study include the design with carefully thought out controls, and the premise of combining their proteomics with behavioral analysis to better understand the biological significance of their proteomics findings. However, the criteria for defining hits and prioritization of hits for behavioral characterizations were major weaknesses of the paper.

Advance:

There have been multiple transcriptomic studies in the worm looking at gene expression changes in the context of behavioral training (Lakhina et al., 2015, Freytag 2017). This study compliments and extends those studies, by examining how the proteome changes

in a different training paradigm. This approach here could be employed for multiple different training paradigms, presenting a new technical advance for the field.

Audience:

This paper would be of interest to the broader field of behavioral and molecular neuroscience. Though it uses an invertebrate system, many findings in the worm regarding learning and memory translate to higher organisms.

I am an expert in molecular and behavioral neuroscience in both vertebrate and invertebrate models, with experience in genetics and genomics approaches.

We appreciate Reviewer 2's thoughtful assessment and constructive feedback. In response to concerns regarding definition and prioritisation of hits, we have revised our approach as detailed above to place more consideration on 'strong' hits present in multiple biological replicates. We have also added new behavioural data for additional mutants that fall into this category (Figures 6 & S7). We hope these revisions strengthen our study and enhance its relevance to the behavioural/molecular neuroscience community.

Reviewer #3 (Evidence, reproducibility and clarity):

Summary:

*In the manuscript titled "Identifying regulators of associative learning using a protein-labelling approach in *C. elegans*" the authors attempted to generate a snapshot of the proteomic changes that happen in the *C. elegans* nervous system during learning and memory formation. They employed the TurboID-based protein labeling method to identify the proteins that are uniquely found in samples that underwent training to associate no-salt with food, and consequently exhibited lower attraction to high salt in a chemotaxis assay. Using this system they obtained a list of target proteins that included proteins represented in molecular pathways previously implicated in associative learning. The authors then further validated some of the hits from the assay by testing single gene mutants for effects on learning and memory formation.*

Major Comments:

In the discussion section, the authors comment on the sources of "background noise" in their data and ways to improve the specificity. They provide some analysis on this aspect in Supplementary figure S2. However, a better visualization of non-specificity in the sample could be a GO analysis of tissue-specificity, and presented as a pie chart as in Figure 2A. Nonneuronal proteins such as MYO-2 or MYO-3 repeatedly show up on the "TurboID trained" lists in several biological replicates (Tables S2 and S3). If a major fraction of the proteins after subtraction of control lists are non-specific, that increases the likelihood that the "hits" observed are by chance. This analysis should be presented in one of the main figures as it is essential for the reader to gauge the reliability of the experiment.

We agree with this assessment and thank Reviewer 3 for this constructive suggestion. In response, we have now incorporated a comprehensive tissue-specific analysis of the learning proteome in the revised manuscript. Using the single neuron RNA-Seq database CeNGEN, we identified the proportion of neuronal vs non-neuronal proteins from each biological replicate of mass spectrometry data. Specifically, we present Table 1 on page 17 (which we originally intended to include in the manuscript, but inadvertently left out), which shows that 87-95% (i.e. a large majority) of proteins identified across replicates corresponded to genes detected in neurons, supporting that the TurboID enzyme was able to target the neuronal proteome as expected. Table 1 is now described in the main text of the revised work on page 16.

In addition, we performed neuron-specific analyses using both the WormBase gene enrichment tool and the CeNGEN single-cell transcriptomic database, which we describe in detail on our response to Reviewer 1 Major Comment 2. To summarise, these analyses revealed enrichment of several neuron classes, including those previously implicated in associative learning (e.g., ASEL, AIB, RIS, AVK) as well as neurons not previously studied in this context (e.g., IL1, DA9, DVC) (summarised in Table S7). By examining expression overlap across neuron types, we identified shared and distinct profiles that suggest potential functional connectivity and candidate circuits underlying behavioural plasticity (Figure 4). Taken together, these data show that the proteins identified in our dataset are (1) neuronal and (2) expressed in neurons that are known to be required for learning. Methods are detailed on pages 50-51.

Other than the above, the authors have provided sufficient details in their experimental and analysis procedures. They have performed appropriate controls, and their data has sufficient biological and technical replicates for statistical analysis.

We appreciate this positive feedback and thank the Reviewer for acknowledging the clarity of our experimental and analysis procedures.

Minor Comments:

There is an error in the first paragraph of the discussion, in the sentences discussing the learning effects in gar-1 mutant worms. The sentences in lines 12-16 on page 22 says that gar-1 mutants have improved salt-associative learning and defective salt-aversive learning, while in fact the data and figures state the opposite.

We appreciate the Reviewer noting this discrepancy. As clarified in our response to Reviewer 1, Major Comment 1 above, we reanalysed the behavioural data to ensure consistency across genotypes by comparing only those tested within the same biological replicates (thus having the same N for all genotypes). Upon this reanalysis, we found that the previously reported phenotype for gar-1 mutants in salt-associative learning was not statistically different from wildtype controls. Therefore, we have removed references to GAR-1 from the manuscript.

Reviewer #3 (Significance):

Strengths and limitations:

This study used neuron-specific TurboID expression with transient biotin exposure to capture a temporally restricted snapshot of the C. elegans nervous system proteome during salt-associative learning. This is an elegant method to identify proteins temporally specific to a certain condition. However, there are several limitations in the way the experiments and analyses were performed which affect the reliability of the data. As the authors themselves have noted in the discussion, background noise is a major issue and several steps could be taken to improve the noise at the experimental or analysis steps (use of integrated C. elegans lines to ensure uniformity of samples, flow cytometry to isolate neurons, quantitative mass spec to detect fold change vs. strict presence/absence).

Advance:

Several studies have demonstrated the use of proximity labeling to map the interactome by using a bait protein fusion. In fact, expressing TurboID not fused to a bait protein is often used as a negative control in proximity labeling experiments. However, this study demonstrates the use of free TurboID molecules to acquire a global snapshot of the proteome under a given condition.

Audience:

Even with the significant limitations, this study is specifically of interest to researchers interested in understanding learning and memory formation. Broadly, the methods used in this study could be modified to gain insights into the proteomic profiles at other transient developmental stages. The reviewer's field of expertise: Cell biology of C. elegans neurons.

We thank the reviewer for their thoughtful evaluation of our work. We appreciate the recognition of the novelty and potential of using neuron-specific TurboID to capture a temporally restricted snapshot of the C. elegans nervous system proteome during learning. We agree that this approach offers a unique opportunity to identify proteins associated with specific behavioural states in future studies.

We also appreciate the reviewer's comments regarding limitations in experimental and analytical design. In revising the manuscript, we have taken several steps to address these concerns and improve the clarity, rigour, and interpretability of our data. Specifically:

- We now provide a frequency-based representation of proteomic hits (Table 2), which helps clarify how candidate proteins were selected and highlights differences between trained and control groups.
- We have added neuron-specific enrichment analyses using both WormBase and CenGEN databases (Table S7 & Figure 4), which help identify candidate neurons and potential circuits involved in learning (methods on pages 50-51).
- We have clarified the rationale for using qualitative proteomics in the context of TurboID, in addition to acknowledging the challenges of integrating quantitative mass spectrometry with biotin-based enrichment (page 39). Additional methods for improving sample purity, such as using integrated lines or FACS-enrichment of neurons, could further refine this approach in future studies. For transparency, we did attempt to integrate the TurboID transgenic line to improve the strength and consistency of biotinylation signals. However, despite four rounds of backcrossing, this line exhibited unexpected phenotypes, including a failure to respond reliably to the established training protocol. As a result, we were unable to include it in the current study. Nonetheless, we believe our current approach provides a valuable proof-of-concept and lays the groundwork for future refinement.

By addressing the major concerns of peer reviewers, we believe our study makes a significant and impactful contribution by demonstrating the feasibility of using TurboID to capture learning-induced proteomic changes in the nervous system. The identification of novel learning-related mutants, including those involved in acetylcholine signalling and cAMP pathways, provides new directions for future research into the molecular and circuit-level mechanisms of behavioural plasticity.

Reviewer #4 (Evidence, reproducibility and clarity):

Summary:

In this manuscript, authors used a learning paradigm in C. elegans; when worms were fed in a saltless plate, its chemotaxis to salt is greatly reduced. To identify learning-related proteins, authors employed nervous system-specific transcriptome analysis to compare whole proteins in neurons between high-salt-fed animals and saltless-fed animals. Authors identified "learningspecific genes" which are observed only after saltless feeding. They categorized these proteins by GO analyses and pathway analyses, and further stepped forward to test mutants in selected genes identified by the proteome analysis. They find several mutants that are defective or hyper-proficient for learning,

including *acc-1/3* and *lgc-46* acetylcholine receptors, *gar-1* acetylcholine receptor GPCR, *glna-3* glutaminase involved in glutamate biosynthesis, and *kin-2*, a cAMP pathway gene. These mutants were not previously reported to have abnormality in the learning paradigm.

Major comments:

(1) There are problems in the data processing and presentation of the proteomics data in the current manuscript which deteriorates the utility of the data. First, as the authors discuss (page 24, lines 5-12), the current approach does not consider amount of the peptides. Authors state that their current approach is "conservative", because some of the proteins may be present in both control and learned samples but in different amounts. This reviewer has a concern in the opposite way: some of the identified proteins may be pseudo-positive artifacts caused by the analytical noise. The problem is that authors included peptides that are "present" in "TurboID, trained" sample but "absent" in the "Non-Tg, trained" and "TurboID, control" samples in any one of the biological replicates, to identify "learning proteome" (706 proteins, page 8, last line - page 9, line 8; page 32, line 21-22). The word "present" implies that they included even peptides whose amounts are just above the detection threshold, which is subject to random noise caused by the detector or during sample collection and preparation processes. This consideration is partly supported by the fact that only a small fraction of the proteins are common between biological replicates (honestly and respectably shown in Figure S2). Because of this problem, there is no statistical estimate of the identity in "learning proteome" in the current manuscript. Therefore, the presentation style in Tables S2 and S3 are not very useful for readers, especially because authors already subtracted proteins identified in Non-Tg samples, which must also suffer from stochastic noise. I suggest either quantifying the MS/MS signal, or if authors need to stick to the "present"/"absent" description of the MS/MS data, use the number of appearances in biological replicates of each protein as estimate of the quantity of each protein. For example, found in 2 replicates in "TurboID, learned" and in 0 replicates in "Non-Tg, trained". One can apply statistics to these counts. This said, I would like to stress that proteins related to acquisition of memory may be very rare, especially because learning-related changes likely occur in a small subset of neurons. Therefore, 1 time vs 0 time may be still important, as well as something like 5 times vs 1 time. In summary, quantitative description of the proteomics results is desired.

We thank the reviewer for these valuable comments and suggestions.

We acknowledge that quantitative proteomics would provide beneficial information; however, as also indicated by Reviewer 1 (in cross-comment), it is practically challenging to perform with TurboID. We have included discussion of potential future experiments involving quantitative mass spectrometry, as well as a comprehensive discussion of some of the limitations of our approach as summarised by this Reviewer, in the Discussion section (page 39). However, we note that our qualitative approach also provides beneficial knowledge, such as the identification of functional protein networks acting within biological pathways previously implicated in learning (Figure 2), and novel learning regulators ACC-1/3, LGC-46, and F46H5.3.

We agree with the assessment that the frequency of occurrence for each candidate we test per biological replicate is useful to disclose in the manuscript as a proxy for quantification. This was also highlighted by Reviewer 2 (Major Comment 1). As detailed above in response to R2, we have now separated candidates into two categories: 'strong' (present in 3 or more biological replicates) and 'weak' candidates (present in 2 or fewer biological replicates). We have also added behavioural data after testing 9 of these strong candidates in Figures 6 & S7.

We have also added Table 2 to the revised manuscript, which summarises the frequency-based representation of the proteomics results, as suggested. This is described on pages 22-23.

Briefly, this shows the range of candidates further explored using single mutant testing. Specifically, this data showed that many of the tested candidates were more frequently detected in trained worms compared to high-salt controls. This includes both strong and weak candidates, providing a clearer view of how proteomic frequency informed our selection for functional testing.

(2) There is another problem in the treatment of the behavioural data. In Experimental Procedures, authors state that they excluded data in which naive or control groups showed average CI < 0.6499, and/or trained groups showed average CI < -0.0499 or > 0.5499 for N2 (page 36, lines 5-7). How were these values determined? One common example for judging a data point as an outlier is > mean + 1.5, 2 or 3 SD, or < mean - 1.5, 2 or 3 SD. Are these values any of these standards, or determined through other methods? If these values were determined simply by authors' decision, it could potentially introduce a bias and in the worst cases lead to incorrect conclusions. A related question is, authors state "trained animals showed a lower CI (~0.3)" where in the referred Figure 1B, the corresponding data shows averages close to 0. Why is the inconsistency? The assay that authors use is close to those described in the previous literature (Kunitomo et al., <http://dx.doi.org/10.1038/ncomms3210>). In this previous paper, it was described that animals conditioned under no salt with food show negative CI and are attracted to the low salt concentration area. Quantitative analysis of behavioural patterns showed migration bias towards lower salt concentrations (negative chemotaxis). Essentially the same concept was reported by Luo et al. (<http://dx.doi.org/10.1016/j.neuron.2014.05.010>). The experimental procedure employed in the current work is very similar with those by the Japanese group, with a notable difference: the chemotaxis assay plate included 50mM NaCl in Kunitomo et al, while authors used chemotaxis plate without added NaCl (p35, line 18). The latter is expected to cause shallow gradient towards the low-salt area, which may be the reason for the weak negative CI in the trained animals. In any case, the value of CI itself is not a problem, and authors' current assay is valid. The only concern of mine is the potential of author-introduced cognitive bias, possibly affecting, for example, whether a certain mutant has a significant defect or not. What happens if the cut-offs of -0.0499 and 0.5499 are omitted and all data were included in the analyses? What are the average CIs of N2 in all performed experiments for each of naive, control and trained groups?

Thank you for pointing this out. As mentioned by both Reviewer 1 and Reviewer 4, the original manuscript states the following: "Data was excluded for salt associative learning experiments when wild-type N2 displayed (1) an average CI ≤ 0.6499 for naïve or control groups and/or (2) an average CI either < -0.0499 or > 0.5499 for trained groups."

To clarify, we only excluded experiments in rare cases where N2 worms did not display robust high salt attraction before training, or where trained N2 did not display the expected behavioural difference compared to untrained or high-salt control N2. These anomalies were typically attributable to clear contamination or starvation issues that could clearly be observed prior to counting chemotaxis indices on CTX plates.

We established these exclusion criteria in advance of conducting multiple learning assays to ensure an objective threshold for identifying and excluding assays affected by these rare but observable issues. However, these criteria were later found to be unnecessary, as N2 worms robustly displayed the expected untrained and trained phenotypes for salt associative learning when not compromised by starvation or contamination.

We understand that the original criteria may have appeared to introduce arbitrary bias in data selection. To address this concern, we have removed these criteria from the revised manuscript from page 50.

Minor comments:

(1) Related to Major comments 1), the successful effect of neuron-specific TurboID procedure was not evaluated. Authors obtained both TurboID and Non-Tg proteome data. Do they see enrichment of neuron-specific proteins? This can be easily tested, for example by using the list of neuron-specific genes by Kaletsky et al. (<http://dx.doi.org/10.1038/nature16483> or <http://dx.doi.org/10.1371/journal.pgen.1007559>), or referring to the CenGEN data.

We thank this Reviewer for this helpful suggestion, which was echoed by Reviewer 3 (Major Comment 1). As indicated in the response to R3 above, the revised manuscript now includes Table 1 as a tissue-specific analysis of the learning proteome, using the single neuron RNASeq database CeNGEN to identify the proportion of neuronal proteins from each biological replicate of mass spectrometry data. Generally, we observed a range of 87-95% of proteins corresponded to genes from the CeNGEN database that had been detected in neurons, providing evidence that the TurboID enzyme was able to target the neuronal proteome as expected. Table 1 is now described in the main text of the revised work on pages 16 & 17.

(2) The behavioural paradigm needs to be described accurately. Page 5, line 16-17, "C. elegans normally have a mild attraction towards higher salt concentration": in fact, C. elegans raised on NGM plates, which include approximately 50mM of NaCl, is attracted to around 50mM of NaCl (Kunitomo et al., Luo et al.) but not 100-200 mM.

We thank the Reviewer for pointing this out. We agree that clarification is necessary. The revised text reads as follows on page 5: "C. elegans are typically grown in the presence of salt (usually ~ 50 mM) and display an attraction toward this concentration when assayed for chemotaxis behaviour on a salt gradient (Kunitomo et al., 2013, Luo et al., 2014).

Training/conditioning with 'no salt + food' partially attenuates this attraction (group referred to 'trained')."

Authors call this assay "salt associative learning", which refers to the fact that worms associate salt concentration (CS) and either presence or absence of food (appetitive or aversive US) during conditioning (Kunitomo et al., Luo et al., Nagashima et al.) but they are looking at only association with presence of food, and for proteome analysis they only change the CS (NaCl concentration, as discussed in Discussion, p24, lines 4-5). It is better to attempt to avoid confusion to the readers in general.

Thank you Reviewer 4 for highlighting this clarity issue. We clarify our definition of "salt associative learning" for the purpose of this study in the revised manuscript on page 6 with the following text:

"Similar behavioural paradigms involving pairings between salt/no salt and food/no food have been previously described in the literature (Nagashima et al. 2019). Here, learning experiments were performed by conditioning worms with either 'no salt + food' (referred to as 'salt associative learning') or 'salt + no food' (called 'salt aversive learning')."

(3) page 32, line 23: the wording "excluding" is obscure and misleading because the elo-6 gene was included in the analysis.

We appreciate this Reviewer for pointing out this misleading comment, which was unintentional. We have now removed it from the text (on page 21).

(4) Typo at page 24, line 18: "that ACC-1" -> "than ACC-1".

This has been corrected (on page 37).

(5) Reference. In "LEO, T. H. T. et al.", given and sir names are flipped for all authors. Also, the paper has been formally published (<http://dx.doi.org/10.1016/j.cub.2023.07.041>).

We appreciate the Reviewer drawing our attention to this – the reference has been corrected and updated.

I would like to express my modest cross comments on the reviews:

(1) Many of the reviewers comment on the shortage in the quantitative nature of the proteome analysis, so it seems to be a consensus.

Thank you Reviewer 4 for this feedback. We appreciate the benefit in performing quantitative mass spectrometry, in that it provides an additional way to parse molecular mechanisms in a biological process (e.g., fold-changes in protein expression induced by learning). However, we note that quantitative mass spectrometry is challenging to integrate with TurboID due to the requirement to enrich for biotinylated peptides during sample processing (we now mention this on page 39). Nevertheless, it would be exciting to see this approach performed in a future study.

To address the limitations of our original qualitative approach and enhance the clarity and utility of our dataset, we have made the following revisions in the manuscript:

(1) Candidate selection criteria: We now clearly define how candidates were selected for functional testing, based on their frequency across biological replicates. Specifically, "strong candidates" were detected in three or more replicates, while "weak candidates" appeared in two or fewer.

(2) Frequency-based representation (Table 2): We appreciate the suggestion by Reviewer 4 (Major Comment 1) to quantify differences between high-salt control and trained groups. We now provide the frequency-based representation of the candidates tested in this study within our proteomics data in Table 2. This data showed that many of the tested candidates were more frequently detected in trained worms compared to high-salt controls. This includes both strong and weak candidates

We hope these additions help clarify our approach and demonstrate the value of the dataset, even within the constraints of qualitative proteomics.

(2) Also, tissue- or cell-specificity of the identified proteins were commonly discussed. In reviewer #3's first Major comment, appearance of non-neuronal protein in the list was pointed out, which collaborate with my (#4 reviewer's) question on successful identification of neuronal proteins by this method. On the other hand, reviewer #1 pointed out subset neuron-specific proteins in the list. Obviously, these issues need to be systematically described by the authors.

We agree with Reviewer 4 that these analyses provide a critical angle of analysis that is not explored in the original manuscript.

Tissue analysis (Reviewer 3 Major Comment 1): We have used the single neuron RNA-Seq database CeNGEN, to identify that 87-95% (i.e. a large majority) of proteins identified across replicates corresponded to genes detected in neurons. These findings support that the TurboID enzyme was able to target the neuronal proteome as expected. Table 1 provides this information as is now described in the main text of the revised work on page 16.

Neuron class analyses (Reviewer 1 Major Comment 2): In response, we have used the suggested Wormbase gene enrichment tool and CeNGEN. We specifically input proteins from the learning proteome into Wormbase, after filtering for proteins unique to TurboID trained animals. For CeNGEN, we compared genes/proteins from control worms and trained worms to identify potential neurons that may be involved in this learning paradigm.

Briefly, we found highlight a range of neuron classes known in learning (e.g., RIS interneurons), cells that affect behaviour but have not been explored in learning (e.g., IL1 polymodal neurons), and neurons for which their function/s are unknown (e.g., pharyngeal neuron I3). Corresponding text for this new analysis has been added on pages 16-20, with a new table and figure added to illustrate these findings (Table S7 & Figure 4). Methods are detailed on pages 50-51.

(3) Given reviewer #1's OPTIONAL Major comment, as an expert of behavioral assays in C. elegans, I would like to comment based on my experience that mutants received from Caenorhabditis Genetics Center or other labs often lose the phenotype after outcrossing by the wild type, indicating that a side mutation was responsible for the observed behavioral phenotype. Therefore, outcrossing may be helpful and easier than rescue experiments, though the latter are of course more accurate.

Thank you for this suggestion. To address the potential involvement of background mutations, we have done experiments with backcrossed versions of mutants tested where possible, as shown in Figure 6. We found that F46H5.3(-) mutants maintained enhanced learning capacity after backcrossing with wild type, compared to their non-backcrossed mutant line. This was in contrast to C30G12.6(-) animals which lost their enhanced learning phenotype following backcrossing using wild type worms. This is described in the text on pages 24-26.

(4) Just let me clarify the first Minor comment by reviewer #2. Authors described that the kin-2 mutant has abnormality in "salt associative learning" and "salt aversive learning", according to authors' terminology. In this comment by reviewer #2, "gustatory associative learning" probably refers to both of these assays.

Reviewer 4 is correct. We have amended the wording appropriately on page 31 to clarify our meaning to address Reviewer 2's comment.

"Although kin-2(ce179) mutants were not shown to impact salt aversive learning, they have been reported previously to display impaired intermediate-term memory (but intact learning and short-term memory) for butanone appetitive learning (Stein and Murphy, 2014)."

(5) There seem to be several typos in reviewer #1's Minor comments.

"In Page 9, Lines 17-18" -> "Page 8, Lines 17-18".

"Page 8, Line 24" -> "Page 7, Line 24".

"I would suggest to remove figure 3" -> "I would suggest to remove figure 2"

"summary figure similar to Figure 4" -> "summary figure similar to Figure 3"

"In the discussion Page 24, Line 14" -> "In the discussion Page 23, Line 14"

(I note that because a top page was inserted in the "merged" file but not in art file for review, there is a shift between authors' page numbers and pdf page numbers in the former.) It would be nice if reviewer #1 can confirm on these because I might be wrong.

We appreciate Reviewer 4 noting this, and can confirm that these are the correct references (as indicated by Reviewer 1 in their cross-comments)

Reviewer #4 (Significance):

(1) Total neural proteome analysis has not been conducted before for learning-induced changes, though transcriptome analysis has been performed for odor learning (Lakhina et al., <http://dx.doi.org/10.1016/j.neuron.2014.12.029>). This guarantees the novelty of this manuscript, because for some genes, protein levels may change even though mRNA levels remain the same. We note an example in which a proteome analysis utilizing TurboID, though not the comparison between trained/control, has led to finding of learning related proteins (Hiroki et al., <http://dx.doi.org/10.1038/s41467-022-30279-7>). As described in the Major comments 1) in the previous section, improvement of data presentation will be necessary to substantiate this novelty.

We appreciate this thoughtful feedback. We agree that while the neuronal transcriptome has been explored in Lakhina et al., 2015 for *C. elegans* in the context of memory, our study represents the first to examine learning-induced changes in the total neuronal proteome. We particularly agree with the statement that “for some genes, protein levels may change even though mRNA levels remain the same”. This is essential rationale that we now discuss on page 42.

Additionally, we acknowledge the relevance of the study by Hiroki et al., 2022, which used TurboID to identify learning-related proteins, though not in a trained versus control comparison. Our work builds on this by directly comparing trained and control conditions, thereby offering new insights into the proteomic landscape of learning. This is now clarified on page 36.

To substantiate the novelty and significance of our approach, we have revised the data presentation throughout the manuscript, including clearer candidate selection criteria, frequency-based representation of proteomic hits (Table 2), and neuron-specific enrichment analyses (Table S7 & Figure 4). We hope these improvements help convey the unique contribution of our study to the field.

*(2) Authors found six mutants that have abnormality in the salt learning (Fig. 4). These genes have not been described to have the abnormality, providing novel knowledge to the readers, especially those who work on *C. elegans* behavioural plasticity. Especially, involvement of acetylcholine neurotransmission has not been addressed. Although site of action (neurons involved) has not been tested in this manuscript, it will open the venue to further determine the way in which acetylcholine receptors, cAMP pathway etc. influences the learning process.*

Thank you Reviewer 4, for this encouraging feedback. To further strengthen the study and expand its relevance, we have tested additional mutants in response to Reviewer 3's comments, as shown in Figures 6 & S7. These results provide even more candidate genes and pathways for future exploration, enhancing the significance and impact of our study.

<https://doi.org/10.7554/eLife.108438.1.sa0>